Battle of Bengal

The TMC is fighting off anti-incumbency sentiment

The West Bengal Assembly election, to be held in two phases on April 23 and 29, is shaping up as a two-party race between the incumbent All India Trinamool Congress (TMC) and the Bharatiya Janata Party (BJP). The Left and the Congress have been pushed to the margins in the intense combat. The Bharatiya Gorkha Prajatantrik Morcha (BGPM) is contesting the three Darjeeling hill seats as an ally of the TMC. The TMC, which has been in power for three terms, has replaced 74 sitting Members of the Legislative Assembly to deflect anti-incumbency and bring in fresh faces to the field. In 2021, the TMC won 215 seats with a 48.02% vote share while the BJP secured 77 seats with 38%, a sharp rise from when it bagged just three seats and a 10.2% vote share in 2016. Chief Minister and TMC leader Mamata Banerjee is facing off against Leader of the Opposition Suvendu Adhikari, from the BJP, in Bhabanipur, where she is the sitting MLA. In 2021, she had lost to Mr. Adhikari in Nandigram by a margin of 1,956 votes, even while the TMC won yet another term. The run-up to the election in 2026 has been marred by controversies around the Special Intensive Revision (SIR) of the voter list. West Bengal's final electoral roll, which was published on February 28, saw the electorate shrink from 7.66 crore to 7.04 crore, with over 62 lakh names deleted. A further 60 lakh names remain under adjudication even as campaigning is underway, creating unprecedented uncertainty.

The standoff over SIR has overshadowed other debates. Having been in power for three terms, there is much that the TMC should be held accountable for, but SIR turned out to be a gift for Ms. Banerjee, which she has used to amplify her charges against the BJP that it is a party hostile to West Bengal and its voters. The BJP began 2026 with a reasonable case in its favour – the rape and murder of a trainee doctor in Kolkata, in 2024, had shaken public confidence in the Mamata government. But the party's limitations – and even refusal – to negotiate the cultural and linguistic diversity of India are markedly visible in the State, and it is placing its hopes on the trimmed voter list and the manoeuvrability of the electoral process. Its State unit is a house divided, while the TMC appears to have consolidated itself further, with Ms. Banerjee and her nephew Abhishek firmly in command. The BJP's upward trajectory may have hit the ceiling. The Congress is contesting without any alliances. The Left is not a power contender, but its vote share in select seats could influence margins in a tight race. In a long campaign period, there may be some surprises in store in West Bengal, as the BJP and the TMC both fight a battle for survival.

Grim future

India's economic fundamentals need a realistic reassessment

The economy has recently been yielding some discomfiting data, even before the war in West Asia began. The most recent of these was the February 2026 data on the Index of Eight Core Industries. This release showed that growth in the overall Index had dropped to a three-month low in February, half the rate of what it saw in January. This was not a case of a high base effect, either, with growth in February 2025 having been just 3.4%. The sector-wise data reveal further issues. The domestic crude oil industry has been contracting for six consecutive months. In fact, the sector has contracted in 20 out of the last 24 months. The natural gas sector, too, has contracted for the last 20 consecutive months. As context, from about the middle of last year, it had been increasingly clear that there would be some sort of escalation between the U.S. and Iran, even if the exact nature of the war now could not have been predicted then. It is no secret that India is highly dependent on energy imports, especially from the region that would have faced the brunt of any such escalation. As a matter of fact, a large part of the contraction in domestic production was likely because imports were coming cheap. Yet, as tensions increased, it might have been more prudent to have increased domestic oil and gas production over the last eight months so as to at least build reserves, if not to also temporarily reduce import dependence. Permanently reducing such dependence will take time, but a temporary domestic production push – as is belatedly underway – might have considerably alleviated the current supply crunch. Hindsight is 20-20, but that does not excuse a lack of foresight or an inability to learn. The Pradhan Mantri Ujjwala Yojana of 2016 should have triggered a concerted policy to secure LPG supplies and reserves.

This poor core sector performance comes on top of the new series of GDP data showing that the Indian economy is actually smaller than previously thought. Further, between 2022-23 and 2025-26, the contributions to GDP of key engines such as private consumption, capital formation, exports and imports have all fallen. The share of 'change in stocks' has nearly doubled, however, meaning that there is production but not enough sales. Soon, that production too will reduce to fall in line with the subdued demand. All of this leads into the impact of the current crisis. With commercial fuel sources curtailed, oil prices at more than $100 a barrel, and global economic uncertainty at a high, economists and rating agencies are already downgrading India's growth outlook to about 6.5%. Perhaps, India's much-vaunted macroeconomic fundamentals and resilience need a more realistic reassessment.

A curated selection of articles on the Editorial and Opinion pages for World TB Day, March 24

The evolving diagnostic landscape for tuberculosis

Dr. Soumya Swaminathan

is Chairperson of the M.S. Swaminathan Research Foundation and National Science Chair, Anusandhan National Research Foundation (ANRF)

Strengthening the tuberculosis testing infrastructure with an expanded diagnostic toolbox will improve outcomes and support TB elimination efforts

Two weeks ago, in the run-up to World Tuberculosis (TB) Day (which falls on March 24), the World Health Organization (WHO) formally recommended the use of new near point-of-care (NPOC) molecular tests for diagnosing TB. WHO also endorsed the use of tongue swab samples for TB testing and sputum pooling strategies to potentially improve testing efficiencies at scale. These are the latest developments in what has been an unusually remarkable decade for the TB diagnostic landscape, a decade when new technologies have not only emerged but have also been tested, swiftly recommended and utilised to advance global efforts to eliminate TB.

Probably the best example is the rapidly expanding use of portable chest X-ray (CXR), in tandem with artificial intelligence (AI) solutions for TB screening. In India, the National Tuberculosis Elimination Programme (NTEP) has made available hundreds of portable CXR machines that are being utilised for the Pradhan Mantri TB Mukt Bharat Abhiyaan, in an effort to take active TB screening into the community.

Until recently, X-ray access was limited to hospital settings and dependent on the availability of technicians and radiologists to record the X-ray and interpret the findings. Today, active case-finding efforts are in full swing through mobile vans equipped with portable CXR with AI, making it more convenient. While this takes diagnostic services into the community, we must ensure on-the-spot collection of sputum samples for those with any CXR findings to significantly reduce attrition in the diagnostic cascade. Further, those with CXR lesions other than TB (for example, lung cancer) must get the referral and treatment services they need. Systematic microplanning will also help to refine these screening efforts to focus on those who are most vulnerable, particularly in urban and tribal settings.

The use of AI also presents a credible option for opportunistic screening. Tens of thousands of X-rays are performed daily across India in public and private clinics and hospitals for various reasons. Installing AI algorithms in digital X-ray machines that can swiftly identify suspicious lesions, whether for TB or other respiratory disorders, can reduce delays in diagnosis. Building health system capacity, particularly at the service delivery level, to use AI effectively is an important first step.

The diagnostic landscape is shifting

India has long relied on sputum smear microscopy as the primary test for diagnosing TB, despite its low sensitivity and inability to identify drug-resistance. In 2016, India began scaling up molecular testing – starting with the Cartridge-based Nucleic Acid Amplification Test (CBNAAT) and later adopting the indigenous Truenat in 2020. Today, many more Indian products are available, and decentralised access to molecular testing is central to the TB programme.

However, access to up-front molecular testing remains uneven. To achieve 100% testing through NAAT prior to treatment initiation, we must also strengthen sputum collection and transportation mechanisms, especially for the elderly, people with disabilities, and those who live in hard-to-reach areas. To complete the diagnostic cascade swiftly, we must reduce delays in first-line and second-line drug resistance testing and work with the general health system to leverage diagnostic human resource capacity. Minimising test turnaround time will ensure that people with drug-resistant TB are started on the correct treatment regimen at the earliest.

With WHO's endorsement of near point-of-care Nucleic Acid Amplification Tests (nPOC-NAAT), there are now more options for molecular testing at the primary-care level that are not heavily dependent on infrastructure. The use of non-sputum samples, such as tongue swabs, are especially useful for those who find it difficult to produce a good sputum sample, including children.

It is important to acknowledge that none of these tools is a complete solution in itself. What this growing collection of new tests offers is a comprehensive toolbox for TB diagnosis. Diagnostic network optimisation can help identify which tools to best use, individually and in combination, and where they should be located, to establish a diagnostic cascade that is accessible, affordable and person-centered. India must field-test these tools through robust implementation research that matches the pace of innovation, so that solutions can be tested in specific settings and decisions taken for scale-up.

Indian innovators have made a significant contribution, bringing to the market screening and diagnostic tools for global use, and will continue to play an important role in taking them to scale across the country. Over the years, we must examine the evidence to evolve clear diagnostic algorithms that are operationally feasible, involving the use of both AI-enabled CXR and molecular tests for diagnosing TB and identifying any antibiotic resistance.

Research and innovation priorities

Simultaneously, we must streamline innovation assessment and procurement pathways so that all evidence is rigorously reviewed by the Indian Council of Medical Research (ICMR), comprehensive health technology assessments address clinical, social, economic, and ethical concerns, and only recommended tools are procured and distributed in both public and private sectors. Several areas still require research and innovation.

First, as India is scaling up testing for TB infection and access to TB preventive therapy (TPT), we need more cost-effective and easy-to-use biomarkers that can identify and predict those at high risk for disease progression. Studies have shown that 'test and treat' approaches are more likely to convince people with TB infection to start on TPT, if they are at higher risk for disease, particularly in the private sector.

Second, the National TB Survey shows that asymptomatic TB is a serious issue, making symptom-based screening alone insufficient. Increased access to chest X-ray is a step in the right direction but we also need faster, less invasive diagnostic tools that use saliva and other non-sputum samples. More feasibility studies are required to understand utility and performance in real-world settings, particularly among individuals with low bacterial load.

Third, diagnosing TB in children remains challenging. Children cannot produce sputum as they often have low bacillary levels. Testing using stool samples has been tried in some countries and we need more implementation research in this area. We need far greater investment – and urgency – in finding the right new tools for diagnosing TB in children.

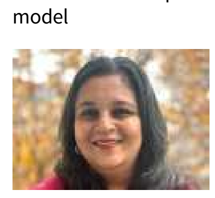
Finally, diagnosing extra-pulmonary TB (EP-TB) – which is almost a quarter of India's TB burden – remains difficult, often inaccessible and very expensive. Misdiagnosis as well as delayed diagnosis of EP-TB contributes to both catastrophic costs for families and poor outcomes for individuals. There are some pilots being done globally using AI-enabled portable ultrasound devices alongside molecular testing using new tools; we need India-specific evidence on this, including on cost-effectiveness.

Investing in strengthening the diagnostic landscape can be the greatest return on investment for the public health system. The sooner a person with TB is diagnosed, ideally before they are very ill, the more likely they are to recover fully, with fewer long-term post-treatment morbidities. Early diagnosis can not only improve treatment outcomes and long-term lung health but also dramatically reduce transmission within communities. A careful but swift and streamlined public sector expansion of diagnostics can also significantly reduce out-of-pocket expenditure for families affected by TB. An evidence-based, strong and expanded diagnostic toolbox can be a powerful lever to accelerate pathways to TB elimination in India.

A decade of building India's TB Champion movement

Dr. Nalini Krishnan

is Co-Founder and Executive Secretary of the Resource Group for Education and Advocacy for Community Health (REACH), a non-profit organisation working on tuberculosis (TB) for 27 years and a pioneer of the TB Survivor to Champion model



Anupama Srinivasan

is Deputy Director, Resource Group for Education and Advocacy for Community Health (REACH)

Survivors become 'TB Champions', supporting treatment, awareness and community engagement

In early 2017, at a workshop in New Delhi, Cedric Fernandes, a tuberculosis (TB) survivor and advocate from Maharashtra, said to a room full of other TB survivors, "It has been very lonely having this disease. I am fed up with the loneliness and I hope all of you are too." Addressing the organising team at the workshop, he gently admonished us: , "I am so glad that someone is finally bringing us together ... but what took so long?"

India diagnoses and treats over 25 lakh people with TB every year, which means that there are hundreds of thousands of individuals such as Cedric, who are coming to terms with the disease and fighting to overcome it. Their path, from diagnosis to cure, is often fraught with challenges in an ecosystem where stigma and social exclusion continue to thrive within communities. TB remains one of the most stigmatised illnesses in both urban and rural India; stigma and discrimination uniquely impact women, men, transgender communities, children and adolescents, the elderly and other socially vulnerable groups.

Challenging the belief

One of the early criticisms of the global TB response was that it remained stubbornly biomedical, focusing solely on clinical interventions whilst excluding communities and devaluing lived experience. Back in 2016, when we first began to imagine and build our wish list for how TB-affected communities – people with TB, TB survivors and their families – could play a greater role, there were many sceptics. Most of all, we were told that TB survivors, unlike people living with HIV, would want to move on with their lives once they had completed treatment, and would not be interested in becoming peer supporters or advocates. Ten years later, the evolution of the TB Champion movement in India has firmly dispelled this notion. TB Champions are survivors who are passionate, motivated and committed individuals, firm in their conviction that they have an important role to play in India's response to TB.

India's National TB Elimination Programme (NTEP) provides access to high-quality free diagnostics and treatment. Over the last few years, diagnosis and case-finding have accelerated, and treatment outcomes have steadily improved, which means that more people are being cured and mortality has reduced, with fewer people dying of TB. Newer strategies – Artificial Intelligence-enabled screening, active case-finding within communities, shorter and less toxic treatment regimens, expanded nutrition support, differentiated care approaches – have been adopted. But even the most robust health system cannot achieve elimination of any disease on its own, without active community participation. This is where TB Champions are uniquely poised to draw on their own personal experiences of TB to be powerful communicators and advocates for changes.

'Empower, support, educate'

In the early years of working with TB survivors, our focus was on developing the 'Survivor to Champion' training curriculum, which has since been formally adopted by the NTEP.

The involvement of TB Champions has reaffirmed a conviction that we have held from our inception – that a person with TB needs emotional support and motivation to get them through the treatment period, as much as they need high-quality clinical care. TB Champion-led peer counselling, through both one-to-one communication and support groups, has helped to improve treatment literacy and empowered people with TB with the knowledge they need. We have found a greater level of confidence and comfort among people with TB who have been supported by TB Champions and a significant reduction in self-stigmatisation.

Even today, myths and misconceptions about TB persist. TB Champions have played a critical role in dispelling these and in improving knowledge of TB, the symptoms of the disease and where to seek care. They literally reach the last mile through regular community meetings, where they talk about the disease, share their own stories and reassure people that TB is a treatable and curable disease. This has been a powerful anti-stigma strategy, enabling people to come forward to seek diagnosis and treatment.

Building networks, looking ahead

In addition, over the last decade, TB Champions in several States have come together to form survivor-led networks. These collective entities are naturally designed to be a powerful bridge between those seeking care and those providing services, especially among social vulnerable groups. Importantly, networks have become a source of strength for TB survivors, many of whom continue to experience the physical, social and economic toll of the disease even after being cured. However, the long-term future and sustainability of networks remains uncertain, given that they are dependent on external resources for funding. The next challenge is to test and build self-sustaining socio-economic models for survivor-led networks.

Over the last 10 years, we have been humbled by the trust that thousands of TB Champions from across the country have placed in us. They have travelled long distances to attend our workshops, sharing their TB journeys and stories, their pain and successes. They have become empowered local leaders, unafraid to use the TB Champion moniker to refer to themselves. They remain staunchly committed to working with their communities – supporting people with TB and their families, talking about stigma, meeting their panchayat leaders – and doing all of this for one simple, powerful reason – 'so that no one else should suffer like I did'.

LETTERS TO THE EDITOR

Erratic policy

The dramatic post by U.S. President Donald Trump, on a 'five-day ceasefire of sorts', must have stunned Benjamin Netanyahu's Israel. The 'second coming' of Mr. Trump has been marked less by strategic doctrine and more by disruptive impulse. Alliances once seen as bedrock as NATO, have been treated as transactional, their value weighed against immediate domestic optics rather than long-term geopolitical balance. Trade agreements and multilateral pacts have been entered and exited with disconcerting ease, reinforcing a perception of the U.S. as an uncertain partner. In conflict theatres, this inconsistency has been stark. Support extended to Ukraine appeared forceful yet episodic, leaving allies unsure of endurance. Europe, once a strategic anchor, has often been addressed with impatience rather than partnership. Even Israel, a beneficiary of unequivocal backing, has found that patronage can pivot abruptly.

R. Narayanan,
Navi Mumbai

PNG transition issue

I wish to draw the attention of government officials to a troubling issue being faced by many consumers registered for piped natural gas (PNG) connections with GAIL (India) Limited. Under the recent government mandate (LPG Amendment Order 2026), households with functional PNG connections must surrender their domestic LPG cylinders. However, GAIL officials and distributors are reportedly asking consumers to surrender their LPG connections before the PNG line has been installed or activated.

Parthasarathy P.,
New Delhi

Letters emailed to letters@thehindu.co.in must carry the postal address.

From disease-specific to person-centred care

By using TB as an entry point, integrated healthcare delivery can optimise and improve public health efficiency

Anupama Srinivasan

Deputy Director of REACH

Dr. Ramya Ananthakrishnan

Director of REACH

Tuberculosis (TB) rarely presents as a standalone disease in an individual. Many people with TB have other comorbidities or disease conditions that they must simultaneously cope with, while being on treatment for TB. Therefore, treating TB alone is not enough. Adopting integrated care approaches by transitioning from disease-specific to person-centred care can accelerate India's efforts to achieve both universal health coverage and TB elimination. Simply put, we must treat the person and not the disease.

Take the example of Divya (name changed), a 48-year-old woman who was diagnosed with TB in 2023. She had been living with diabetes and hypertension for several years and it is highly likely that her diabetes played a role in her contracting TB. Crucially, she had poor glycemic control with a HbA1c over 10, which had a direct impact on her treatment for TB.

Infectious diseases like TB; non-communicable diseases such as diabetes and chronic respiratory diseases; malnutrition; and social and economic vulnerabilities inevitably intersect in individuals and families, adversely impacting outcomes and the overall quality of life. Recognising the interconnected nature of these multiple vulnerabilities is the first step towards delivering integrated care.

Treating TB and diabetes

As in the case of Divya, an important opportunity for integration is between TB and diabetes, given India's growing diabetes burden. Over 15 years ago, the National TB Elimination Programme (NTEP) identified diabetes as a key comorbidity for TB, and rolled out formal bidirectional screening guidelines for TB-diabetes. Every person with TB must be screened for diabetes and people with diabetes must ideally be screened for TB once a year. Studies have shown that people with diabetes are more likely to develop active TB disease than those without; equally, people with TB also have diabetes are more likely to have poor TB treatment outcomes.

In Chennai, in a cohort of over 9,000 people with TB for three years, the authors found that over one-third (34%) had diabetes, and of this group, 41% had poor glycemic control which is the biggest challenge. Monitoring the blood sugar levels of a person with TB and diabetes throughout the TB treatment period is therefore essential, as is expanding clinical management and the counselling that is provided from TB-specific guidance to also focus on overall lifestyle, physical activity, diet and nutrition etc.

TB patients seen inside the Govind Ballabh Pant Hospital on March 24, 2025. ANI

Integrating respiratory care

Another important opportunity for integration is at the intersection of TB and Chronic Respiratory Diseases (CRD) such as asthma or Chronic Obstructive Pulmonary Disease (COPD), at the screening stage itself. Only a fraction of people with respiratory symptoms are eventually diagnosed with TB – the rest are likely to have viral flu, bronchitis, asthma etc. In REACH's ongoing TB-CRD integrated care pilot in two districts each of Bihar and Tamil Nadu, nearly 3,000 people with COPD and asthma were identified and linked to care, among the 26,000 people who were screened for TB but found TB-negative. In addition, integrated care can also change the course of post-treatment follow-up, given that many people who complete TB treatment continue to have respiratory impairments.

Other areas of intersection include undernutrition and risk factors such as smoking and alcohol use. This multi-disease approach is already envisaged in the framework of the Ayushman Arogya Mandirs for delivering comprehensive care at the community and primary care levels.

Delivering integrated care at scale can have its share of challenges. At the health system level, it is a likely stressor on already strained human resources, with the brunt of the workload falling on community health workers. In addition, many of India's large health programmes are vertical and not always designed to talk to each other, in terms of both human resources and data; this will require both creative and pragmatic solutions.

Integrated care delivery will require adequate and additional human resources, a careful readjustment of the health worker to care recipient ratio and investment in periodic training.

Holistic wellness

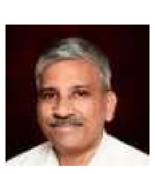
Given that the NTEP is among India's more robust, well-structured public health programmes, using TB as an entry point to identify and address closely associated morbidities such as diabetes, CRDs, undernutrition etc. will help optimise healthcare delivery and improve public health efficiency. More importantly, it could reduce delays in diagnosis, minimise the need for visits to several health facilities, and help overcome barriers to accessing a continuum of care. With empathetic person-centred design, integrated healthcare delivery can be a win-win for health systems, individuals and families affected by TB.

The need to integrate nutrition in TB care

For most patients with TB who are severely underweight, nutritional support is an essential and not optional part of treatment

Dr. Madhavi Bhargava

is a public health specialist and teaches at Yenepoya Medical College, Mangalore. Led the RATIONS trial in India



Dr. Anurag Bhargava

is a physician-epidemiologist and teaches at Kasturba Medical College, Mangalore. Led the RATIONS trial in India

In 2026, India still has the highest number of new cases of Tuberculosis (TB) globally and a decreasing but still unacceptable number of about three lakh deaths due to TB. Almost 10% of successfully treated patients have recurrence within two years. An answer to all these challenges may partly lie in addressing a leading cause and consequence of TB in India, which is undernutrition, closely related to poverty, food insecurity and a predominantly cereal-based diet with deficient protein intake.

On the occasion of World TB Day, evidence from the Reducing Activation of Tuberculosis by Improvement Of Nutritional Status (RATIONS) trial could offer insight into the importance of nutritional interventions. RATIONS was done in Jharkhand with support from the Indian Council of Medical Research and was the first such trial of a food-based intervention. It aimed at reducing new TB cases in TB-affected families, and improving treatment outcomes for patients with TB.

Ramlal, a 19-year-old tribal boy in the trial, weighed only 26 kg and was almost bed-bound at the time of his diagnosis of lung TB. Along with TB drugs, he received a monthly food basket. Over the next few weeks, he started walking and could cycle to work at the end of the treatment, weighing 42 kg. Laxmi, weighing 34 kg, had TB two years ago and had stopped medication after she developed jaundice within two weeks of starting anti-TB drugs. Now, as a result of the trial, she could adhere to the medications and also breastfeed her four-month-old.

Preventing deaths due to TB

Undernutrition, especially severe undernutrition, is common in patients in India. It is a major reversible risk factor for death, drug toxicity, and recurrence after successful treatment. For most patients with TB who are severely underweight, nutritional support is an essential and not optional part of treatment. Poor families are food-insecure and become even more so due to the disease. In the RATIONS trial, a monthly 10 kg food basket of cereals, pulses, milk powder, and oil led to a weight gain of almost 4.5 kg in patients. Even a 5% weight gain in the first two months was protective against death, reducing the risk by more than 60%.

PM TB Mukt Bharath kits, which include basic nutritious food items, distributed to patients in Chittoor, Andhra Pradesh in 2023. FILE PHOTO

In India, the National TB Elimination Programme's (NTEP) initiatives in addressing undernutrition in TB care are welcome. For example, the monthly direct benefit transfer under the Ni-Kshay Poshan Yojana increased from ₹500 to ₹1,000 and similarly the Ni-Kshay Mitra Yojana, although not universal, provides in-kind support to patients with TB. However, the plan to introduce energy-dense nutrition supplements for two months to underweight patients with TB needs to be considered with caution, given higher expenses, and the mystification of an adequate balanced diet apart from the risk of monotony and poor acceptability. One in 7 patients with TB in India may have diabetes which also needs to be taken into consideration.

A new avenue

While infection with the TB germ is necessary to develop the disease, it is not sufficient to produce active TB. Almost 90% of those infected, with good immunity, never develop the disease in their lifetime. Undernutrition is the most common cause of poor immunity in India and globally. It contributes to nearly 40% of new cases in India, and improved population-level nutritional status offers an alternative approach to reducing TB incidence.

Diets of the poor in India are particularly deficient in quality protein. TB-affected families experience vulnerabilities of poverty, food insecurity, and undernutrition. The 1.5 kg per month of pulses per family member, along with extra rice, worked like a vaccine in the RATIONS trial, with new TB cases in these families coming down by almost half.

The World Health Organization released new guidelines last year recommending nutritional interventions in settings of poverty and undernutrition, drawing significantly from the evidence of the RATIONS trial.

Improving nutrition security by including pulses in the public distribution system, and expanding the Ni-Kshay Mitra scheme to include TB patients' households will have long-term benefits in reducing the TB burden in India.

The way ahead

India is a global leader in implementing comprehensive nutrition assessment and support initiatives through the NTEP. Research and evidence from India have influenced global policy. Some of the next positive steps will be to record and report nutritional assessment at diagnosis; at two months; and at the end of treatment. This will help to identify those at risk of death, monitor early weight gain, and assess nutritional recovery. A differentiated TB care model in line with the TN-KET intervention in Tamil Nadu can identify and prioritise severely underweight patients. This will require local contextualisation of food baskets, with due attention given to comorbidities, supported by transport and inpatient care for the sick.

The inverse care law also works in TB and nutrition – the most vulnerable persons such as those in difficult-to-reach areas, the tribals, and the migrant workers, should not be left out.

The views expressed are personal.

Mental health of TB patients should not be overlooked

Studies from India have shown that at least one-third of people with TB experience symptoms of depression or anxiety

Dr. R. Thara

Co-Founder and Vice-Chair of the Schizophrenia Research Foundation (SCARF)

At least one in seven Indians experience a mental health condition at some point in their lives. In people with infectious diseases like Tuberculosis (TB) or HIV, this increases to anywhere between one-third to a half of those affected. TB takes a toll on not only physical but also mental health, either in the form of definite diagnosable conditions like depression, anxiety or psychoses or as emotional suffering worsened by social stigma. TB is fundamentally a disease of inequity, perpetuated by poverty and simultaneously contributing to increasing poverty. Unemployment, the lack of livelihood, food insecurity and malnutrition are social drivers for both TB and mental health conditions.

Studies from India have shown that at least one-third of people with TB experience symptoms of depression and anxiety, which often occur together as well. The mental health burden is estimated to be even higher among people with drug-resistant TB, with upto two-thirds having mental health issues. There are anecdotal accounts from TB survivors on suicidal ideation as well as documented instances of people with TB dying by suicide. There is not enough evidence about the prevalence of severe mental illnesses such as schizophrenia or bipolar disorders in people affected by TB.

The stigma associated with TB remains pervasive, and includes self-stigma, stigma within families, at workplaces and within communities. Many TB survivors have spoken of the guilt and shame they feel on being diagnosed with TB, despite knowing that it is an airborne disease that can affect anyone. For those with pulmonary TB, the guilt is driven primarily by the fear of transmitting the disease within their families, particularly to children. Those with depression or anxiety and TB must cope with an increased burden of stigma associated with both conditions, which can further worsen mental health.

TB and mental health

Given the high prevalence of common mental health disorders in the general population, it is possible that depression or anxiety could have been pre-existing and unidentified, even prior to the TB diagnosis; there have not been enough studies on this aspect. Depression or anxiety could delay care-seeking for TB. Long-standing untreated depression can also weaken the body's immune responses which in turn could lead to active TB disease – this is yet again another under-researched aspect.

A doctor offers flowers to a TB patient in Agartala on World TB day. PTI

Post diagnosis, untreated mental health conditions can adversely affect TB treatment outcomes. A person with TB who is depressed, and has not received mental healthcare, is more likely to take TB treatment irregularly or discontinue treatment prematurely. This can lead to treatment failure, relapse, or drug-resistance.

Co-existing alcohol and tobacco use, in the absence of access to de-addiction programmes, are also crucial factors affecting the treatment for TB.

Screening and identifying

Integrating routine screening for all people with TB for depression and anxiety is the starting point. Building awareness about common mental health disorders among healthcare workers, equipping them with the knowledge and skills to provide mental health first-aid and ensuring early screening at the right point in the TB care cascade are the other essential steps. Primary healthcare workers and community supporters, who are often the first to notice any psychological distress, should be trained to elicit information about sleep, appetite, mood, stress etc. which can help early identification of mental health conditions.

Given that TB itself could trigger depression, mental health screening must ideally be done at multiple time points during the TB treatment period.

From an individual perspective, being treated for TB and depression at the same time can be a stressful experience. While India's TB programme already provides decentralised access to free anti-TB drugs and financial aid for nutrition, strengthening the availability of drugs to treat depression or anxiety through the district mental health programme is vital. While antidepressants or antipsychotics can be used alongside anti-TB drugs, clinicians must monitor for any adverse drug reactions or interactions. Psychological interventions, including cognitive-behavioural therapy and participation in peer support groups, can also help reduce isolation. To provide these services, qualified and trained mental health professionals are needed, particularly in rural and tribal areas.

Looking out for everyone

Mental health interventions cannot be limited to people with TB. The task of caring for people with TB can place considerable strain on families. Studies in India have shown that up to 80% of caregivers may exhibit high levels of Expressed negative Emotions (EE) and high stress levels due to their caregiving responsibilities. Often, high EE leads to increased caregiver burden and poorer treatment outcomes. In India's family-centred ecosystem, where most of the caregiving needs are met by family members, this can lead to critical or over-involved caregiving because of emotional, financial, and social strains.

Psychoeducation is important not just for persons with TB but also for their families and caregivers.

A blended and syndemic TB-mental health approach that considers mental healthcare as an essential part of TB care, rather than a standalone service, can significantly improve TB outcomes, reduce the burden on caregivers and improve long-term mental health.

Text&Context

THE HINDU

NEWS IN NUMBERS

Rejection rate of Right to Information applications in 2024-25

3.26 in per cent. The rejection rate of Right to Information (RTI) applications has decreased to 3.26 per cent in 2024-25, the Centre has said. If an applicant receives no reply to an RTI within the prescribed time or is aggrieved by the public information officer's decision, they may file appeals before the CIC, Union Minister Jitendra Singh said. PTI

Number of people killed due to rains in Kenya in March

81 Torrential rains continued to pound Kenya as flash floods killed at least 81 people this month, authorities said Sunday. Flash floods also destroyed around 1,200 hectares of farmland and swept away crops. Over 3,000 families have been forced from their homes in the community of Nyakach, with some sheltering in eight evacuation centres, locals said. AFP

Number of AC e-buses to be deployed in 116 cities by 2027

10,000 On Monday, the government informed the Rajya Sabha that 10,000 air-conditioned e-buses would be provided to 116 cities across 26 States and UTs by the end of 2027 under the PM-eBus Sewa Scheme, and a new scheme would be launched thereafter to provide another 35,000 e-buses. . PTI

Number of airlines that exited India in the last decade

11 As many as 11 airlines exited the market in the last 10 years due to financial stress, non-availability of aircraft and other internal issues, according to the government. Minister of State for Civil Aviation Murlidhar Mohol told the Rajya Sabha on Monday that the aviation sector is deregulated, and airlines operate based on commercial considerations. PTI

Persons displaced in Ethiopia's Tigray over the past three years

1 In million. More than 1,300 people have died from hunger in displacement camps in Ethiopia's volatile Tigray region over the past three years, a local official told AFP on Monday. A civil war between the federal government and forces from the northern Tigray region in 2020-2022 killed at least 600,000 people according to African Union estimates. AFP

COMPILED BY THE HINDU DATA TEAM

Path ahead for Nepal's new leadership

After securing a historic mandate in Nepal's polls, Balendra Shah and the RSP face the task of navigating the West Asia conflict and managing ties with India, China and the U.S. While capitalising on old-guard fatigue proved an effective electoral strategy, the new leadership now faces a test in international relations

WORLD INSIGHT

Bibek Raj Kandel

In the noisy, crowded landscape of Nepali politics, the rise of Prime Minister-designate Balendra Shah and his party, the Rastriya Swatantra Party (RSP), represents a rare political anomaly. While veterans of the 2008 republic traded barbs, the rapper-engineer-turned-mayor bypassed traditional campaigning for a "monastic" silence. His victory in Nepal's post-Gen Z parliamentary election on March 5 secured a historic mandate for an alternative force he joined a mere six weeks before the polls. Throughout the campaign, Shah spoke for barely thirty minutes, avoided media interviews and notably never once asked for a vote.

His unapologetic critiques of the political establishment during his tenure as Kathmandu's mayor, mirrored a generation exhausted by stale ideological party politics. In a nation with a median age of twenty-five, Shah's reputation as a disciplined, clean reformist promising better governance became a viral mandate. His calculated silence mirrored these frustrations, positioning him as the ultimate outsider for an electorate eager for results.

Beyond domestic politics

While capitalising on domestic old-guard fatigue served Shah as a winning electoral strategy, Nepal's hard geopolitical reality remains stubbornly unchanged. In Nepal, political shifts rarely remain purely domestic, often prompting debates about foreign influence given its geography wedged between the rivalries of India and China.

Yet, Shah, however has sought to counter this by projecting an image of a staunch nationalist. As mayor, his symbolism was deliberate: hanging a "Greater Nepal" map in his office as a direct retort to the "Akhand Bharat" mural in India's new Parliament House, and

Balendra Shah with children and supporters during a door-to-door election campaign in Nepal. AFP

briefly banning Indian films. Simultaneously, he signaled caution toward Beijing by dropping a China-backed industrial park from his election manifesto. By distancing himself from large-scale geopolitical projects, Shah reframed the narrative, asserting a sovereignty that felt local, visible, and unapologetically independent.

Balancing ties with India and China

Historically, Nepal's politics followed rigid ideological scripts. The Nepali Congress, the country's oldest liberal force, leaned toward Delhi, while communist factions like CPN- UML maintained proactive affinities with Beijing. And at times, the ideological rhetoric from Kathmandu stretched far beyond the Himalayas, from debates over Venezuela's regime change to contentious political statements on the Ukraine war, issues largely peripheral to Nepal's own priorities. Under Shah, this era of predictable ideological signaling may finally be fading. His minimalist approach and by speaking less about the world's ideological battles, Shah's personality itself can be potent strategic asset to Nepal, but it is immediately replaced by a different kind of geopolitical pressure that needs sustained diplomatic communique.

India remains Nepal's most consequential partner, linked by an open border, "Roti-Beti" social bonds, and accounting for a significant share of its trade, supplying virtually all of its petroleum, and emerging as the primary market for Nepal's burgeoning hydropower exports. Meanwhile, China has deepened its footprint through major infrastructure financing, such as $216 million Pokhara International Airport. Intended as a regional gateway, its underutilisation is viewed in Kathmandu as a casualty of the broader India-China friction, particularly New Delhi's hesitancy to facilitate air routes for Chinese-financed infrastructure.

This is perhaps inaugural diplomatic crucible for Balendra Shah. Having invested heavily, Nepal cannot afford for such massive infrastructure to remain a "white elephant." The "monastic" outsider must now navigate a landscape where India is indispensable and China is influential. Meanwhile, Washington has pivoted from an aid partner to a high-stakes strategic interest partner, with recent post-election congratulations explicitly hinting at "shared security goals."

But the first "baptism by fire" for the Shah administration may well lie in the volatile West Asia. With millions of Nepalese migrants' lives and critical energy lifelines tied to the Persian Gulf caught in an escalating US-Israel-Iran War, India's logistical depth as a regional first responder offers an indispensable synergy for contingency planning to Kathmandu. This is where Balen's nationalist doctrine must meet the hard reality of strategic pragmatism. The new leadership remains untested in conventional diplomacy.

An opportunity for India

This moment offers a rare opening for New Delhi to recalibrate. India must move past the coercive shadow of the 2015 blockade and what is widely perceived in Kathmandu as outdated impulse for political micromanagement. New Delhi must recognise this transition as a new opening for modern partnership that respects the domestic rise of 'Nepal First' politics. Shah's mandate mirrors India's own tectonic shift in 2014; in Shah, Nepal has found its "strongman" archetype, a leader whose personal charisma and promise of technocratic reform have dismantled a decades-old establishment.

In the end, Nepal's geopolitical reality remains unchanged even as its politics transform at home. India's proximity will always matter most, China's influence will remain structural reality, and world powers like the U.S. will continue to pursue its strategic interests with renewed rigor. Ultimately, Nepal's voters were not adjudicating between global strategies, but seeking domestic renewal. For Balendra Shah and the RSP, bashing the "old guard" is a potent domestic strategy, but it carries zero currency in the cold-eyed theater of international relations.

(Bibek Raj Kandel is an analyst and AsiaGlobal Fellow at the University of Hong Kong and a graduate of Harvard Kennedy School.)

THE GIST

In the landscape of Nepali politics, the rise of Prime Minister-designate Balendra Shah and the RSP, represents a rare political anomaly. His reputation as a disciplined, clean reformist promising better governance became a viral mandate

Shah's personality itself can be potent strategic asset to Nepal, but it is immediately replaced by a different kind of geopolitical pressure that needs sustained diplomatic communique

Nepal's political shift opens a strategic window for India

The new government in Nepal extends an opportunity to fashion a forward-looking bilateral relationship

K.V. Rajan
Atul K. Thakur

The election results in Nepal have been described as a political earthquake. There has been an emphatic and comprehensive rejection of old leaders and established parties. A younger generation of professionals and tech savvy figures, enjoying the support of Gen Z activists is set to take over.

Challenges ahead

By giving the Rastriya Swatantra Party (RSP) a two thirds majority, Nepali voters have granted Balendra Shah and his government, a powerful mandate for Nepal's complete transformation. It now has a huge responsibility to answer wide-ranging expectations--enough jobs for the youth, reversing the migration for work abroad, stimulate economic growth, and ensure good governance. It needs to be noted, however, that while voters have demonstrated their impatience with the old order and its decades- old insensitivity to their aspirations, this is not a positive vote for a clear-cut new agenda for reform, political or economic, since it was never spelt out and placed before them.

There is a real possibility of frustration and disillusionment that the new government will have to deal with as it settles down. The first warning shots were fired by the caretaker Prime Minister Sushil Karki even before the election, when she reminded the political class that the agitations of September 2025 had erupted because of the frustrations of people insisting on good governance, and a recurrence was inevitable if the situation lapsed into the same old pattern.

Hopefully the people of Nepal will give the new leaders enough time to address the country's problems.

Restructuring India-Nepal ties

For now, Nepal deserves every encouragement possible. India has been quick to extend it, without being loud or patronising. India had not been an issue during the election campaign. Its relationship with Nepal in recent years has focused on the right priorities – development, infrastructure, digital connectivity, energy. It has played its cards well and can continue to capitalise on the existing goodwill as the new leaders in Nepal seek to respond to development needs of the people.

Restructuring of the India-Nepal relationship has been long overdue. For far too long it has been trapped in the shadows of the legacy of British India days. Hopefully India and Nepal will seize every opportunity to fashion a forward-looking relationship based on today's realities and the immense potential for expanding cooperation. For this it will be necessary for policy makers on both sides to discard old mindsets, address long standing irritants with fresh approaches, and prioritise people-centric policies which can be delivered to keep pace with people's expectations and needs.

India also needs to look at the recent developments in Nepal as part of a wider regional phenomenon since happenings in Bangladesh, Sri Lanka and elsewhere also fall into the same pattern–agitations led by frustrated youth incidentally toppling pro-India political figures, demanding faster development and better governance. Labeling new political leaders being thrown up everywhere as anti-India just because of the legacies of the past does not seem justified, as seen from the pragmatic readiness shown in Bangladesh and Sri Lanka to cooperate with India, by parties and leaders once seen as unfriendly. Nepal could be a good partner for India in the evolving geopolitical scenario, if both countries try seriously to fashion a clear cut sub-regional strategy for rapid growth.

(K.V. Rajan is former Indian Ambassador to Nepal and Atul K. Thakur is a policy professional. They are the authors of 'Kathmandu Chronicle: Reclaiming India-Nepal Relations'. Views are personal.)

THE GIST

The people of Nepal have granted Balendra Shah and the RSP a powerful mandate for Nepal's complete transformation. The government now faces the challenge of addressing the issues of unemployment and providing good governance

India has been quick to extend encouragemnet to the new leadership. Hopefully India and Nepal will seize every opportunity to fashion a forward- looking relationship based on today's realities

ABSTRACT

How do we know climate science is credible?

On what aspects did a recent paper in the Science of Climate Change journal question the foundations of climate change, and how robust are its arguments?

Vasudevan Mukunth

On March 10, a journal called Science of Climate Change published a paper calling into question the foundations of climate change. The paper concluded that after accounting for some sources of uncertainty in the climate data, the 'correct' changes in the oceans' heat content and Earth's energy imbalance are practically zero. In other words, the oceans are not warming, Earth's surface is not accumulating heat, and global warming is not happening.

The paper is more sophisticated than it seems at first sight and makes three claims with different levels of merit.

There is value in addressing them in detail because doing so reveals how we know that climate science is credible.


Thinning ice: A penguin stands on an iceberg in Yankee Harbour, Antarctica in 2018. REUTERS

How are changes in ocean heat measured?

The paper's foremost claim is that temperature is an intensive property – meaning its value does not depend on the mass of the material – and thus scientists cannot average it in a meaningful way when estimating the amount of heat oceans hold.

Scientists have already addressed this claim. First, by the same logic, we cannot measure average air temperature, average atmospheric pressure, average sea level rise, and so on.

Second, and more importantly, scientists do not just measure and average the temperatures of different water bodies to determine the heat content. They also calculate the thermal energy. Temperature is nothing but the average kinetic energy of the atoms or molecules in a body. And thermal energy is the total kinetic energy of the water molecules. This is an extensive quantity – it depends on the number of molecules – and can be averaged. Its value has been increasing over time as well, and it also clarifies that the way scientists are handling the temperature data is correct.

Next, the paper takes issue with the Argo floats data. According to the U.S. National Oceanic and Atmospheric Administration, "Argo is an international program that collects information from inside the ocean using free drifting [devices called] profiling floats. These floats drift with the ocean currents and move up and down between the surface and a mid-water level. The floats are distributed over the global ocean to measure temperature and salinity in the upper 2,000 m."

According to the paper, the data the floats collect have some gaps, which has created uncertainties in the final processed data that have gone underreported. To the paper's credit, scientists have already raised and addressed the numbers it has cited on phenomena called mesoscale aliasing and deep ocean ignorance in the research literature. However, the paper's authors inflate these uncertainties and add them up in unscientific ways. For instance, some of the errors they add up have the same underlying cause, so adding them as if they were separate errors ends up counting the same causes more than once.

To avoid this pitfall, oceanographers run the whole calculation in different ways and check whether they keep getting roughly the same answer. They check whether the calculation predicts temperatures at locations where they actually have measurements to compare with – and whether the overall estimate holds up when they remove data from the calculations. This way, they make sure their methods are robust and do not over-count uncertainties.

Finally, if scientists have an independent estimate of total sea level rise (from altimetry satellites that use radar, say) and an independent estimate of how much new water has been added (from the GRACE satellites that use gravity), they can estimate how much the ocean has 'expanded'. Then they compare this figure with Argo data about the ocean's heat content. If the two match – as they do – it would mean Argo arrived at the same result as altimetry and GRACE (Gravity Recovery and Climate Experiment) satellites but from completely different starting points.

What is 'Balancing' and 'Filling'?

Finally, the paper says the CERES-Argo cross-calibration is "circular". This is probably the most rhetorically effective piece for 'climate doubters' because it genuinely sounds damning. However, that is really because it misrepresents what a particular 'adjustment' does.

CERES is a suite of scientific instruments in Earth orbit operated by NASA. Its name stands for 'Clouds and the Earth's Radiant Energy System'. The instruments measure incoming solar radiation and outgoing shortwave radiation (which includes visible light) and longwave radiation (mostly heat) at the top of the atmosphere. By subtracting the incoming rate of energy from the outgoing rate, scientists can say how much heat is being 'left behind' in the planet's atmosphere and on the surface.

Now, the CERES instruments have been calibrated such that they are accurate to around 1% for shortwave radiation and 0.75% for longwave. This implies an absolute uncertainty of roughly 2 W/m2 in the net energy flux. The paper says the raw CERES uncertainty is around 3-5 W/m2, which is slightly inflated in favour of the paper's claims.

To address this, a process called EBAF – short for 'Energy Balanced And Filled' – makes a one-time change to shortwave and longwave fluxes at the top of the atmosphere to ensure the global mean net flux for July 2005 to June 2015 is consistent with the value measured by Argo: 0.71 W/m2.

'Balancing' and 'Filling' are separate adjustments. CERES instruments cannot see through clouds, so 'Filling' patches gaps in the map where data is missing. However, the paper treats the whole EBAF product as if it is just the adjustment for calibration, i.e. 'Balancing'.

Nonetheless, as things stand, the paper's circularity argument is partly correct: CERES data is 'corrected' using Argo data while Argo-derived estimates of the oceans' heat content are validated using CERES data.

What does CERES do?

But then the paper goes wrong. The 'Balance' of EBAF only adjusts the mean energy flux. It does not interact in any way with the increases and drops in the temperature data over time. That comes from the CERES instruments' raw data and is what contains evidence of the warming trend.

Specifically, the CERES instruments continuously measure the radiation coming from the Sun into the atmosphere and the outgoing longwave and shortwave radiation at the top of the atmosphere. Then computers produce a monthly global mean net flux value for every month from March 2000 to the present. These raw monthly data are off from the 0.7 W/m2 measured by Argo, so EBAF 'adds' or 'subtracts' some flux from every monthly value in the record to bring the long-period mean in line with the Argo estimate.

The monthly values after this adjustment are thus higher or lower than the raw values by exactly the same amount in every single month. This means the difference between any two months – March 2005 and March 2015, say – is unaffected by EBAF. For example, if EBAF adds 3.6 W/m2 to the measured values, the difference between 4 and 5 in the raw data is the same as the difference between 7.6 and 8.6 in the processed data, which is 1 W/m2.

The idea that Earth's energy imbalance has been increasing over the satellite record is based on these differences, which Argo data is unconcerned with. As a result, the circularity objection proves less than what the paper has claimed it does.

What sustains credibility?

For added measure, scientists have also estimated Earth's energy imbalance (outgoing minus incoming) using atmospheric reanalyses, deep ocean temperature records from research vessels, and physical models informed by observed surface warming – all of which have been consistent with the CERES-Argo figures. If the imbalance were actually zero, the independent estimates would all have to be wrong for independent reasons. And the odds of that are extremely low.

In fact, the answer to whether the paper could be right is not that one of its authors is a "clarinet instructor" (which he was at the time he worked on the paper), that its rhetoric is often obscurantist, that some of its other authors have done questionable things in the past or even that it has not passed peer-review by a "prestigious" journal. It is that the paper does not perform any independent tests of the data the way credible studies do.

That is in fact the bedrock of the credibility of climate science as a whole. And any efforts that claim to overturn that must also convincingly explain how independent checks arrived at the same result while being flawed in independent ways.

FROM THE ARCHIVES

Know your English

S. Upendran

"What did you do on New Year's Eve? Did you have fun?"

"Oh, yes! I had a great deal of fun! My friends and I danced till 4:00 in the morning. And as you can imagine, I was dead beat after that. I found it...."

"....you were beat? Who beat you? And why?"

"Nobody beat me! All that dancing had me beat. You see, when you say that you're beat, it implies that you are extremely tired. It's one of the meanings of the word 'beat'."

"I see. It's an informal expression, I suppose?"

"That's right! It's not used in formal contexts. Here's an example. The college students were dead beat after the long trek."

"Vyomekesh looked dead beat after doing his karate exercises."

"That's a good example. Agarkar looked beat after bowling his tenth over."

"I don't really agree with you. He doesn't usually look beat after bowling ten overs. It's just that he always looks as if he wants to sit down and cry."

"That's true. He always looks sad, doesn't he?"

"Most of our cricketeers do. And the way they are playing, they are making everyone in the country sad, too. Tell me, is there another word which means 'tired'? An informal word, I mean."

"Sure, there is. 'Bushed' is another word used in informal contexts to mean 'tired'. After playing the third set, Ramdas was bushed."

"When Mala asked me if I was interested in going to the movies, I told her I was too bushed to go. How does that sound?"

"Sounds good."

"Hey, doesn't the word 'stressed out' mean the same thing as 'bushed'?"

"Not really. You see, ..."

"....what do you mean? Doesn't 'stressed out' mean tired too?"

"Well, yes. But there is a difference between being 'stressed out' and being 'bushed'. You see, when you say you are 'bushed' or 'beat', what you are implying is that you are physically exhausted."

"I see. And when you say that you are 'stressed ou" it means that you are mentally exhausted. Am I right?"

"I guess you could say that. 'Stressed out' is more mental than physical. For example, studying for the GRE has really stressed out Sukesh."

"Talking to my uncle for five minutes stresses me out."

"I am sure my nephews say the same thing."

"When I was young, sitting in maths class used to stress me out."

"I still have nightmares thinking about my maths class."

"My friend Suresh gets asthma thinking about maths."

"Gets what?"

"A..s..t..h..m..a. Asthma!"

"The 'th' in the word 'asthma' is not pronounced."

"It isn't?"

"....the 'as' in the first syllable is like the word 'ass'. The final 'a' is like the 'a' in 'china', 'about', and 'amount'."

"The 'th' is not pronounced at all and the stress is on the first syllable. Correct?"

"Excellent. Some people pronounce the 's' in the first syllable like the 'z' in 'zip', 'zest', and 'zoo'."

Published in The Hindu on January 4, 2000

Word of the day

Boisterous:
noisy and lacking in restraint or discipline

Synonyms: rambunctious, unruly

Usage: *The crowd grew boisterous as the match reached its final moments.*

Pronunciation:
https://newsth.live/ZQEJN5

International Phonetic Alphabet: /ˈbɔɪstəɹəs/

For feedback and suggestions for Text & Context, please write to **letters@thehindu.co.in** with the subject 'Text & Context'

THE DAILY QUIZ

Please send in your answers to dailyquiz@thehindu.co.in

On the anniversary of the execution of three young revolutionaries on March 23, 1931, revisit one of the most debated moments in India's freedom struggle

Prathmesh Kher

QUESTION 1
The execution of Bhagat Singh, Sukhdev Thapar and Shivaram Rajguru followed their conviction in which case related to the assassination of John Saunders in Lahore?

QUESTION 2
The assassination of Saunders was intended as retaliation for the police assault on which nationalist leader who later died of injuries sustained during a lathi charge?

QUESTION 3
In which prison were the three revolutionaries executed?

QUESTION 4
Which pact between nationalist leaders and the British government was signed shortly before the executions in 1931?

QUESTION 5
What was the title of the editorial published after the executions, by B. R. Ambedkar in the Marathi weekly Janata, on April 13, 1931?


Identify this Indian revolutionary, commander-in-chief of the Hindustan Socialist Republican Association, who died in a gun battle with the British police at Alfred Park in 1931. PUBLIC DOMAIN

Questions and answers to the previous day's daily quiz: 1. The World Meteorological Organization is headquartered in this European city. **Ans: Geneva, Switzerland**
2. Expand GRIB, the concise data format commonly used in meteorology. **Ans: General Regularly-distributed Information in Binary form**
3. To a meteorologist, an 'okta', a scale of measurement is. **Ans: The amount of cloud cover at any given location**
4. The term for rain that falls from a cloud but evaporates before reaching the ground, seen as streaks under a cloud. **Ans: Virga**
5. Parts of Kerala witnessed 'blood rain' as recently as 2012. Name its prime cause. **Ans: Arial spores of the orange-coloured 'Trentepohlia' algae, often triggered by atmospheric conditions**
6. The everyday two-word term for strong, high-altitude wind band that pilots use to save time and fuel on long flights. **Ans: Jet Stream**
7. The recurring natural phenomenon characterised by fluctuating ocean temperatures in the equatorial Pacific is the ENSO. Expand ENSO. **Ans: El Niño-Southern Oscillation**
Visual: Name the Indian scientist who in his capacity as the chairman of the Intergovernmental Panel on Climate Change (IPCC) received the Nobel Peace Prize, along with Al Gore, in 2007. **Ans: Rajendra Pachauri**
Early birds: Piyali Tuli | K.N. Viswanathan | Prem Nath Tiwari | Sukdev Shet | Ashok Tripathi

New way helps superconductor break temperature record

While superconductivity is easy to achieve at extremely low temperature, bringing it to room temperature is a 'holy grail' of physics; researchers have achieved much higher temperatures in recent years, up to -13 °C, but only by applying pressure equivalent to that near the earth's core

Vasudevan Mukunth
CHENNAI

For more than a century, physicists have looked for a material that conducts electricity with zero resistance at room temperature – practical enough to transform how the world generates and uses energy. But for a long time, the highest temperature at which a material became superconducting at room pressure was -140 °C. Some other materials become superconducting close to room temperature, but only under extraordinary pressure.

In a study published in *Proceedings of the National Academy of Sciences* on March 9, scientists have reported raising the temperature by 18°C, and under only room pressure. They used a new technique called pressure quenching to achieve this.

The result has broken a record that had stood since 1993, when the same material the team used – a copper oxide called Hg1223 – first demonstrated superconductivity at -140°C, itself a landmark that launched decades of research.

'Sounds reasonable'

Superconductors that work at room temperature and normal pressure could one day carry electricity through power grids without losing any energy to resistance – a problem that currently wastes billions of dollars' worth of power every year.

They could also enable faster MRI machines, more efficient electric motors, better transportation systems, and cheaper renewable energy infrastructure.

The team was led by physicists Liangzi Deng and Ching-Wu Chu at the University of Houston.

"The claim sounds reasonable to me," aid Sumit Mazumdar, professor of physics, chemistry, and optical sciences at the University of Arizona, in an email to *The Hindu*.

"The authors carefully study high-pressure structures, and when they find first order transitions (which involves latent heat), they do what they call pressure-quenching so that the changed high pressure structure persists even when the pressure is removed," he added.

Intense pressure

Since 1993, the record for ambient pressure superconductivity has been stuck at -140 °C. While superconductivity is easy to achieve at extremely low temperature, bringing it to room temperature is a "holy grail" of physics.

Researchers have achieved much higher temperatures in recent years, up to -13 °C, but only by applying pressure equivalent to that near the earth's core.

These high-pressure states have also disappeared the moment the pressure was released, making them useless for practical technologies like lossless power grids or high-speed trains.

The problem was not that scientists lacked the materials. Under extreme pressure, Hg1223 itself superconducts at -109 °C .

Some compounds rich in hydrogen have also shown hints of becoming superconductors at near room temperature. But they have all required intense pressure as well.

Reproducible data

The Houston team's insight was to stop looking for new materials and to focus on preserving a known high-pressure state in Hg1223.

For this, the scientists developed the pressure-quench protocol (PQP) – a three-stage process designed to freeze a high-pressure superconducting state in place by removing pressure fast enough and at a low enough temperature.

They loaded a small crystal of Hg1223 into a diamond anvil cell and compressed it to up to 30 billion pascals (GPa), raising its superconducting temperature to over -123°C , while keeping track of its superconducting properties. Then they cooled the sample to -269 °C , just above absolute zero.

Finally, they rapidly released the pressure. Because the material was so cold, the atoms lacked the energy to relax back into their normal structure, effectively trapping the material's desirable electronic properties at the normal atmospheric pressure.

Across five tests with different pressures and quenching conditions, the team consistently retained the transition temperature between -122 °C and -134 °C. They achieved the higher -122 °C by quenching from approximately 19 GPa at -269 °C. They could also reproduce it, a sign that it is likely real rather than an artifact in the data.

Ching-Wu Chu holds a diamond anvil cell in a University of Houston lab. UNIVERSITY OF HOUSTON

The 122 °C reading is 18 °C higher than the previous record held by the same material in its relaxed state.

The scientists also used synchrotron X-ray diffraction at the U.S. Argonne National Laboratory to find that while the material retained its original crystal structure after quenching, there were new defects in the structure and additional internal strain.

These defects helped lock the crystal into a state where it mimicked the effects of pressure even after the pressure was removed. The team confirmed that roughly 78% of the material's volume became superconducting, so it wasn't a surface or filamentary effect.

If superconductivity is filamentary, it means electricity is only traveling through microscopic channels, called filaments, within the material. These filaments can only carry a very small amount of electrical current before they break and return to a normal state.

If the bulk is superconducting, as the researchers reported, the material can be expected to reliably carry large currents, which is required to transport power on run high-speed trains.

The quenched phase was stable for at least three days when stored in liquid nitrogen. Warming above -73 °C reduced the effect while room temperature (23 °C) partially reversed it. One experiment also produced a hint of superconductivity at -101 °C but the team wasn't able to reproduce it.

'Woodstock of physics'

The study's lead investigator, Ching-Wu Chu, is a pioneer of high-temperature superconductivity. In 1987, Prof. Chu and his team made a landmark discovery when it synthesised a material called yttrium barium copper oxide (YBCO) that became a superconductor at -180 °C.

Until then, scientists only had materials that became superconducting below -269 °C, which meant they had to be cooled using expensive liquid helium.

However, -180 °C was above the boiling point of liquid nitrogen (-196 °C), which meant scientists could cool YBCO using this much cheaper coolant, paving the way for more research and potential applications.

That same year, the American Physical Society held a special session on high-temperature superconductivity that ran past midnight and drew thousands of attendees – an event that has since been called the 'Woodstock of physics'. Prof. Chu was one of the central figures there and presented his team's results to a packed and electrified audience. The session captured the excitement of a field that, in that moment, seemed to be on the edge of transforming technology.

> The method, if it works out in other cases, has great promise for a room-pressure superconductor. Not yet for room-temperature. The latter is not essential for technological applications
>
> **SUMIT MAZUMDAR**
> Professor at the University of Arizona

His work earned him numerous honours, including the National Medal of Science in 1994.

Now, beyond breaking a 33-year-old record, according to the new study's authors, their work may also open the door for scientists to 'stabilise' exotic electronic properties in other materials that arise under pressure and make them available at ordinary conditions.

"The method, if it works out in other cases, has great promise for a room-pressure superconductor. Not yet for room-temperature," according to Prof. Mazumdar. "The latter is not essential for technological applications. The entire issue therefore revolves around what the authors call pressure-quenching [and] whether structural changes that occur under pressure can really be retained post-quenching."

He added that while this issue is beyond his expertise, it is not impossible.

"This result will be widely noted in the superconductivity community, and I expect that many experimentalist groups will apply the same methodology to other candidate materials," Brookhaven National Laboratory, New York, physicist Ivan Božović, who wasn't involved in the study but reviewed it for the journal, told *Physics*.

The paper's authors themselves wrote: "We … believe that the record-breaking results at ambient pressure reported here represent only the beginning of an extremely fruitful scientific journey."

Recent controversies

High-temperature superconductivity research has been shadowed by controversy for much of the past decade and understanding that backdrop has come to matter when evaluating new claims in the field.

These episodes have made the research community acutely sensitive to the claims of new studies, particularly whether a given material's resistance can really drop to zero – the hallmark of superconductivity – and if the data showing that is because the instruments are actually measuring the sample.

The most prominent case involved Ranga Dias at the University of Rochester in the U.S. Dias published papers claiming room-temperature superconductivity in carbonaceous sulphur hydrides in 2020 and nitrogen-doped lutetium hydrides in 2023. Dias and his team retracted both papers, published in *Nature*, after allegations that the magnetic susceptibility measurements had been manipulated to show signs of superconductivity.

Multiple independent labs also failed to reproduce the findings. Dias contested the retractions and the dispute has already become one of the most acrimonious data-integrity controversies in physics.

In 2023, a group of researchers from South Korea said they had made a lead-based material called LK-99 that was a superconductor at room-pressure and room-temperature.

The result generated intense public interest but soon, other scientists who recreated the experiment couldn't find signs of superconductivity in LK-99.

Later studies revealed that some of the measurements the South Korean team had made were 'polluted' by unexpected ferromagnetic behaviour in LK-99.

The episode further exhausted the public appetite for superconductivity news.

'Decades-old reputation'

Both Dias's work and the new University of Houston study use diamond anvil cells. However, unlike the materials in Dias's work, Hg1223 is well-understood and has a record going back to the 1980s.

The team also focused on a unique state of the material – one that arises under intense pressure – rather than claim a new phase, experts said.

The study also directly addressed the primary criticism of the Dias episode: whether superconductivity is a bulk property or filamentary.

As noted earlier, the team found that some 78% of the material was superconducting. Dias and others didn't perform these tests.

"The dubious claims … came from people who were practically unknown prior to their claims of novelty," Prof. Mazumdar said. "The principal author Chu in this case came close to winning the Nobel Prize [and] is the recipient of the US National Medal of Science. His current paper's ref. 22 goes back to 1968.

"He is not going to risk his decades-old reputation writing fraudulent stuff."

mukunth.v@thehindu.co.in

A Hudsonian godwit in the Monomoy National Wildlife Refuge, Massachusetts, U.S. AFP

Hope builds for nations to save bird that flies 30,000 km

Agence France-Presse

Chasing an endless summer, one shorebird species undertakes a gruelling annual journey from the Arctic to the tip of South America and back -- a feat increasingly fraught with peril.

The Hudsonian godwit (*Limosa haemastica*) is one of the world's most remarkable travellers, but its population has plunged 95% in four decades due to a complex mix of environmental changes across multiple countries.

It is one of 42 species proposed for international protection at a meeting of parties to the UN Convention on the Conservation of Migratory Species (CMS) that starts in Brazil on March 23.

Iconic creatures like the snowy owl -- of *Harry Potter* fame -- striped hyena and hammerhead shark are also on the list deemed in danger of extinction and needing conservation by the countries they pass through.

> **The Hudsonian godwit (*Limosa haemastica*) is one of the world's most remarkable travellers, but its population has plunged 95% in four decades due to a complex mix of environmental changes**

Migratory birds are facing "rapid and dramatic declines," said Nathan Senner, an ecologist and ornithology professor at the University of Massachusetts Amherst who has studied the Hudsonian godwit for 20 years.

Scientists are still unravelling the mysteries of the shorebird, which can fly up to 11,000 km in one stretch without stopping to eat, drink or sleep.

And it is only part of the 30,000 km that the godwit travels every year from their breeding grounds in the Arctic to Patagonia where they spend the southern summer.

In order to do this "epic flight," they need "really predictable, abundant food resources" at every step of the journey, Senner told *AFP*.

That predictability is crumbling. In the Arctic, shifting spring timing attributed to climate change has created a mismatch between when chicks hatch and the peak availability of insects they feed on.

One of the puzzles Senner is currently working on is why Hudsonian godwits have begun migrating later by six days than they did a decade ago.

Something "has either disrupted the cues that they use to time their migrations or their ability to successfully and rapidly prepare for the migration," he said.

In southern Chile, a boom in salmon and oyster farming has led to a build-up of infrastructure and the presence of people in the intertidal zones where they feed. And in the United States, changes in farming practices are making the shallow water wetlands that the godwits rely on rarer and less predictable -- meaning they spend more time looking for a place to stop and feed.

"I think that is emblematic of lots of species, that most species can respond to one kind of change, but not a whole bunch of them all at the same time," said Senner.

"Climate change is taking a heavy toll on species that rely on a 'geological clock' for their survival; many are disappearing," Rodrigo Agostinho, president of Brazil's environmental agency (Ibama) said.

These are some of the issues CMS parties will tackle at their meeting in Brazil's biodiversity-rich Pantanal, one of the world's most important global meetings for wildlife conservation.

These countries are legally obliged to protect species listed as at risk of extinction, conserve and restore their habitats, prevent obstacles to migration and cooperate with other range states.

For feedback and suggestions for 'Science', please write to **science@thehindu.co.in** with the subject 'Daily page'

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Plane truths

Airline ticket pricing should be transparent and fair

The Centre is justified in protecting passenger interests, by mandating that airlines make available at least 60 per cent of flight seats 'free' — that is, with no add-ons to the ticket price. The additional charges for such premium seats — which are typically towards the front of the aircraft or alongside windows and the aisle —sometimes account for between 5-10 per cent of the total fare. By also directing airlines to seat passengers on the same PNR together, the government has removed a pain point for travellers.

The new rules may not significantly impact finances of airlines as ancillary revenues — including seat prices, food and beverages and excess baggage — do not exceed 10 per cent of total revenues, in general. But they make a world of difference to passengers. There are two consumer concerns here: the price being paid; and whether this is disclosed upfront or concealed in hidden charges, only to be revealed when the 'payment' window comes up, as for example, convenience fee. Indeed, it is compelling to argue that in a free market, the government should not meddle with pricing decisions of business, but leave it to market forces instead. However, the question here is whether free market forces are really working in a duopoly market, one in which IndiGo is way ahead of Air India. To be sure, there have been both demand and supply side factors driving up prices. Geo-political disruptions have upset flight schedules and costs, while passenger traffic too has been rising sharply. Even in a more competitive market, prices would have risen; the question is, whether episodic price spurts are justified, and whether the consumer is duly informed about hidden charges while purchasing the ticket.

India is not alone in the matter of dealing with flyer-unfriendly practices. In 2024, the US' Department of Transportation came down on 'drip' pricing practices of airlines there and required the display of all fees at one go, though this was challenged in court. The UK issued similar norms on transparent pricing under a 2022 Act. The European Court of Justice recently ruled that airlines must refund the full amount paid — including booking fees — rather than slice up charges. The Centre should ask airlines and booking platforms to display the full cost of a ticket at the beginning of the booking process, rather than conceal it in convenience fees or seat pricing.

Rising passenger traffic conceals the fact that consumers are not a homogeneous lot — they could be corporate/institutional travellers, tourists, or budget-strapped individuals and families. Therefore, having more free seats is only fair. However, price regulation should be a light-touch affair — to monitor any unusual spike in prices, or hidden charges, while at the same time allowing demand and supply to adjust. In the long run, the best deal for consumers is to have more airlines. A level-playing field for smaller airlines is being attempted, but there's a long runway ahead.

OTHER VOICES.

The Guardian

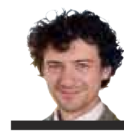
Development funding: UK's meanness is shortsighted

Progress is possible. Over two decades, global child mortality plummeted. There were many reasons for a 39% reduction in deaths in lower and middle income countries between 2001 and 2021, but a significant one was overseas development aid, which supported everything from sanitation to vaccination programmes to food security. That shift has slowed, and – like similar advances – is likely to reverse if aid budgets continue to be slashed. Researchers warned last month that continuing cuts could result in more than 22 million avoidable deaths in the next five years, with a quarter of those among children under five. The UK's decision to slash aid by 40% is part of a global trend: G7 spending will be 28% lower this year than in 2024. Donald Trump has dismantled USAID; Germany, France and others are chopping their budgets. But Britain's case is particularly dismaying. A bipartisan consensus saw David Cameron, building on Gordon Brown's work, make Britain the first G7 country to hit the internationally agreed aid target of 0.7% of gross national income (GNI). LONDON, MARCH 22

THE WALL STREET JOURNAL. EUROPE

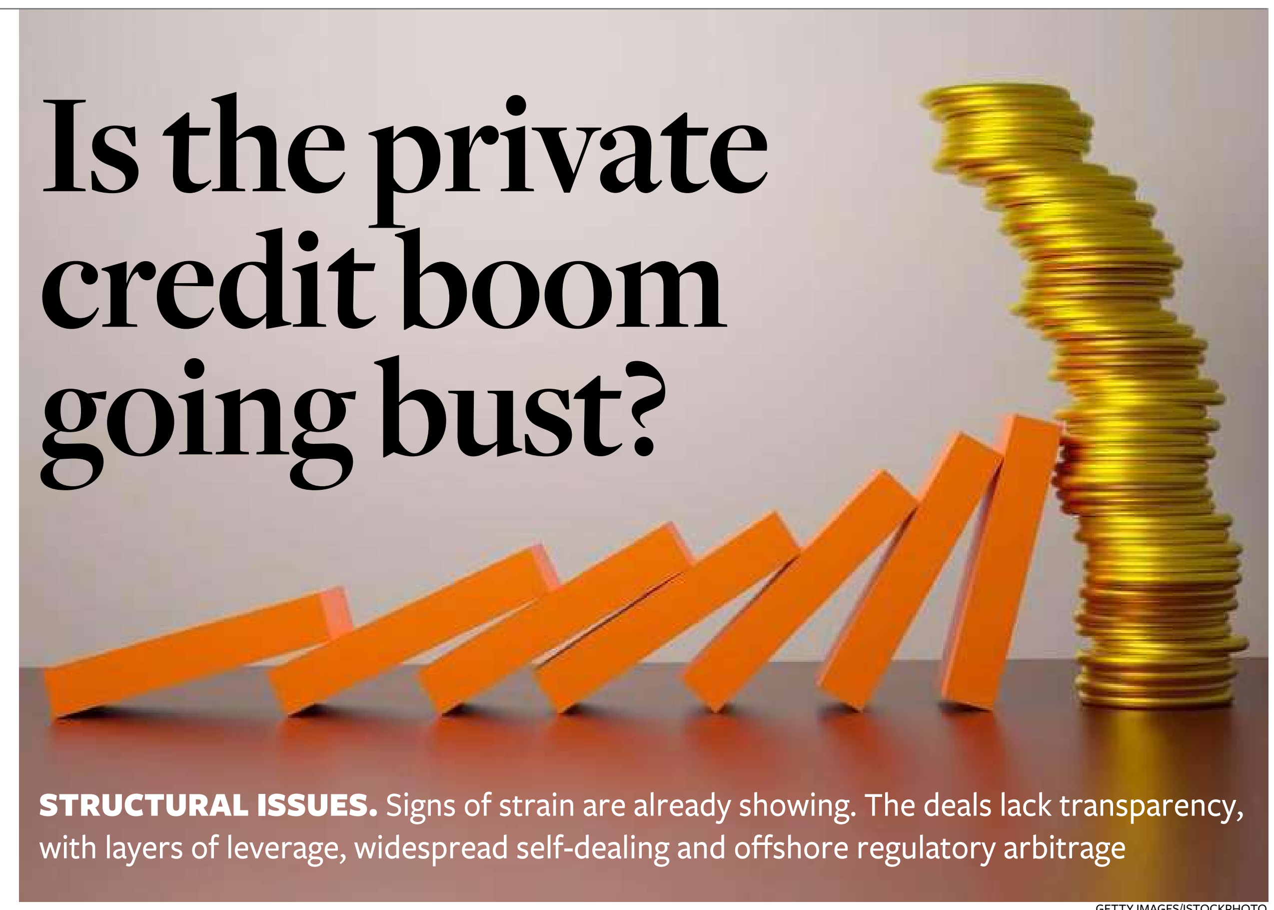
The Tragedy of Robert Mueller

Robert Mueller, who died Friday at age 81, served his country as a Marine and as the second-longest-serving director of the Federal Bureau of Investigation. The tragedy of his legacy is that he agreed to lead the probe into whether Donald Trump colluded with Russia to win the 2016 election. Robert S. Mueller III was born in Manhattan in 1944, the son of a DuPont executive. He attended Princeton, but when a classmate was killed in Vietnam in 1967, he joined the Marines. He was wounded and returned to the battlefield. This noble contrast with so many others of his generation couldn't be greater. NEW YORK, MARCH 22



BRIAN JUDGE

Is the private credit boom going bust?



STRUCTURAL ISSUES. Signs of strain are already showing. The deals lack transparency, with layers of leverage, widespread self-dealing and offshore regulatory arbitrage

GETTY IMAGES/ISTOCKPHOTO

A defining feature of financial markets is that the most important information is often held by those least likely to reveal it. The system's inner workings are invisible to outside observers — which makes it all the more striking when leading Wall Street executives start sounding the alarm.

In October, JPMorgan Chase CEO Jamie Dimon warned of "cockroaches" lurking in the private credit market. His remarks quickly reverberated across the industry, with UBS Chairman Colm Kelleher pointing to the "looming systemic risk" posed by poorly regulated private credit, which he likened to the rating-agency failures that helped trigger the 2008 financial crisis. At its core, private credit is lending that takes place outside the traditional banking system. Loans are made by investment funds, often managed by firms that specialise in corporate buyouts.

Unlike bank loans or publicly traded bonds, these deals lack transparent pricing, an active secondary market, and meaningful regulatory oversight. Terms are negotiated privately, valuations are set internally, and the investors whose money is at risk — often pensioners and insurance policyholders — have virtually no ability to verify any of it.

Despite — or perhaps because of this opacity — private credit grew rapidly after the 2008 crisis, as banks pulled back from riskier lending and private credit funds rushed to fill the vacuum. Today, the industry manages roughly $3.5 trillion.

The rise of private credit is often framed as a story of financial innovation, with traditional banks retreating and nimbler private lenders stepping in. But a closer look reveals a troubling, eerily familiar pattern: layers of leverage, widespread self-dealing, offshore regulatory arbitrage, and a feedback loop that ties the savings of millions of retirees to risky bets on AI data centres, leveraged software companies, and bundled loans marketed as safe investments.

Much of this activity is driven by private equity. When private-equity firms acquire companies, they typically rely on debt provided by private credit funds. But the initial buyout is just the beginning, as portfolio companies often undergo multiple rounds of borrowing to finance additional acquisitions, re-financings, dividend recapitalisations, and restructurings.

A single deal can therefore generate several separate transactions. In many cases, the lender and the private-equity sponsor are affiliated, allowing the same parent company to collect fees on both sides of the deal while deploying other people's money — namely, retirees who believe their savings are safe.

Over the past decade, large alternative-asset managers have taken this model a step further by acquiring life-insurance companies. Apollo's 2022 merger with Athene, which has roughly $400 billion in assets and serves more than 535,000 policyholders, helped set the template: insurers provide a steady stream of capital, and affiliated asset managers channel it into private credit, with profits generated at every step, from insurance float to deal fees and carried interest. Nearly every major alternative-asset manager has since followed suit.

Traditional banks making similar loans operate under strict capital requirements. Unlike private credit funds, they must hold reserves against potential losses, submit to regulatory scrutiny that can force write-downs, recognise losses before borrowers default, and undergo stress tests whose results are publicly disclosed. Against this backdrop, it is hardly reassuring that roughly one-third of the US life-insurance industry's $6 trillion in assets is now invested in private credit.

> **Whether turmoil in private credit markets could trigger a broader financial crisis is impossible** to know in advance, but that uncertainty is itself part of the risk

BERMUDA TRIANGLE STRATEGY

To boost returns, these firms employ what industry analysts call the "Bermuda Triangle" strategy, whereby a single sponsor controls three interlinked entities: a life insurer, an asset manager, and an offshore reinsurer. The insurer gathers premiums; the asset manager channels those funds into private credit deals it originates and prices; and the reinsurer — typically based in Bermuda or the Cayman Islands — assumes the insurer's liabilities under looser capital requirements than those imposed by US regulators.

The system is already showing signs of strain. Software companies, long a favoured target for private-equity buyouts because of their steady revenues and low capital expenditures, are being repriced as AI threatens widespread disruption. Private credit funds' exposure to the sector is an estimated $600-750 billion, fuelling liquidity pressures. Blue Owl recently restricted withdrawals from a $1.7 billion retail fund, and Blackstone's BCRED has recorded its largest net outflows since its inception. The shares of publicly traded alternative-asset managers have also fallen sharply.

These developments highlight a deeper structural problem. When the same firm originates a loan, holds it in a fund it manages, values it using its own models, and reports that value to an insurer it owns, the result is unlikely to reflect what an independent buyer would pay.

Whether turmoil in private credit markets could trigger a broader financial crisis is impossible to know in advance, but that uncertainty is itself part of the risk. Before 2008, few observers understood how vulnerabilities in the US subprime-mortgage market could cascade through the global financial system. There is no reason to assume that we have a better view of the risks in private credit markets today.

To be sure, none of this means a financial crisis is inevitable. But it does suggest that risk has migrated into less transparent structures, subject to lighter regulation yet still tied to government guarantees. The gains accrue to Wall Street, while the potential losses may once again fall on retirees and state insurance systems. In that sense, the echoes of 2008 are becoming hard to ignore.

The writer is Research Director of the Program on Finance and Democracy at the University of California, Berkeley. Copyright: Project Syndicate, 2026. www.project-syndicate.org

A national water vision for Viksit Bharat

Water governance must become climate-resilient with women playing a central role in it

Nilanjan Ghosh

India is already approaching the threshold of water stress. The paradox is stark: a country endowed with major river systems is confronting intensifying scarcity, conflict, and ecological stress. The explanation lies not in the lack of water but in how it is governed.

Water in India refuses to obey political boundaries. Rivers traverse States; groundwater interacts with surface flows; and water simultaneously serves agriculture, industry, cities, and ecosystems.

Yet governance remains fragmented across ministries, sectors, and States. Decisions taken upstream — diversions, dams, pollution — generate downstream consequences that are rarely internalised. A single ecological system is governed by disconnected administrative silos.

Climate change is only aggravating the problem. Variability in rainfall, more frequent floods and droughts, and shifting hydrological cycles are rendering static governance frameworks obsolete.

Agricultural incentives, particularly minimum support prices, have entrenched water-intensive cropping patterns. Food security has been pursued through paddy and wheat, but at the cost of groundwater depletion and river basin stress.

For decades, India's response has been guided by the "arithmetic hydrology" paradigm — a reductionist belief that water challenges can be solved through supply augmentation. Large dams, canals, and extraction systems have expanded availability in the short run, but often at the cost of long-term sustainability. Sediment flows have been disrupted, ecosystems degraded, and flood risks heightened. The "arithmetic hydrology" paradigm has also promoted the simplified linear thinking of "more water, more crops, more growth". However, if only today's water problems could have been expressed in the framework of the ordinate and the abscissa, they would have been solved by linear algebra only – they would have ceased to be complex problems.

The moment now calls for a decisive change. India needs a coherent National Water Vision aligned with its Viksit Bharat aspirations. Here are the 12 tenets of the National Water Vision.

12 TENETS

First, the river basin should be the unit of governance. Basin-level governance institutions must replace fragmented, State-centric approaches.

Second, water must be recognised as a dynamic socio-ecological flow rather than a static stock. Surface water, groundwater, sediments, biodiversity, and energy are parts of interconnected systems, with disruptions in one

WATER. Looming crisis PTI

element cascading across the system. Governance must embrace this complexity.

Third, India must operationalise Integrated Water Resource Management at the basin scale by aligning agriculture, urban development, industry, and ecosystems within a common framework.

Fourth, empowered River Basin Authorities, multidisciplinary expertise, and transparent data systems should replace the prevailing model of conflictual federalism evident in recurring inter-State disputes.

Fifth, water must be valued, as a free or underpriced resource is subjected to systematic overuse and misallocation.

Sixth, this creates space for market-based instruments like water credits, tradable permits, and water derivatives, which can align incentives for conservation and efficient allocation.

Seventh, environmental flows must be placed at the centre of policy. Rivers are living systems that sustain agriculture, livelihoods, and biodiversity. Ignoring these functions undermines development itself.

Eighth, India must shift from supply augmentation to demand management. Efficiency gains, crop diversification, and behavioural change can deliver water security without perpetually expanding infrastructure.

Ninth, governance must be data-driven. Basin-level information systems integrating hydrological, ecological, and socio-economic data are critical for informed decision-making.

Tenth, the private sector must be part of the solution. ESG-linked water stewardship, water credits, and responsible industrial practices can mobilise capital and innovation.

Eleventh, water governance must become climate-resilient. Adaptive systems that incorporate uncertainty and risk are essential.

Finally, while governance needs to be inclusive and strike an optimal balance between bottom-up and top-down approaches, women must be considered central to water management.

The twelfth tenet of the proposed National Water Vision coincides with the theme for World Water Day 2026: Water and Gender, which explicitly states, "it's time to centre women and girls in water solutions".

The writer is Vice-President – Development Studies at the Observer Research Foundation

LETTERS TO EDITOR

Send your letters by email to bleditor@thehindu.co.in or by post to 'Letters to the Editor', The Hindu Business Line, Kasturi Buildings, 859-860, Anna Salai, Chennai 600002.

Solar security

This refers to 'Realising solar power's potential for energy security' (March 23). The solar energy scheme, PM Suryaghar Muft Bijali Yojana, despite being cost-effective for the beneficiaries, is not attractive, due to bureaucratic lethargy and lack of public awareness.

The infrastructure to establish the panels and their accessories remains unutilised or underutilised. The beneficiaries need to be motivated and provided with the necessary support to establish the panels for the generation of solar power from their rooftop.

There should be more awareness programmes to push the schemeTimely payment of the subsidy must be ensured.

VSK Pillai

Changanacherry

It is with reference to the news article 'Realising solar power's potential for energy security' (March 23). Solar energy is crucial, sustainable and increasingly cost-effective energy source.

It provides long-term price stability, acting as a hedge against rising unpredictable costs of fossil fuels.

While solar energy generation is limited to daylight hours and affected by weather conditions, the advanced battery technologies enable us its use, 24/7 making it a reliable long-term solution.

By generating more solar power, India could reduce its reliance on imported coal, oil and gas and insulate itself from supply disruptions caused by geopolitical factors. Solar energy is an inexhaustible domestic resource that adds diversity to the energy mix.

P Victor Selvaraj

Palayamkottai (TN)

Blessings in disguise

This refers to the article 'Trump, Netanyahu and the stag hunt' (March 23). The example of the hunt for the rabbit and deer is apt. The concluding para about the possibility of positive externality of an adverse event turning out a blessing in disguise also stands to logic.

A country that could evade responsibility for Hiroshima to Venezuela will have to face trial in the humanity's court sooner than all of us think. There are no escape routes left.

MG Warrier

Mumbai

Futile move

Apropos "Demographic fixation" (March 23); the draft policy in Andhra Pradesh reflects a troubling cycle of demographic engineering. Incentives for larger families may appear supportive, yet they echo earlier coercive measures that undermined women's autonomy and health. International experience shows that financial inducements rarely succeed in raising fertility, while structural inequities remain unaddressed.

The real challenge lies not in manipulating birth rates but in creating conditions where families feel secure. Strengthening healthcare, education, and employment opportunities would empower citizens far more than pronatalist schemes. Labour shortages can be met through migration and better workforce planning.

K. Chidanand Kumar

Bengaluru

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Hazy signals from Iran war

US' statements and actions are often contradictory

Sridhar Krishnaswami

If the objectives of starting the Iran war had been clear there would not have been a problem in deciphering the messages coming out of the US.

After joining hands with Israel and bombing nuclear facilities in Iran on February 28, Washington shifted its sights to regime change once Ayatollah Ali Khamenei was assassinated and the top line of the clerics wiped out in missile strikes by Israel.

But the ballgame soon changed when Teheran began striking Gulf nations that were seeing supporting American war efforts.

SUBTLE SIGNS. Trump looking for a way out of the war mess REUTERS

IN THE DARK

While the conflict is into its fourth week, the international community is unaware of any steps being taken to de-escalate the situation but is witnessing conflicting statements and actions.

While US claims Iran is the "head of the snake for global terrorism", yet sanctions were lifted on its oil that were at sea — apparently about 140 million barrels — to facilitate the easing of oil prices that were roiling energy markets. However, Washington also put out word that steps would be taken to prevent Teheran from making money out of the sales. No one is sure though, how this is going to be done. Even Republican lawmakers don't seem happy. "Bombing Iran with one hand and buying Iran oil with the other," quipped one.

The conflicting messages are not confined to economics and trade. Militarily, President Trump announced that he was "winding down" the Iran war but almost in the same breath threatened to hit Iran hard should it obstruct transit through the Strait of Hormuz; reports indicate some 2,500 Marines were en route to the Gulf to augment the existing 50,000 or so already there.

Of course the additional boots does not mean that a land invasion is imminent or even on the cards. And the Pentagon is asking for an additional $200 billion for its Iran war efforts, a demand that is not sitting well with Congress.

While saying the war is "nearly complete" with little or nothing of Iran's military facilities left, the continued American and Israeli assault on Iranian infrastructure is baffling allies in Europe and Asia.

The Israeli hit on Iran's South Pars gas field brought forth a swift response on Qatar's LNG hub Ras Laffen. This threatens to further escalate retaliatory strikes to include the scores of desalination plants that are in the range of Iran's drones and missiles. Analysts believe that weaponisation of water will have disastrous consequences for the region.

The Trump administration claimed that it was not aware that Israel was going to strike South Pars; Tel Aviv, however, avers that Washington was in the loop.

In the maze of chatter are also subtle signals that President Trump is trying to wriggle out of the mess he has created.

And the first of the steps is in his saying that Israel has been told not to hit Iranian energy infrastructure. "I told him (Prime Minister Netanyahu) 'Don't do that, and he won't do that'". While Trump is still maintaining that he is not planning to land troops in Iran, the Israeli Prime Minister is wondering how the entire thing could be wrapped up without the "ground component".

BACK CHANNEL TALKS

Bringing the conflict to an end would face a larger problem if Tel Aviv keeps targeting individuals in the existing power structure in Iran. Even for back channel talks there must be moderates left to take decisions.

Leaving a vacuum at the top will only encourage attempts by fringe radical elements to grab power, the end result of which will not be in the interests of the international community.

The last thing Washington would want is a Teheran that is even more hell bent on going ahead with its nuclear weapons programme, intensifies its anti-Israel stance, or unleashes brutality on its people that would put previous regimes to shame.

The writer is a senior journalist who has reported from Washington DC on North America and United Nations

Petroyuan— a geopolitical signal

DOLLAR'S SWAY. Yuan is constrained by lack of capital account convertibility and PBOC control on exchange rate

RAM SINGH

Amid the Hormuz choke, Iran has signalled conditional openness to energy shipments by permitting oil cargo passage only if transactions are invoiced in Chinese-yuan (¥), a move widely seen as an attempt to challenge the long-standing dominance of the petrodollar system. This development has revived debate over whether the Petroyuan could emerge as an alternative currency for global energy trade.

However, replacing the dollar requires several structural attributes of an international currency, including deep and liquid financial markets, full capital convertibility, global investor trust, and reserve currency dominance, conditions that the renminbi still only partially meets. Therefore, decoding these structural constraints is essential to evaluate whether such shifts could realistically reshape global energy trade and what implications they may hold for India's energy security and financial stability.

PETROYUAN: THE CONSTRAINTS

Petroyuan is a signal, not a system, its growing use in energy trade reflects geopolitical intent to challenge dollar dominance, but structural constraints limit its emergence as a true global monetary alternative. First, from a monetary perspective, the ¥ remains only partially internationalised. According to the IMF, the Chinese-¥ accounts for 2-3 per cent of global foreign exchange reserves, compared with about 56 per cent held in US dollars (see table).

SWIFT data show that the ¥ accounts for 3-4 per cent of global payment transactions, even though China now invoices over 60 per cent of its trade in ¥, while the dollar still dominates with more than 40 per cent of global payments. A key constraint lies in China's continued reliance on capital controls.

Unlike the US dollar or euro, the ¥ is not fully convertible on the capital account, and its exchange rate remains managed by the PBOC. For energy exporters holding large trade surpluses, limited convertibility reduces the attractiveness of ¥ reserves. Moreover, wider ¥ use can constrain PBOC's monetary and liquidity management, particularly during inflationary cycles.

Petrodollar vs Petroyuan: Hype vs Reality

Dimension	Petrodollar (USD)	Petroyuan (RMB)
Monetary credibility & convertibility	Fully convertible currency; global reserve dominance (56% of global FX reserves). Independent central bank, Federal Reserve	Partially convertible currency with capital controls. RMB accounts for 2-3% of global reserves; exchange rate managed by PBOC
Financial market depth	Deep and liquid financial markets; US Treasury market >$26 trillion. Global investors can park oil revenues safely in dollar assets	Financial markets expanding but limited global participation; foreign holdings in Chinese bonds remain relatively low (<10%)
Forex & global payments role	USD dominates global payments (>40% via SWIFT) and >80% of global oil trade is dollar-denominated. India invoices over 85% trade in $	RMB accounts for 3-4% of global payments; limited use in global commodity pricing. Very limited use in India except in the case of some oil trade settlements with Russia
Institutional & regulatory framework	Strong rule of law, transparent financial regulations, investor protection, open capital markets	Regulatory opacity, strong state influence over banks and capital flows, limited investor protection perception
Energy trade & recycling mechanism	Oil exporters recycle petrodollars into US assets (Treasuries, equities, bonds). Large and liquid asset markets sustain system	Limited avenues to recycle large yuan surpluses globally; yuan-denominated oil futures exist but limited adoption
Geopolitical & strategic networks	Supported by US security alliances with Gulf oil producers and dominance of global financial infrastructure (SWIFT, dollar clearing)	China expanding influence via Belt and Road, currency swaps, and CIPS, but geopolitical trust and adoption remain limited

Source: Author's compilations based on IMF data inputs

Second, constraints within China's financial system limit the yuan's global role. Reserve currencies require deep and liquid markets capable of absorbing large international capital flows. The US Treasury market alone exceeds $26 trillion, offering a vast pool of safe assets for global investors and sovereign wealth funds.

Although China's government bond market is sizeable, around $20 trillion, foreign participation remains low, near 10 per cent of outstanding bonds. Capital controls, regulatory restrictions, and concerns over legal transparency discourage investors from holding large ¥-denominated assets. Moreover, China's financial system remains dominated by state-owned banks and policy-driven lending, weakening global confidence in market-based capital allocation.

Third, foreign-exchange recycling mechanisms for the ¥ remain limited. Under the petrodollar system, oil exporters particularly Gulf states, reinvest surplus revenues into US financial markets, especially US Treasuries and other dollar-denominated assets that provide liquidity, transparency, and legal certainty. China lacks a comparable ecosystem capable of absorbing large global energy surpluses. Although Shanghai hosts ¥-denominated oil futures, their trading volumes remain far smaller than benchmark Brent and WTI contracts.

> **India should strengthen the international role of the rupee through deeper domestic financial markets, and bilateral currency-swap arrangements**

Geopolitical and institutional factors further reinforce dollar dominance. The US financial system is supported by global infrastructure such as SWIFT messaging networks, dollar clearing systems, extensive banking linkages and 'rule-of-law' unlike China's 'rule-by-law'.

Moreover, many energy exporters maintain deep strategic and security ties with the US, encouraging continued dollar invoicing. Despite China's currency swaps and Belt and Road settlements, most global oil contracts remain dollar-priced.

WAY FORWARD

Even as a geopolitical signal, Petroyuan-based invoicing warrants a nuanced policy debate in India, given its implications for energy security, monetary stability, and strategic autonomy. Given the dollar's liquidity, transparent pricing, and stable recycling, India should largely remain within the dollar-based system, while allowing selective ¥ invoicing to manage transactional flexibility, sanctions exposure, and growing unpredictability in US policy. Further, given the ¥'s partial convertibility, capital controls, shallow financial integration and state-controlled policy initiatives, it should be treated as a tactical tool, not a systemic alternative, with vigilance against excessive dependence on China's state-controlled financial ecosystem.

Second, India must simultaneously strengthen its export capabilities. Without a resilient balance of payments supported by strong export growth, India's ambition of rupee internationalisation will remain largely aspirational and confined to sanction-driven corridors.

In this context, India may explore emerging digital settlement architectures such as the mBridge platform, enabling corridor-specific payment arrangements with trusted partners. It is paramount given that India's existing rupee-trade settlement is also subject to risks of sanctions (Russia, Iran).

Third, strengthening foreign-exchange resilience remains essential. Maintaining robust and diversified reserve holdings including gradual integration of yuan in India's currency reserve amid opening of Chinese investment can be considered as a hedge to manage volatility arising from geopolitical shocks, commodity price fluctuations, or currency realignments in global energy markets amid escalating transactional opacity.

Fourth, India must deepen energy diplomacy beyond 'supplier diversification' to 'supplier integration' involving them as stakeholders in expanding its strategic reserve infrastructure. As India scales geo-economically and geo-strategically, its rise will certainly be tested under strategic rivalry and deterrence and, correspondingly, it must gradually scale crude oil and gas reserves to initially 90 days to become a member of IEA and eventually to 120-150 days.

Finally, India should gradually strengthen the international role of the rupee through deeper domestic financial markets, expanded bilateral currency-swap arrangements, and more resilient cross-border payment infrastructure. Such a balanced strategy, preserving dollar stability while cautiously expanding currency flexibility, will best safeguard India's energy trade, financial stability, and long-term strategic autonomy.

The writer is Professor and Head, IIFT New Delhi. Views expressed are personal

India's economy resilient despite West Asia headwinds

Nirmal K Minda

India's economy continues to grow rapidly, with GDP expected to rise by 7.6 per cent in FY 2026, and exports likely to stay steady at 6 per cent. Looking ahead, growth is expected to remain above 7 per cent in FY 2027, even as global growth is projected to fall below 3 per cent.

Despite geopolitical and tariff challenges, India has grown by more than 7 per cent over the past three years. The Purchasing Managers' Index at 56.9 for manufacturing and 58.1 for services was the highest among the top economies, the US, China and Germany, in February 2026.

India's export trajectory remains steady, increasing by around 6 per cent from April to February 2025-26, reaching $791 billion, up from $748 billion in the same period last year.

Exports are strongly supported by engineering goods, electronics, chemicals, gems and jewellery, and agricultural products. India has numerous diversification opportunities in advanced, emerging, and developing economies.

On the gas and energy front, India maintains a buffer through strategic petroleum reserves and significant commercial stocks, offering an energy cushion of 50-60 days that surpasses the global average. Additionally, India has diversified its crude import sources, drawing supplies from numerous countries, including Russia, the US, Canada, Brazil, and Mexico.

At the current juncture, strict oversight of supply chains must be enforced to prevent traders or intermediaries from hoarding petrol, diesel, and LPG.

Maintaining reliable fuel supply and public confidence are essential to avoid unnecessary market panic. A coordinated policy approach that includes supply monitoring and strategic stock management will help India navigate global energy fluctuations while safeguarding economic stability.

ECONOMY. Robust outlook REUTERS

However, a prolonged US-Israel-Iran conflict could raise logistics costs and lead to price volatility, putting pressure on the transport, logistics, manufacturing, and consumer goods sectors.

MSMEs, which form the backbone of our journey towards Viksit Bharat, require support. Helping them become more resilient to global shocks will strengthen overall growth and job creation. Their working capital needs during global headwinds should be addressed.

The plug-and-play industrial development under the Bharat Industrial Development Scheme (BHAVYA) marks a historic breakthrough in India's manufacturing sector. A seamless, end-to-end ease-of-doing-business model, from approvals to operations, will meet the aspirations of a new India and contribute to the vision of Viksit Bharat.

The writer is President, ASSOCHAM

thehindu businessline.

TWENTY YEARS AGO TODAY.

March 24, 2006

SBI Act changes okayed

The Union Cabinet gave its nod for legislative changes that will enable State Bank of India to sell shares in its associate banks. The changes will also facilitate SBI to split the shares of its subsidiaries. SBI has seven associate banks of which only three are listed.

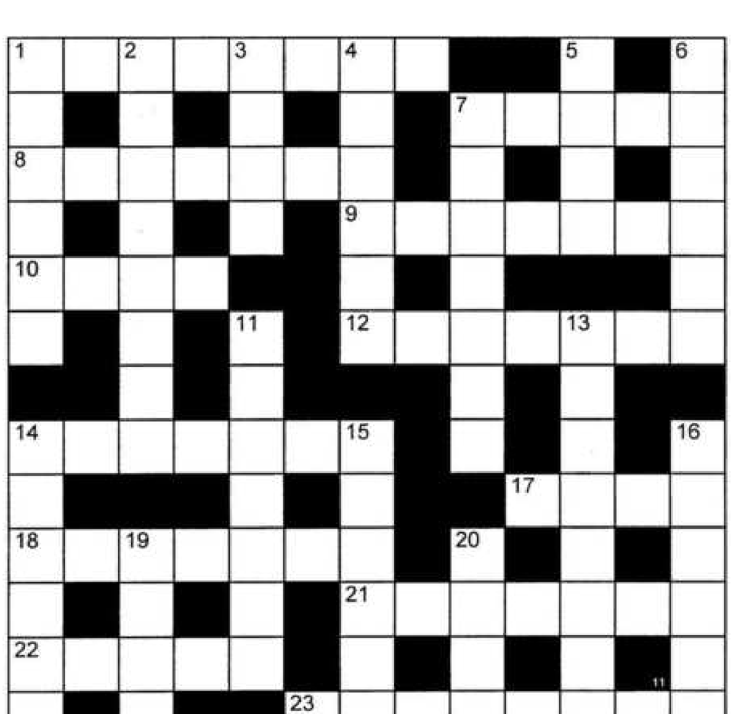
Sonia resigns from Lok Sabha, NAC

The Congress President, Ms Sonia Gandhi, on Thursday resigned from her position as a Lok Sabha MP from Rae Bareilly and as Chairperson of the National Advisory Council (NAC), thereby opening a political Pandora's Box on the issue of legislators holding office of profit. Ms Gandhi's resignation follows Opposition demand for her disqualification as an MP in view of her holding the post of NAC Chairperson, which the Opposition alleged was an office of profit.

Jet-Sahara deal may go through

Barring a last-minute hitch, the Jet-Sahara buyout deal is all set to go through, though there could be a re-negotiation on the end price. There were hectic parleys throughout Thursday between the top officials of Jet Airways, followed by a meeting with the Sahara representatives.

BL TWO-WAY CROSSWORD 2642



EASY

ACROSS

1. Ember-days (4-4)
7. Haughtiness (5)
8. Hanging droppings of frozen water (7)
9. Handcuff (7)
10. Drag, pull with violence (4)
12. Delicate shades of difference (7)
14. Get out of way to escape legal process (7)
17. Use teaspoon (4)
18. Departure of aircraft (4-3)
21. 40 poles (7)
22. Clasp firmly (5)
23. Looked down on with contempt (8)

DOWN

1. Sudden fear (6)
2. Influence producing response (8)
3. Deal out in small portions (4)
4. Obedient follower with no initiative (3-3)
5. Flat, circular body (4)
6. Coat lapel (6)
7. Man-of-war's tender boat (7)
11. Lengthen out (7)
13. Has the governing of (8)
14. Tricks, capers (6)
15. Thick-napped coarse woollen cloth (6)
16. Unanimated (6)
19. Draw together, contract (4)
20. Pleasure excursion (4)

NOT SO EASY

ACROSS

1. Rapid periods when one may deprive oneself of food (4-4)
7. A number of lions showing their amour-propre (5)
8. Droppings that are frozen (7)
9. Handcuff a fellow with torn lace (7)
10. What thief gets from a dacha ultimately (4)
12. Can use name with various shades of meaning (7)
14. Make off with rating the second energy departs (7)
17. A disturbance in which one is confined (4)
18. Departure of aircraft for a parody (4-3)
21. For lung it can be an eighth of a mile of displacement (7)
22. Hang on tightly to a sort of peach (5)
23. Looked down on the speeds I'd achieved (8)

DOWN

1. Scare caused by combat is about right (6)
2. It may suit slum to be something rousing one to action (8)
3. A hand-out that will satisfy the French (4)
14. He'll agree to say men are wrong (3-3)
5. Record is to be found in the situation of Washington (4)
6. Extension of collar one will almost move back (6)
7. A boat that can be made up in wood (7)
11. A non-amateur will yearn to spin it out (7)
13. Has the regulation of knobs, wheels and levers (8)
14. Can sit around and get up to tricks such as these (6)
15. With 'flu, fed on a different sort of coarse wool cloth (6)
16. Frozen with cold if grid is adapted to it (6)
19. Give up some stinking work with wool (4)
20. Will make one lose one's footing on the journey (4)

SOLUTION: BL TWO-WAY CROSSWORD 2641

ACROSS 7. Indefatigable 8. Permanently 12. Dotage 14. Novice 16. Strode 18. Result 19. Flower vases 23. Lets well alone

DOWN 1. Snap 2. Peer 3. Damage 4. Pigeon 5. Vast 6. Clay 9. Enthral 10. Leisure 11. Cent 12. Dose 13. God 15. Owe 17. Eleven 18. Revile 19. Flew 20. Oast 21. Sole 22. Song

CLARITY & CREDIBILITY HT 1924

OPINION

The Hindustan Times
ESTABLISHED IN 1924

{ OUR TAKE }

Prospect of truce in West Asia

The global community should try and turn the US's pause in the war into lasting peace

US President Donald Trump has announced a postponement of attacks on Iranian power plants and energy infrastructure for five days, "subject to the success of ongoing meetings and discussions". The announcement follows his earlier 48-hour ultimatum to Iran to either open the Strait of Hormuz or face further strikes. Iran had said that any such action would be reciprocated with similar attacks on other West Asian countries and Israel. Iran, after Trump's announcement, has denied any direct talks with the US. To be sure, it has been recognised that regional efforts to restore peace and open the Strait of Hormuz should continue.

Whether this development really marks an end of hostilities — it is unlikely that Israel will continue the war if the US withdrew — or is just a temporary pause is anybody's guess at the moment. Markets are willing to be optimistic. Crude oil prices plunged, and international equity markets jumped following the US president's announcement.

Hours before Trump spoke about a temporary ceasefire, Prime Minister (PM) Narendra Modi gave his first detailed statement in the Lok Sabha on the ongoing crisis. While an end of hostilities, and more importantly reopening of the Strait of Hormuz, will really help everyone, including India, there is a widespread consensus that the economic after-effects of the war are not going away anytime soon. The biggest reason for this is the damage to energy infrastructure in West Asia and the time required to reboot the shut production facilities. The PM rightly said that India's first response to the crisis was to safeguard the well-being of the 10-million-strong Indian diaspora in the region. He also spoke at length about minimising the economic pain of the war because of disruption to the energy supply chain and larger petrochemical industrial complex. India's response, the PM said, is working on securing supplies from both West Asia and other regions, taking a differentiated approach on whether the higher costs of inputs are passed on to users or absorbed by the exchequer (for fertilisers, it will be the latter, he said) and rationalising the availability of scarce inputs among different stakeholders.

Modi also reiterated India's commitment to restoring peace by dialogue rather than siding with either of the parties in the war. India, along with other like-minded countries, should do all it can to ensure that the five-day ceasefire is the end of the current war.

In Kerala poll season, party fidelity loses out

Kerala votes on April 9, and political parties have been quick to announce their lineups. What is unusual this time is the high presence, by Kerala standards, of turncoats in the candidate lists. This should be particularly disturbing for the ruling Left Democratic Front: Two former CPI MLAs and a former CPM MLA, joined the BJP when denied seats (all three are now BJP candidates); two others are contesting as candidates of the Congress-led United Democratic Front; and besides these five, two senior CPM functionaries, in the party's red fort of Kannur, rebelled against the party and joined the fray as independents (they are now backed by the UDF).

Parties splitting and/or shifting alliances are not unusual in Kerala's coalition politics, but the kind of party hopping among politicians visible in this election season is unusual. For instance, former Communist legislators joining the BJP over the denial of seats has almost no precedence in Kerala. Such a dilution of ideological fidelity suggests multiple undercurrents. One, politics has become more transactional. Elections have become expensive, and parties love candidates who can mobilise material resources. Two, political parties are more focussed on the winnability of candidates rather than their commitment to ideology or party. Three, the party system has broken down. Ambitious individual politicians are increasingly willing to pursue personal goals instead of staying obedient party members. This trend is triggering rebellion in parties. Enterprising politicians recognise that the electorate does not see ideological clarity or political fidelity as an asset. Four, the BJP is finding cross-community acceptance and has been successful in attracting leaders from other parties.

That's the new Kerala story in the making.

Five priorities for India to eliminate TB

India's fight against tuberculosis tells a story of real and hard-won progress

Soumya Swaminathan

Over the past decade, India has cut new tuberculosis (TB) cases by 21%, almost twice the pace of the rest of the world, a testament to enduring political commitment, increased funding, and home-grown innovation in detection and care. India is now finding more TB patients than ever before. Now, with over 26 lakh cases recorded in 2025, the country is closing the long-standing gap of "missing" patients, reaching those who were always there but not diagnosed. In TB, being found is the first step to being cured.

As we acknowledge this hard-fought win, we must maintain this progress and accelerate towards global TB goals. Before we can eliminate TB (this could take decades), we must prevent all TB deaths and reduce catastrophic costs to families. Five priorities can help define the path ahead.

First, reducing TB deaths must become a central focus. It remains concerning that more than 3,00,000 people continue to die from TB each year, despite the disease being curable. A tailored and individualised approach — known as Differentiated TB Care — can help identify high-risk patients early and address their specific needs, such as comorbidities including diabetes, HIV, and malnutrition.

Timing is critical. Evidence shows that most TB patients who die do so within two months of diagnosis, with nearly half of these deaths occurring in just the first two weeks. Tamil Nadu's TN-KET (TB Death-Free Project) demonstrates the impact of screening all TB patients with a simple five-sign tool to identify those at highest risk: Within six months of its launch in several districts, overall TB deaths fell by nearly 10%, while early fatalities declined by almost 20%.

The national programme's move to adopt this approach is a welcome step, but it must now become the standard norm across the country. We need beds in every district to be earmarked for sick TB patients and intensive nutritional rehabilitation and alcohol de-addiction services, when needed.

Second, we must close the diagnostic gap; arguably, one of the most critical links in TB care. Studies suggest that nearly half of all TB cases are subclinical or asymptomatic — meaning individuals show no visible symptoms while continuing to transmit the disease. India's strategy of expanding community-based screening is, therefore, both timely and essential. Encouragingly, several Made-in-India innovations can accelerate this effort, with TB testing moving to the doorstep. Portable AI-powered X-rays and simple non-invasive tongue or nasal swabs could enable testing closer to communities. And the vast lab network built during Covid-19 can now work for TB, with new Indian-made machines (Open PCR systems) that cut testing costs by 80%.

Third, focused action is needed for vulnerable populations, particularly tribal communities. TB rates among Scheduled Tribes are about 50% higher than among non-tribal populations. Addressing this disparity requires directed institutional attention.

Innovative financing mechanisms can support these efforts. District Mineral Foundation (DMF) funds were created to support communities affected by mining, many of whom live in tribal areas with a high TB burden, representing a largely untapped opportunity. With a national corpus of ₹1.03 lakh crore, nearly 60% of which remains unspent, even a small reallocation toward TB screening and diagnostics could have a significant impact. The Chamarajanagar DMF in Karnataka, for example, has invested in TB diagnostics, including molecular tests and X-ray services, offering a promising example for replication.

Fourth, India must adopt specific strategies to address TB in urban settings. Cities are emerging as major TB hotspots due to dense populations, high migration, and other risk factors. National plans alone cannot adequately address the diversity of urban challenges. Greater engagement with urban local bodies (ULBs) is, therefore, essential.

It remains concerning that more than 3,00,000 people continue to die from TB each year, despite the disease being curable. HT ARCHIVE

Just as Panchayati Raj institutions have been rallied in rural areas, ULBs can drive local ownership and accountability in cities. Integrating TB as a priority within municipal health planning is key, and with these bodies receiving more funds than ever through central grants, they are well-placed to lead TB efforts in urban areas. The experience of the Swachh Bharat Swachh Survekshan initiative demonstrates how local governments, when mobilised around a national priority, can deliver exceptional results. The capacity exists; it now needs to be directed towards TB as well.

Dedicated subcommittees within the national programme could help ensure that interventions are tailored to the environmental, social, and health system realities of tribal and urban low-income areas.

Finally, true elimination will depend on continuous investment in next-generation research — particularly new vaccines, diagnostics, and drugs, as well as epidemiological and operational research. India is well placed to take a leading role in advancing TB research, not only for its own population but also for countries across the Global South.

We are witnessing a renewed national focus on strengthening India's research ecosystem, with the establishment of the Anusandhan National Research Foundation (ANRF). The ICMR could lead a TB Accelerator by setting priorities and standards, making available greater resources, establishing public-private-philanthropic partnerships and global collaboration, which would help accelerate breakthroughs, develop affordable tools, and shape solutions for high-burden settings worldwide.

India today has the momentum, the capacity, and the political will to end TB. The task now is to translate this momentum into a time-bound, accountable roadmap so that progress is not just retained, but accelerated.

Soumya Swaminathan is chairperson, MS Swaminathan Research Foundation and principal advisor, NTEP, ministry of health and family welfare. The views expressed are personal

{ SIMPLY ECONOMICS }

Pramit Bhattacharya

Time of great global demographic u-turn

In the late 19th century, the British biologist Francis Galton came up with a new field of population studies called eugenics (literally meaning "good birth"). Drawing on the work of his half-cousin Charles Darwin, Galton argued that the human race could be improved by encouraging the fittest members to have more children.

The eugenics movement spread across the West in the early part of the 20th century, with the focus gradually shifting from promoting desirable traits to removing undesirable traits.

In the US, this took the form of population control for minority groups (including forced sterilisations for African-Americans and native Americans). The movement took an even more horrific turn in Germany, with concentration camps designed to segregate, and eventually kill, those deemed undesirable by the Nazis.

The association with Nazism proved to be a death knell for eugenics. But the population control movement survived. In the post-War years, global concerns about food insecurity and environmental degradation stirred a new paranoia about population growth. The "undesirable" elements of the human race were now located in the Third World, rather than in the West.

The ideas of an 18th century economist and clergyman were used to drum up support for population control campaigns. In his 1798 book, *An Essay on the Principle of Population as it Affects the Future Improvement of Society*, Robert Thomas Malthus had warned that population growth would soon outstrip global food supply. The Industrial Revolution proved Malthus wrong. It raised productivity across all spheres of the economy, including agriculture.

Undeterred by the historical record, neo-Malthusians of the mid-20th century revived fears of an impending food crisis. Back in the day, Reverend Malthus had favoured sexual abstinence to limit population growth. In a concession to modernity, neo-Malthusians recommended the use of contraceptives.

Some neo-Malthusians such as the Nobel Prize-winning biologist Paul Ehrlich went further, advocating a "series of financial rewards and penalties designed to discourage reproduction". In his widely-cited 1968 book, *The Population Bomb*, Ehrlich painted a vivid picture of the overflowing streets of Delhi to convey the urgency of defusing the ticking "population bomb". The teeming millions of India and China became the new targets of population control, replacing African-Americans and the Jews. UN agencies and American philanthropic foundations helped spread the new gospel of population control.

Governments in developing countries such as India and China jumped aboard the population control bandwagon. Weak economic growth and high population growth had led to anaemic per capita income growth in these countries. Policymakers felt it offered a way to raise incomes per head and lower the strain on government finances at the same time.

The population control drive took a coercive turn with China implementing the one-child policy in the late 1970s. Under the cover of the Emergency, India undertook forced sterilisations. The allocation of Lok Sabha seats across states was frozen to penalise states with high birth rates (through the 42nd amendment in 1976).

When the United Nations Population Fund launched the Population Award in 1983, the first winners were the Indian Prime Minister Indira Gandhi and the Chinese family planning commission chief Qian Xinzhong.

Economists and demographers remain divided on the effectiveness of the population control strategy. Total fertility rates (or births per woman) have been declining in India and China (and in many large developing countries) since the 1960s, well before the population control campaign reached its peak.

Socio-economic changes such as rising enrollment of women in educational institutions, declining child mortality rates, and falling poverty may have impacted fertility rates more than State-funded "family planning" programmes.

The faster-than-anticipated decline in fertility rates has stirred new fears about "depopulation". For many years, UN projections had suggested that the global population would keep growing throughout the 21st century. But in 2022, it made a sharp revision, indicating that the global population would peak in the 2080s, and decline after that.

Some demographers and economists believe that governments should now encourage people to have more children. A depopulating world may not necessarily be good for either the environment or economic growth, they argue. The rising share of the elderly might precipitate fiscal crises around the world, and slow down growth. The declining share of youth would slow down innovation, and inhibit humanity's ability to fight the climate crisis, they fear.

While these concerns may be justified, it is not entirely clear what the policy response should be. We still don't know why the pace of decline in fertility rates varies across regions with similar socioeconomic characteristics. Or what has caused the uptick in fertility rates among better educated women in several high-income regions.

For more than a century, the world has witnessed numerous misguided attempts to control population growth, motivated by paranoia and bigotry. We need to avoid further misadventures in the name of population stabilisation.

We need more careful and multi-disciplinary research to understand the drivers of demographic changes in different parts of the world. Radical policy prescriptions based on limited knowledge will not help the world.

Pramit Bhattacharya is a Bengaluru-based journalist. The views expressed are personal

{ S JAISHANKAR } EXTERNAL AFFAIRS MINISTER, INDIA

India and Russia share a special and privileged strategic partnership rooted in trust and mutual respect. Over decades, our mutually beneficial cooperation has advanced regional and global peace, stability and progress

HT

West Asia's shadow on energy security in India

Hemant Mallya

Sabarish Elango

Oil and gas together account for 36% of India's primary energy supply, yet India imports ~90% of its crude oil and 47% of its natural gas. The Strait of Hormuz, now effectively closed, is the transit route for over 50% of India's crude oil imports and 55–60% of its liquified natural gas (LNG) imports. India has limited options to fall back on, despite turning to spot LNG markets and the temporary relief in buying Russian crude. Rising crude prices have already driven up liquified petroleum gas (LPG) prices and could drive up petrol, diesel, and CNG prices as well, with knock-on effects across the economy. India's energy security cannot rely on reacting to crises in concentrated supply chains.

The first priority is to strengthen storage buffers. India's Strategic Petroleum Reserve (SPR) provides only 9–10 days of net crude oil import cover (74, including commercial inventories). This is below the 90-day benchmark recommended by the International Energy Agency. Japan holds nearly 200 days of reserves and South Korea ~207. In these countries, minimum stockholding levels are statutory obligations on both government and industry, whereas India's SPR targets rest on cabinet resolutions and executive orders. The planned expansion by 6.5 million tonnes by 2027 should be accelerated under clear statutory obligation, with defined stockholding requirements for oil marketing companies.

India also currently has no dedicated strategic gas reserves to cushion price-sensitive sectors like fertiliser production and piped gas. During the 2022–23 gas price spike, the fertiliser subsidy bill exceeded $30 billion. Underutilised capacity at India's LNG terminals (operating at 50% utilisation on average) could be used as an interim strategic buffer. GAIL has planned 3–4 billion cubic metres of storage in depleted gas fields. India should build substantially more gas reserves and also future-proof new strategic reserves to be green-hydrogen-ready from the outset.

Electrification of road transport also strengthens energy security. Research by the Council on Energy, Environment and Water (CEEW) finds that electric two- and three-wheelers are already cheaper to own than their petrol- or gas-powered counterparts. They constitute 67% of India's petrol consumption, demand for which is projected to peak by 2032. Diesel demand could likely peak only by the mid-2040s because freight electrification remains costlier.

In the interim, LNG offers a pragmatic bridge for trucks and buses, moderating excessive growth in diesel demand and reducing the risk of stranded refining assets.

Over the longer term, India must reduce structural dependence on imported oil and gas.

First, freight should undergo modal shifts while transitioning to clean fuels. Shifting trucks to electricity or green hydrogen will require sustained domestic R&D on hydrogen and electric powertrains, and the development of hydrogen refuelling and high-capacity charging infrastructure along major freight corridors. Road-based diesel transport is projected to grow sharply through mid-century under current trends.

Second, the industrial energy transition should be leveraged as a major opportunity for strategic substitution. India has nearly 3,700 GW of wind-solar hybrid power generation potential, which reduces intermittency. This could support green hydrogen production and industrial electrification. Transitioning fertiliser production from imported LNG to green hydrogen could release over 15 billion cubic metres of gas for other uses. Electrifying industrial heat and steam applications would also free up additional gas.

Finally, refineries also need a clear national transition strategy. India currently imports 60% of its LPG. Reducing the share of petrol production could unlock additional domestic LPG supply from the excess naphtha currently used to produce petrol.

The crisis in West Asia is a reminder that India's energy security remains tied to geopolitical events far beyond its borders. The energy transition is India's most credible long-term energy security strategy. Treating it as such, rather than primarily as an environmental commitment, is where policy must now go.

Hemant Mallya is fellow, and Sabarish Elango is programme lead at the Council on Energy, Environment and Water (CEEW). The views expressed are personal

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Manufacturing push

Execution will be key to the success of industrial parks

The Union Cabinet's recent approval of the ₹33,660 crore BHAVYA (Bharat Audyogik Vikas Yojana) scheme to develop 100 plug-and-play industrial parks is a significant step towards strengthening India's manufacturing base, promoting job creation and improving facilitating business. Spread over about 33,000 acres and to be implemented in more than six years, these parks will provide pre-approved land, ready infrastructure, and integrated services, supported by robust single-window clearance systems and deregulation measures led by states. Financial assistance of up to ₹1 crore per acre will be provided for core infrastructure such as internal roads, underground utilities, drainage, and common facilities. Estimates suggest it could generate around 1.5 million direct jobs while strengthening domestic supply chains through cluster-based development.

Industrial nodes such as the Dholera Special Investment Region, the Shendra-Bidkin Industrial Area, Vikram Udyogpuri, and the Greater Noida Industrial Area highlight the potential of plug-and-play ecosystems, where pre-approved land, integrated infrastructure and multimodal connectivity enable faster industrialisation. The National Industrial Corridor Development Corporation, which is implementing 20 projects across 13 states, has demonstrated how such models can attract both micro, small, and medium enterprises (MSMEs) and large businesses by reducing entry barriers and improving project readiness. However, challenges such as construction delays and long completion timelines often raise costs and erode investor confidence.

Past experience also offers a cautionary lesson. Many industrial areas across states remain underutilised; for instance, some parks in Karnataka have only about 25 per cent of allotted units operational due to poor connectivity, weak governance, regulatory delays, and lack of demand. Similarly, several special economic zones and industrial corridors have struggled with various issues. Delays in synchronising internal infrastructure with external networks, such as highways, rail freight corridors, and ports, often leave parks operationally constrained. Even when land and basic facilities are ready, firms face bottlenecks in accessing reliable power, water, and digital connectivity, raising operating costs and reducing competitiveness. These gaps highlight that facilitating business is not merely about policy design but about consistent, on-ground implementation and coordination across agencies.

To succeed, BHAVYA must focus on execution. The scheme's emphasis on ensuring seamless last-mile connectivity through integration with PM Gati Shakti would reduce logistics costs and improve competitiveness. Reliable power, water, and digital infrastructure, supported by features such as underground utility corridors, must be guaranteed to ensure uninterrupted industrial operations. The scheme's challenge-based selection process must prioritise high-quality, investment-ready proposals, and promote sector-specific clustering in areas such as electronics, textiles, pharmaceuticals, and green manufacturing. Linking these parks with skilling ecosystems will be required so that employment generation is both inclusive and sustainable. In essence, BHAVYA emphasises ready ecosystems, integrated planning, and ease of doing business. This could help India deepen its manufacturing base, attract investment, and create jobs at scale. However, its success will depend on translating the ambitious design into efficient and functioning industrial ecosystems. It must also be highlighted that the development of industrial parks should not lead the government to lose focus on the rest of the country. An enabling business environment needs to be created across the land.

Protecting biodiversity

Strategy has improved, but outcomes remain uncertain

India recently submitted its Seventh National Report to the Convention on Biological Diversity, offering a detailed stocktaking of its biodiversity commitments at a critical point in the runup to 2030. It is the first full progress assessment since the Kunming-Montreal Global Biodiversity Framework was adopted in 2022. While India has built a wide-ranging institutional and digital-data architecture, marking a shift toward systematised governance and biodiversity conservation, several outcomes still remain uneven. India's updated National Biodiversity Strategy and Action Plan (2024-30), aligned with the Kunming-Montreal framework, includes a biodiversity-monitoring system with 142 indicators mapped to 23 national biodiversity targets (NBTs). On the ground, there has been visible progress in spatial planning and ecosystem monitoring under NBT-1. India's remote-sensing programme, led by the Indian Space Research Organisation (Isro), under the National Natural Resources Management System, has created the capacity for large-scale spatial and temporal assessments of forests, wetlands, and coastal ecosystems, thereby supporting area-based conservation planning.

Environmental governance has also been streamlined through the PARIVESH (Pro-Active and Responsive facilitation by Interactive, Virtuous, and Environmental Single Window Hub) platform, a single-window facility based on the Geographic Information System, or GIS. This system integrates forest, wildlife, and environmental and coastal clearances while tracking compliance. More recently, the expansion to PARIVESH 2.0 reflects an attempt to combine technological efficiency with regulatory oversight. Data-driven approaches are also being put to use to achieve progress under NBT-2 on ecosystem restoration. For instance, Isro's Space Applications Centre has mapped desertification and land degradation across multiple time periods. A pilot land degradation vulnerability assessment conducted across 30 districts, integrating socioeconomic, climate and soil data to identify high-risk areas, is helping restoration efforts. Despite the expansion of restoration programmes, ecological stress continues alongside recovery efforts.

A central concern is the gap between restoration and degradation. While India reports that 24.1 million hectares have been restored or are under restoration against a target of 26 million hectares, nearly 30 per cent of its geographical area continues to face land degradation. Conservation coverage also remains limited, with only a little over 5 per cent of India's land under formal protection. While flagship species such as big cats and rhinos show recovery, the report provides limited data on lesser-known species and other taxa, making it difficult to assess broader biodiversity health. Agricultural ecosystems remain a weak link. Although agroforestry now covers about 8.65 per cent of India's geographical area and trees outside forests contribute significantly to total tree cover, excessive pesticide use and nutrient runoff remain concerns.

Data weaknesses, ranging from non-standardised indicators to fragmented collection systems, complicate the assessment of real progress. Besides, external pressures are intensifying. Climate-related events are affecting ecosystems, compounding existing stresses from land-use change, infrastructure expansion, and urbanisation. Much of this biodiversity loss occurs outside protected areas, where regulatory oversight is weaker. Taken together, it is clear that biodiversity loss increasingly intersects with climate risks, food security, and livelihoods. Thus, monitoring systems should be matched with regulatory enforcement while conservation activities should move beyond protected areas into agriculture, infrastructure, and urban planning.

India's hidden credit constraints

The credit problem may lie less in liquidity and more in how it is priced and regulated

ILLUSTRATION: BINAY SINHA

Even with abundant banking liquidity, banks continue to scramble for deposits. Overall, India's credit ecosystem remains small relative to the size of our economy.

All this reflects a combination of tax incentives, regulatory structures, and monetary conditions that suppress the attractiveness of fixed income and constrain credit expansion. In turn, this has wider implications for our asset markets and external balance.

Sizing the problem

Data from the Reserve Bank of India (RBI), the Securities and Exchange Board of India (Sebi), and the World Bank suggests that India is a global outlier in the relative size of our credit markets. India's domestic credit, across banks, non-bank finance companies, and corporate bonds, stands at just 60 per cent of equity market capitalisation. The global average is 115 per cent.

Even in the United States, with its equity culture, credit is 95 per cent of market capitalisation. In countries like Japan, Germany, and South Korea, this ranges between 125 per cent and 195 per cent. China, with its investment-heavy growth model, is at a staggering 310 per cent.

India's shallow credit base reflects structural constraints in credit formation.

POLICY IN PRACTICE

ANANTH NARAYAN

The pricing of fixed income

Low returns from fixed income weaken the foundations of our credit ecosystem.

First is the fiscal drag. Taxing interest income at marginal income tax rates makes it harder for post-tax returns to clear the expected inflation hurdle. In recent times, this has sparked a migration of household savings away from fixed income into equities. This, in turn, reduces the pool of long-term savings that underpin credit markets.

Second is the impact of monetary intervention. For FY26, the RBI will conduct record open market operation (OMO) bond purchases of ₹8 trillion, effectively absorbing 77 per cent of net central government bond supply through secondary market operations. While intended to inject liquidity and support growth, this anchors risk-free rates below where the market would otherwise clear the deficit.

Economists argue that cooling consumer price inflation justifies lower interest rates. However, in any market economy, prices should ultimately be determined by supply and demand. When rates are influenced by large-scale interventions, their signalling role becomes less clear.

All this results in fixed-income returns being seen as insufficient compensation for inflation and duration risk.

Regulatory constraints

Overlaying this are banking regulatory requirements such as the liquidity coverage ratio (LCR) and the net stable funding ratio (NSFR). These are global post-crisis standards to ensure the resilience of banks. Their system-wide effects merit attention.

LCR requires banks to hold high-quality liquid assets (HQLA) to cover a 30-day "stress" outflow. Regulations prescribe how flight risk is computed. Deposits over 30 days have zero imputed outflow. In contrast, "sticky" retail deposits under 30 days may carry a 7.5 per cent runoff, while "volatile" wholesale deposits may carry a 40 per cent outflow.

To manage LCR, banks prioritise term deposits, followed by retail demand deposits, with wholesale short-term deposits being a last resort. Any adverse shift in the mix forces banks to hold more HQLA, in turn requiring additional deposits. Similarly, NSFR demands "reliable stable funding", or deposits exceeding one year to back long-term lending. If there is insufficient growth in long-term deposits, long-term lending will be curtailed.

The deposit paradox

System-wide, deposits are created by bank lending, government spending funded by banks, or foreign exchange (FX) inflows. They are drained by the repayment of bank loans, an increase in currency in circulation (CIC), or FX outflows.

While loans create their own deposits, the LCR burden is neutralised only by term deposits. As the deposit mix shifts toward shorter-tenor or less stable categories, banks must hold more HQLA, necessitating additional funding.

Deposits from individuals have reduced from 60 per cent of all deposits 10 years ago to 52 per cent now, with no compensatory shift in the extent of term deposits. Alongside, any net FX outflows and CIC spikes drain deposits, and strain LCR management. While the ratio of CIC to deposits has remained under control, there have been FX outflows in recent years.

All this explains the current scramble for deposits, despite ample banking liquidity.

There is also a "fallacy of composition" here. LCR and NSFR address the idiosyncratic risks of a bank run. System-wide, a run usually reshuffles deposits between banks. While buffers enhance the resilience of individual banks, they overcompensate at an aggregate level, constraining the system-wide ability to lend.

External spillovers

With post-tax returns from fixed income struggling to beat inflation expectations, domestic flows into equities have risen. In recent times, this flow has outstripped fresh equity issuance, leading to pockets of overvaluation. This creates an environment conducive for foreign investors to pare exposure, likely contributing to the recent moderation in net foreign investment.

Low domestic interest rates also compress the dollar-rupee forward premium. Especially during times of currency volatility, this encourages the hedging of imports and rupee investments, discourages the hedging of exports, and induces domestic savers to invest overseas. These add pressure on the rupee while further reducing domestic deposits.

The path forward

Addressing these interconnected issues requires a calibrated and multi-pronged approach.

First, a gradual move towards a more asset-agnostic, savings-friendly taxation framework would allow risk-return considerations, rather than tax differentials, to guide asset allocation. This would help stabilise fixed-income flows, support balanced asset valuations, and create room for consistent foreign investments across asset classes.

Second, on LCR and NSFR, we must remain aligned with global best practices but avoid tightening beyond them. Additional buffers, while prudent in isolation, aggregate to a systemic burden that stifles the economy they aim to protect.

Third, monetary and currency interventions must be assessed not only for their immediate objectives, but also for their higher-order effects on savings behaviour, asset markets, and financial intermediation. This is especially crucial if the focus of intervention fluctuates periodically between currency stability and interest rate anchoring.

India's financial system has achieved significant depth in equity markets. The next phase of development requires a comparable strengthening of credit markets across banking and the corporate bonds.

Without this, the paradox of liquidity without sufficient credit will likely persist.

The writer is a former whole-time member, Sebi. The views are personal

War & emerging geopolitical faultlines

The unfolding United States-Iran (US-Iran) war has underscored a familiar but often underappreciated truth in international politics: When military objectives collide with structural economic dependencies, conflicts rarely remain confined to their original intent. What began on February 28 as a calibrated US-Israeli effort at regime decapitation and counter-proliferation has, within weeks, morphed into a far more consequential struggle over the sinews of the global energy order.

This transformation is neither accidental nor surprising. In a region where hydrocarbons are both the foundation of state power and the currency of global interdependence, the targeting of energy infrastructure represents a logical, if deeply destabilising, extension of strategic competition. By mid-March, the Persian Gulf's networks of oil and gas had effectively become the central theatre of conflict, with both sides signalling that economic coercion is now as vital as military dominance. Though US President Donald Trump has decided to "postpone any and all strikes against Iranian power plants and energy infrastructure" for five days, citing "productive conversations" about "a complete and total resolution to the conflict", there is still no certainty about the course of this conflict, especially as Iran is interpreting it as a vindication of its strategy of effectively closing the Strait of Hormuz, thereby throwing the global economy in turmoil.

HARSH V PANT

The speed of escalation has been striking. Iran's decision to shut off the Strait of Hormuz on March 1 was not merely an act of retaliation; it was a calculated demonstration of its enduring capacity to disrupt the global commons. Accounting for roughly a fifth of the global flows of oil and liquefied natural gas (LNG), Hormuz remains the ultimate pressure point in the international energy system. Tehran's intent to weaponise this chokepoint, through tanker interdictions, GPS (global positioning system) disruption, and explicit threats, has forced a reckoning far beyond the region.

Israel's subsequent strike on South Pars, followed by Iran's rapid retaliation against Qatar's Ras Laffan complex, illustrates the emerging logic of reciprocal vulnerability. These are not symbolic targets; they are critical nodes in the global energy architecture. The damage inflicted, particularly on LNG infrastructure, has exposed the fragility of supply chains that underpin economic stability from Europe to East Asia. Parallel strikes across the United Arab Emirates, Saudi Arabia, and beyond reinforce the point: The conflict has expanded horizontally, enveloping the Gulf's entire export ecosystem.

What is particularly noteworthy is the widening aperture of targeting. Critical infrastructure — from ports and desalination plants to data centres and airports — has now entered the conflict matrix. This reflects a broader shift in contemporary warfare, where the distinction between civilian and military assets is increasingly blurred. In the Gulf context, where economic infrastructure is deeply integrated, such actions amplify both immediate humanitarian risks and long-term developmental setbacks.

The economic consequences, as expected, have been swift and severe. Energy markets, already operating in tight conditions, have reacted with volatility. Price spikes, supply disruption, and cascading effects across sectors, from aviation to agriculture, underscore how regional instability can trigger systemic global shocks. The International Energy Agency's stark warning about energy security captures the scale of the challenge, but it also points to a deeper structural issue: The world's continued dependence on a region that remains geopolitically brittle.

Geopolitically, the conflict is accelerating trends that were already underway. Gulf states, long positioned as pillars of stability in global energy markets, now find themselves directly exposed. Their strategic calculus is likely to evolve, balancing security partnerships with the imperative of economic survival. This could translate into greater hedging behaviour, including deeper engagement with alternative power centres such as China and Russia.

At the same time, Iran's use of relatively low-cost, high-impact capabilities highlights the growing efficacy of asymmetric strategies. By raising the costs of intervention for technologically superior adversaries, Tehran is reshaping the deterrence equation. The potential involvement of regional proxies, particularly in the Red Sea, further complicates the operational landscape and risks broadening the conflict's geographic scope.

For major powers, the implications are profound. China's concerns on energy security will inevitably sharpen, potentially driving a more assertive role in safeguarding supply lines or accelerating diversification efforts. Europe, still grappling with the aftershocks of previous energy disruption, faces renewed vulnerability. Russia, while benefiting in the short term from elevated prices, must navigate the longer-term uncertainties of a shifting Gulf order.

Ultimately, this conflict is less about immediate battlefield gains and more about the reconfiguration of global energy geopolitics. Chokepoint vulnerability is likely to accelerate diversification — more renewables, African LNG, or Arctic routes — while underscoring US limits in enforcing maritime security without broader buy-in. If ground operations or further power-plant strikes occur, the conflict could drag into months, entrenching higher baseline prices of energy and a more fragmented world order. The deliberate targeting of infrastructure marks a departure from traditional conflict boundaries, signalling the intent to impose systemic costs on the international economy. It also exposes the limits of unilateral power in managing interconnected global systems.

The central question now is not simply how the conflict will end but what precedents it will set. By making energy and critical infrastructure the battlefield, the war has globalised what began as a bilateral strike. It is testing alliances, inflating economic pain worldwide, and signalling a new era where Gulf hydrocarbons are both weapon and target. The normalisation of infrastructure targeting, the weaponisation of chokepoints, and the deepening of multipolar alignments all point toward a more fragmented and volatile order. In that sense, the 2026 Iran war is not merely a regional confrontation but a defining moment in the evolution of 21st century geopolitics — where energy is both the instrument and the battleground of power.

The author is vice-president for studies and foreign policy, Observer Research Foundation

US engagement in the Indian Ocean

BOOK REVIEW

GUNJAN SINGH

Conflict in West Asia and the disruption of essential energy supplies underscore the critical role of the sea lanes passing through the Indian Ocean region (IOR) and the Persian Gulf. Vivek Mishra's book promises to discuss the trajectory of dominance, diffusion and adaptation of the United States (US) maritime strategy especially focused on the IOR in the post-Cold War period.

The major premise of the book is that by the early 1960s, "Britain's ability and political will to either sustain its dominance in the Indian Ocean or to continue conducting traditional maritime patrols in the region had waned." This was followed by increased American presence in the region. From initial ignorance, the US approach towards the Indian Ocean became one of active presence. The end of the Cold War also paved the way for US attention toward the IOR as the sole challenger to its power, the Soviet Union, had ceased to exist. Mr Mishra writes that the post-Cold War period "...marked a strategic consolidation of US influence – shifting its posture from episodic engagement to sustained regional dominance."

This period also witnessed a change in Washington's goals vis-à-vis its role in the IOR. Now it moved to "crisis response humanitarian relief, multinational joint naval exercises, nation-building, maritime security, promoting US interests abroad etc." Increasing trade relations between the US and Asian nations also warranted Washington's "cautious and compelling involvement" in the IOR. Further, the end of the Cold War brought about a shift in the kind of threats for the US at the seas, piracy and terrorism being two most pertinent. These threats have also forced the US to seek a cooperative and collaborative approach.

Thus, there has been a rise in bilateral and multilateral cooperation between the US navy and the navies of other Indian Ocean littoral states such as India, Pakistan and Australia. The US has also been working with the UAE, Kuwait, Saudi Arabia, Bahrain, Oman, Qatar, Egypt and Jordan, the other littoral states in the Indian Ocean, where it has military bases. The author singles out India as one of the most prominent engagements. "India and the United States have institutionalised cooperation both bilaterally and through multilateral platforms such as the Quad, where the sharing of real-time information and technology is a core metric of operational synergy," Mr Mishra writes. These efforts aim to uphold a free, open and inclusive regional order— one that resists domination by any single power."

Globalisation has also called for increasing cooperation at multiple levels. The author argues, "Supply chains and critical technologies, such as artificial intelligence and biotechnology, are becoming increasingly important to emerging nations in the IOR. Therefore, building strong partnerships and democratic linkages in the Indian Ocean will enable the US to maintain its technological leadership and influence".

The rise of China has definitely played a crucial role in directing US maritime policies and approach toward the IOR. "China's assertive posture in the region presents a strategic challenge for the US, necessitating a collaborative, multilateral approach that combines military and economic strategy," Mr Mishra says. He adds, "For the US, India is the only country that is capable of countering Chinese provocations while maintaining a favourable balance in the Indian Ocean." The rise in the capacity and reach of the People's Liberation Army Navy (PLAN) has also pushed the US to renew its focus towards the IOR. Beijing's Belt and Road Initiative (BRI) and its "energy strategy is likely to drive more [US] engagement with the Indian Ocean".

American Maritime Strategy in the Indian Ocean: Dominance, Diffusion, Adaptation
by Vivek Mishra
Published by Routledge
202 pages ₹1,295

Though the book paints a detailed picture of US measures and policies towards the IOR during and after the Cold War, multiple aspects remain unexplained and unexplored. Mr Mishra does hint at the need and future role of the Quad, without fully engaging with it. The book is a useful addition to the existing literature, but it is repetitive. It provides a lot of data and analysis but most of the book is dedicated to arguments and developments that occurred till 2010. Much has happened since that has altered the dynamic of US engagement. But the reader has to get to the annexure to gain an insight into the author's arguments with respect to current developments. As a result, most of the book is just a build-up to how the US changed its approach toward the IOR in the post-Cold War period. It has, unfortunately, been overtaken by fast-paced events in the past two decades.

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OUR VIEW

PTI

LPG scarcity: We need a full-spectrum response

With second-order impacts of an LPG shortage in evidence, India must employ additional ways to plug supply gaps and double down on fostering a mass transition to electric cooking

Liquefied petroleum gas (LPG) is one of the products facing a global supply crunch today, thanks to the Iran war. The result is more than a cooking gas shortage in India. In order to prioritize the availability of cooking gas for homes, the government reduced supplies to commercial establishments. Although strictures were eased a little, scarcity-hit eateries have shut down in several towns and cities. Other impacts have been in evidence too. A large proportion of the urban poor who live in slums and informal rental housing fail to qualify for LPG supply from oil marketing companies, leave alone subsidized connections under the Centre's Ujjwala Scheme; they mostly buy their LPG in 5kg cylinders filled with gas meant for commercial use and have been the worst affected by the current shortfall. Migrant workers employed by eateries starved of LPG have found themselves out of work. Like other migrant workers, many have struggled to keep kitchen stoves running. Faced with livelihood losses and unaffordable means of cooking food, many are headed back home from their cities of employment. The LPG dislocation has been so sharp, it could take weeks to recover from it even after supplies normalize—for which the war must end and the Gulf's hydrocarbon facilities repaired. India is staring at an LPG scarcity that could stretch from the short to medium term.

One way out is for LPG production at domestic refineries to rise. This can be achieved through tweaks of the usual refining process and the use of catalysts. Of course, refiners would have to forgo some of their high-value petrol output in favour of LPG. There are limits to how far a refinery's output mix can be altered, but the use of lighter crude can also help increase its LPG yield. Yet, such a step-up can make up for only a fraction of our lost imports from the Gulf. India must shop around for LPG from other major producers such as the US, Canada and Russia, even if it means paying a higher price. The higher cost must be passed on to consumers for them to modify their consumption either by reducing gas use or making it more efficient; the poor, of course, would need help. Thankfully, with minor tweaks, LPG stoves can be adapted for piped natural gas (PNG). So, homes in areas where piped gas is available should be shifted to PNG. Urban gas suppliers have reportedly been pushed into action. But this is a stopgap solution, since we also face a shortage of natural gas; the war has clamped most of our imports, which fulfil about half the country's consumption. Over the medium term, thus, India must shift to cooking with electricity derived from domestic coal and renewable energy, as *Mint* has argued earlier.

This switch could start with homes that have reliable electricity and can afford the extra cost of induction stoves and the special utensils that are magnetized by such stoves to generate an electric current in their walls, resistance to which creates heat. To foster electric cooking on a mass scale, we must strengthen grids at the national, state, zonal, divisional, sub-divisional and feeder levels to bear the higher load that induction stoves would call for. Many buildings would need to be rewired, too. Building codes should be amended to require circuitry that can sustain electric cooking. This is the direction in which policy needs to turn. Stopgap measures are needed, but we must not labour under the illusion that our hydrocarbon supply chain will snap back to a sustainable normal.

THEIR VIEW

The bulge in government bonds held by RBI could be put to work

This pile-up has been remarkable and bond earnings can be deployed for infrastructure creation

MADAN SABNAVIS
is chief economist, Bank of Baroda, and author of 'Corporate Quirks: The Darker Side of the Sun'

The Reserve Bank of India's (RBI) balance sheet shows that the amount of government paper (including treasury bills) held as on 28 February was ₹21.34 trillion. It was ₹15.58 trillion in March 2025. With RBI's recently announced open market operations (OMOs), which will probably continue this month, the amount will only increase. The increase so far of ₹5.76 trillion in RBI holdings of government paper is remarkable. An increase of this scale has never been witnessed earlier. Is there an explanation?

Through OMOs, RBI is buying securities from banks to help augment their liquidity. This is usually carried out when liquidity is under pressure, which has been the case this year for a various reasons. First, growth in bank deposits has been weak due to a migration of funds to mutual funds. Second, growth in credit has been faster. Further, RBI has been selling dollars in the market at various moments to stabilize the rupee. This sucks out domestic liquidity, which is then replenished through the purchase of government paper. Note that the rupee weakened by about 7% from 85.60 against the dollar on 2 April 2025 to 91.68 on 6 March 2026.

RBI holdings of government paper have increased; this is tantamount to deficit monetization. But income on securities adds to its surplus, which is transferred to the Centre. Can this surplus be used for loans to private infra projects?

The history of RBI involvement in supporting budgets is quite interesting. Starting from automatic monetization, under which 4.6% of treasury bills were once issued by the government to RBI for funding the deficit, the system has come a long way. There was also private placement of central debt with RBI during liquidity stress; RBI subsequently sold these holdings in the market once conditions improved. In the late 90s, we shifted to the market becoming the sole point of contact for the government, with primary dealers introduced to underwrite government issuances and ensure full subscription. However, with frequent OMOs in recent times, there has been a tendency for that debt to be transferred to RBI and held by it.

The central bank's holdings of central government paper stood at 13.5% of such debt last September, up from 11.2% a year earlier. These holdings tend to increase when RBI intervenes more in foreign exchange and bond markets. In a way, this is the cost of central bank intervention. Larger holdings also increase RBI's income, in turn contributing to the annual surpluses it earns.

This brings RBI surpluses into the frame. These have risen remarkably from ₹50,000 crore in 2017-18 to ₹2.69 trillion in 2025-26. This was after recommendations made by the Bimal Jalan panel were adopted. In the last seven years, RBI's surplus has averaged ₹1.32 trillion and ₹2.4 trillion in the last two. The interesting thing here is that if this transfer of ₹2.69 trillion, which is part of the non-tax revenue of the government, is seen together with RBI's net incremental holdings of government paper, the sum of ₹8.45 trillion is equivalent to 54% of the fiscal deficit for 2025-26. This is fairly strong support by RBI to contain disruptions in the bond market. In the absence of such interventions, the Indian bond market would have been under tremendous pressure and yields would have hardened.

The 10-year government bond yield has been fairly stable in the 6.60% range thanks to aggressive OMO support. Without it, the Centre's borrowing programme would have been a challenge to fulfil, especially at a time when commercial credit has steadily been picking up. RBI's repo rate was lowered to 5.25% last year, but there has been limited movement after its rather aggressive rate cut last June, followed by a 25-basis-point cut in December.

The low repo rate has led to a situation in which banks have not been able to mobilize deposits from their retail customers. Banks have taken recourse to the market for certificates of deposit and bulk deposits, which cost more.

Relatively high 10-year bond yields alongside OMOs indicate that interest rates cannot be reduced any further. The government's net borrowing programme has been pegged at ₹11.7 trillion in 2026-27, around the same as in 2025-26. With the economy likely to grow by over 7% in 2026-27, notwithstanding war effects, growth in credit could be in the 12-14% range. Much will depend on how the stock market performs, as this will influence investment choices by households. In 2024-25 and 2025-26, they showed a proclivity to capital market avenues. If this sustains, RBI's OMOs may again be significant.

Surpluses of RBI support the budget but could also be transferred to a separate fund that can be used specifically for financing infrastructure development. The money can also be used by the government for lending to private enterprise through public financial institutions like public sector banks. This will widen the fiscal deficit but aid private investment, which has been lagging. The interest on such loans can flow back to the government as revenue in future years. It will effectively hand the baton over to private players for infrastructure investment while the government fine-tunes its own efforts in this area.

These are the author's personal views.

10 YEARS AGO

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GLOBAL: Brussels bombers named, main suspect on the run

POLICY: Govt clears construction of one crore houses for rural poor

TELECOM: R-Com, Aircel extend merger talks by two months

TAXATION: I-T department seeks details of income from interest

SEEKING CHANGES

NDA plans new push for labour reform

Govt plans to introduce five new bills in second half of budget session, a move that may irk unions, opposition

MOUNTING TROUBLES

Infibeam IPO fully subscribed despite lukewarm investor reception

QUICK EDIT

Air India for sale?

The evolution of Deepika Padukone

JUST A THOUGHT

> Government does not create jobs, it only creates the conditions that make jobs more or less likely.
>
> MITCH DANIELS

MY VIEW | MUSING MACRO

What India must do to free its youth of the degree trap

AJIT RANADE

is senior fellow with Pune International Centre

In 2011, a young male graduate earned about ₹21,800 a month; a non-graduate ₹9,000. By 2023, the graduate earned ₹19,573 while the non-graduate ₹10,507. Graduate earnings have fallen in nominal terms over 12 years. Adjusted for inflation, they have collapsed. This data, buried in Azim Premji University's fifth *State of Working India* report (2026), tells a stark story.

The headline story from the report is that 67% of India's unemployed youth aged 20-29 are graduates; i.e., about 11 million people. In 2004, graduates made up just 32% of unemployed youth. The share of graduates in the youth population has itself risen from 10% in 2004 to 28% in 2023. But unemployment among them has risen much faster. The ILO-IHD *India Employment Report 2024* had already flagged a graduate unemployment rate of 29%, compared to just 3% for those who cannot read or write. This trend of rising graduate youth unemployment has been known for a decade now.

The reasons lie on both the demand and supply sides of the labour market. On the latter, we have rational waiting combined with an irrational lottery. Educated youth can now afford to hold out for a job worth taking. With household incomes having risen across income groups over the past two decades, even families of modest means can sustain a waiting son or daughter longer than before. This explains the counter-intuitive pattern: the more educated you are, the higher your unemployment rate.

The wait, however, takes a specific form. A multitude of graduates are preparing for government competitive examinations. Many state governments even sponsor and subsidize preparation for public service commission entrance exams. A meticulous study by Kunal Mangal drawing on Tamil Nadu Public Service Commission (TNPSC) data from 2013 to 2019 shows that in 2018-2019, a single TNPSC group 4 recruitment drew 13.7 million applicants, nearly four times the next largest recruiter in the state. About 80% of all unemployed individuals in Tamil Nadu were also preparing for a TNPSC exam.

When Tamil Nadu imposed a partial hiring freeze between 2001 and 2006, vacancies fell 86%, but job applications actually rose 7%. Exam preparation was actively crowding out private employment.

This pattern is seen across states. Why this addiction to government jobs? The reasons are structural and rational. A government post offers iron-clad job security, wages well above comparable private sector roles (for instance a government-employed driver may earn four times higher), generous health and pension benefits, high social status and authority even at the lowest rungs of the bureaucracy. For women, especially, a government posting is the one job a future in-law is likely to approve of. These repeat-attempt candidates do not miss the private job market because what it offers is a treadmill, not a career. They can always hop on that mill later, with no forgone benefits because the pay for entry-level jobs there stagnates.

> **Cut the Gordian knot of a youth obsession with government jobs amid fiscal limits and slow private job creation**

On the demand side, the diagnosis is darker. Between 2012 and 2019, India's GDP grew 6.7% annually but employment grew 0.1%. Manufacturing, the backbone of mass job creation, has seen its share of the workforce stagnate at 12-14% for over two decades, while output grew 7.5% per year. India produced 5 million graduates annually between 2004 and 2023; only 2.8 million found any employment each year, and a mere 1.7 million entered salaried work. India largely missed the export-led, labour-intensive growth path that transformed East Asia.

Of the 83 million jobs added between 2021-22 and 2023-24, nearly half were in agriculture, with low incomes, limited productivity and no career progression. The *Economic Survey 2024-25* notes that only 8.25% of graduates are employed in roles aligned with their qualifications; nearly half are in elementary or semi-skilled work. These are not jobs that offer 'learning by doing,' productivity or skill growth. They are dead ends with a degree requirement.

Labour laws compound the problem. Employers, reluctant to hire permanent workers they cannot easily let go, have worked around rigid job-protections through contract labour. These allow lower pay, fewer protections and easier lay-offs. The automotive sector blazed this trail; many others followed. The result is a profound casualization of the workforce: churn without growth, employment without advancement. By contrast, in the US, a 4% unemployment rate coexists with enormous job mobility. There could be a 20% churn, with workers cycling through several jobs in a year, gathering skills and wages with each.

One policy priority should be to reduce search friction and skill mismatches. Portable apprenticeship programmes that do not require apprentices to be made permanent could ease both sides of the market simultaneously. Employer-driven rather than certificate-driven skilling would help close the gap. Better labour market information and job portals with easy search could reduce the mismatch between jobs and job seekers.

Another priority is to reduce the exceptional premium on government employment. This requires reducing perks, widening the staffing net, introducing fixed-term contracts and more outsourcing. Unemployment insurance or cash doles must be matched with job-search verification.

Graduates waste years on a government job lottery while private job creation moves at a snail's pace and welfarism pushes its fiscal limits—we need to cut this Gordian knot.

THEIR VIEW

Why chaos in Washington can't hold California's economy back

The Democrat-led state foresaw trouble and took policy action early to insulate its prosperity from chaotic federal governance

GAVIN NEWSOM & LENNY MENDONCA
are, respectively, governor of California; and senior partner emeritus, McKinsey & Company, and former chief economic and business advisor to the state of California.

ISTOCKPHOTO

There is a paradox at the heart of the US economy. As the federal government imposes sweeping tariffs, signals hostility to multilateral trade and unnerves long-term investors with erratic policymaking, one state has emerged as the most reliable US address for global capital: California, the world's fourth-largest economy and America's second-largest exporting state with $188 billion in annual goods exports. In 2024, California's GDP grew by 6%, outpacing the US, China and Germany.

This is not a partisan provocation, but a statement grounded in hard data and direct experience: We built the institutional architecture that has made the state resilient in the face of federal policy volatility. As governor and as the former chief economic and business advisor who helped design California's international investment strategy (an initiative now run by Dee Dee Myers), we undertook trade missions to Europe, Asia, Canada and Latin America; direct government-to-business outreach; and long-term investment compacts.

Now that strategy is paying off. Nearly 19,000 foreign-owned enterprises operate in California, supporting more than 814,000 jobs and paying an estimated $89 billion in wages to Californians—a figure that grew even as federal trade policy turned inward. The US Bureau of Economic Analysis confirms that California consistently ranks among the top two or three states on foreign direct investment.

To be sure, the headwinds President Donald Trump's administration has created must be acknowledged. Tariffs raise input costs across supply chains and policy unpredictability suppresses long-term capital commitments that drive investment. No state, however large, is fully insulated from national macroeconomic conditions.

California's 61,000-plus exporters face real exposure from retaliatory tariffs, as do the farmers and ranchers whose top export markets—Mexico, Canada and China—have been targeted by federal trade policy. But for the international firms that remain committed to the US market, the question is not whether to invest, but where. Here, California offers something that Washington currently cannot: predictability. Clear emissions standards create demand certainty for clean-energy investment and well-established labour protections reduce long-term litigation risk. With its commitment to a green transition, California stands out for international makers of electric vehicles (EVs) and other supplies across the battery supply chain.

As an executive from UK-based Octopus Energy recently put it, California's "supportive policy and world-class entrepreneurship" make it "an ideal place to back long-term investment partnerships." That assessment accompanied a nearly $1 billion commitment to invest in California's cleantech sector. Many other international investors have set up shop in California for the same reasons. Our clean-energy sector employs more than a half-million workers and is growing three times faster than the overall economy.

Three sectors illustrate California's pattern of success most clearly. In cleantech and climate infrastructure, the state's Advanced Clean Cars regulations, cap-and-trade system and nation-leading renewables standards create demand certainty for investors in solar, battery storage, hydrogen and EV infrastructure need. Having added more than 30,000 megawatts of new clean energy and storage capacity since 2019, California now generates 67% of its electricity from clean sources and it is building toward 100% by 2045. Such an ambitious, yet achievable, target creates a durable investment runway that extends far beyond any single federal administration.

Likewise, in life sciences and biotech, California's world-leading research universities, regulatory expertise and cluster effects in the Bay Area and San Diego continue to attract European and Asian pharma and medical-tech capital at record rates. In AI and digital infrastructure, California's venture ecosystem and strong lead in talent and compute capacity make it the default destination for international tech investment—a position re-inforced by federal ambivalence towards AI governance that has created a regulatory vacuum that California is beginning to fill.

These sectors and many others are attracting substantial capital for innovation and the scaling of companies that will define our future. Some business leaders are sceptical of California's regulatory environment and they are not wrong to say that the cost of doing business in California remains high. Reforms of the permit process are in progress but incomplete, even as housing affordability constrains the labour market—a challenge that demands sustained attention.

But such problems are not solved overnight. Our Jobs First Economic Blueprint acknowledges the work that needs to be done. Meanwhile, foreign investors are making comparative judgements: The relevant question is not whether a jurisdiction is perfect, but whether it offers the best available combination of market scale, institutional stability, workforce quality and regulatory predictability for a given investment strategy. On that measure, for a widening range of investment categories, California's advantages remain formidable.

Our investment framework was always intended to be durable across administrations and economic cycles. And now the international relationships, sector-specific compacts and regulatory certainty we have established are proving resilient in a turbulent national environment, vindicating our strategy. California's experience holds lessons that extend well beyond our state. In a global economy characterized by rising geopolitical friction and policy volatility, predictability and openness are competitive advantages. Jurisdictions that understand this will attract the capital, talent and partnerships that determine long-term prosperity. California did not wait for Washington. It drew its own conclusions, built its own institutions accordingly and is reaping the rewards.

MINT CURATOR

China and India are poised to score big on weight-loss drugs

Generic makers rushed in after Ozempic went off-patent in both

JULIANA LIU
is a columnist for Bloomberg Opinion's Asia team, covering corporate strategy and management in the region.

Low-cost producers of generic versions could expand into export markets. AP

The simultaneous patent expiry of Ozempic's active ingredient in China and India on Friday was a watershed moment. Until now, these weight-loss drugs have been available largely to people with means. The launch of affordable generic versions will be a levelling force in healthcare, with global consequences.

That's because the world's two most-populous countries aren't just consumers. They also have pharmaceutical firms that are capable of exploiting this moment to reshape the market for metabolic therapies, affecting the earnings of incumbents Novo Nordisk and Eli Lilly. It would be driven by demand from large populations suffering from chronic ailments like obesity and diabetes that can be treated with these medicines that mimic the effect of the natural hormone glucagon-like peptide-1 (GLP-1), which regulates appetite and blood sugar.

Public health will benefit most. According to the *Lancet*, China had 402 million people living with obesity five years ago. That number will grow 56% by 2050 to nearly half the population. India had 180 million individuals with the disease in 2021, with the figure expected to more than double to 450 million in about 25 years. Obesity is known to increase the risks of diabetes, coronary artery disease, hypertension, stroke and other conditions.

While Novo Nordisk has slashed the price of Wegovy (which uses semaglutide, the same ingredient in Ozempic), in China and India in anticipation of the patent loss, its offerings are still too expensive for most people. In China, where weight-loss drugs fall outside insurance coverage, price will be the decisive factor in whether millions begin medical treatment. Many diabetes patients are managing their condition with less effective older medications. Switching to generic semaglutide could result in superior glycemic and weight control, reducing the risk of long-term complications, Nadim Anwer, pharma analyst at GlobalData, told me. Competition could bring prices down to $50 a month, which is much cheaper than in the West.

China has more than 10 companies, including CSPC Pharmaceutical Group Ltd. and Huisheng Biopharmaceutical, developing generic versions. Local media say it will take another half year for the medicines to be available. India is expected to have dozens of options within months.

Even if the drugs are a game changer, they're not a silver bullet. And they can be tricky to get right. The medicines have been associated with loss of muscle mass and even malnutrition. There's a question mark about depression. Countries need to train health workers to deliver long-term care to support the treatment and invest in prevention by encouraging exercise and proper nutrition, according to Johanna Ralston, CEO of World Obesity Federation. China and India's chronically short-staffed, under-funded healthcare systems may struggle to do that, but they must commit to comprehensive care if the drugs become widely available.

The commercial opportunity in these drugs is clear. Even though China did not originate GLP-1 treatments, it dominates the research pipeline, with more than 100 drugs in development, around 40% of the global total, according to Citeline's Pharmaprojects. Over the past two years, therapies originating from China have comprised the majority of global GLP-1-related dealmaking by both value and deal volume. In fact, Pfizer will soon enter the market for weight-loss drugs after years of trying—by licensing a Chinese product after its own efforts fell flat.

This is not a surprise for those watching the trajectory of China's pharmaceutical development. Over the past five years, drugmakers have started introducing experimental therapies and signing multi-billion-dollar licensing deals to bring them to global markets. Established companies are looking for shortcuts to expand their offerings—since many are facing an upcoming 'revenue cliff' with lucrative patents on blockbuster drugs expiring.

And because the patent for semaglutide has already expired in Canada and is poised to do so in Brazil and Turkey, Chinese and Indian generics could become significant players if they receive regulatory approval. With China already the world's biggest supplier of active pharma ingredients, the components in drugs that produce the intended therapeutic effects, its suppliers can make money in those markets, too.

Ozempic went viral for helping people lose weight after its launch as a diabetes injection in 2018. The sheer effectiveness of this class of drugs has advanced the idea that obesity is a chronic disease. Now affordable generics won't just make them more widely available, but also give Asian drugmakers a bigger slice of the growing business.

MY VIEW | A VISIBLE HAND

India's LPG adoption success story is being choked dry

NARAYAN RAMACHANDRAN

is chairman, InKlude Labs. Read Narayan's Mint columns at www.livemint.com/avisiblehand

Growing up in Kerala, it was a weekly ritual to spark the firewood and get ready for an oil bath. The weather was warm enough during the week to use the plentiful water from a running tap or well. The firewood had to be kept dry and was allocated pride of place in the outhouse to be brought out in small batches for weekly baths or cooking. In an unexpected twist, many parts of India are having to return to this tradition of burning firewood.

Liquefied petroleum gas (LPG) is a fuel commonly used in Indian households and commercial establishments for cooking purposes. It is most often a combination of propane and butane. An almost-fully-burning clean fuel, it has replaced firewood and kerosene in most Indian kitchens. LPG is produced as a byproduct of natural gas liquids (NGL) extraction or as a distillate product from the oil refinery process. The cost of extraction as a byproduct is much lower than as a distillate, which explains to some degree why India still relies on LPG imports rather than on domestic oil distillation for its full annual requirement.

The global demand for LPG is over 340 million tonnes annually. India's consumption of LPG, almost entirely for cooking, is approximately one-tenth of the world's consumption at around 31.3 million tonnes. The household sector, utilizing 14.2kg cylinders, accounts for the overwhelming bulk, approximately 87% of total consumption. The commercial sector (restaurants mainly) uses 19kg, 47.5kg or 425kg cylinders and accounts for 13% of total usage.

For nearly 40 years after the late 60s, India saw a gradual uptake of LPG as a kitchen fuel. Most households continued to use a mix of coal, firewood, kerosene and LPG during that time. The 2011 census showed that 29% of Indian households were using LPG; it was 62% in 2016 and has reached 'near saturation' levels today. India has a little over 300 million households and now has 330 million gas connections, which implies full penetration. The government's scheme to convert the population to clean burning fuel has been a success. If you add the fact that its LPG subsidy has been specifically aimed at poor households using an Aadhaar-based direct benefit transfer (DBT) mechanism, it can be deemed a spectacular success. The online process for ordering refills and a transparent scorecard for LPG distributors that uses a star rating system has resulted in much greater empowerment of LPG consumers.

This grand success has an Achilles heel. Nearly 60% of India's LPG needs are imported and nearly 90% of that is sourced from the Gulf. Our biggest source countries are the UAE, Qatar, Saudi Arabia and Kuwait. Bringing these imports using ships called VLGCs (Very Large Gas Carriers) requires transit through the Strait of Hormuz. Each VLGC carries about 50,000 tonnes of LPG. India needs about 400 such ships a year, a little more than one a day, to fulfill its import demand expected to reach 20 million tonnes. The Strait of Hormuz, named for the Zoroastrian divinity Ahura Mazda, has become a major chokepoint in the ongoing war between the US and Iran.

The country's switch to clean cooking gas was impressive but it's now time for kitchens to go electric

The near-term strategy for Indian homes is to practise LPG conservation. Using pressure cookers and communal cooking can reduce cooking time substantially. Using electricity for peripheral cooking tasks like water boiling and making tea diversifies the use of energy. Cooking with induction stoves saves about 30% of the heat for the same cooking task. Only when direct heat is used for charring, like that used to make *rotis*, is gas a better alternative. Beyond measures at the household level, India must continue its diplomacy with Iran to get more ships through Hormuz. Every three ships that pass through the strait gives India two days of LPG supply. India should also increase purchases from Saudi Arabia, Russia and the US. Saudi Arabian LPG could be contracted through its Red Sea ports. Red Sea transit will require passage through the Bab el-Mandeb Strait. This strait is sandwiched between Yemen and Djibouti and could also be targeted if Houthis in Yemen join the fight, although Iran itself struck a Saudi hydrocarbon target on the Red Sea coast after its South Pars gas-field was hit.

Longer term, India must scale its domestic production and increase strategic storage of LPG. India's storage capacity is placed at about 150,000 tonnes, or two days of demand, and it should expand to at least 30 days of demand. As India proceeds towards its net-zero emissions target set for 2070, it is important to promote electricity-based alternatives. This can and should be achieved concurrently with the grid being decarbonized.

I ran a scenario analysis through Claude with the following assumption: blended adoption between rural and urban households rises from 8% today to 25% in 2030 to 45% in 2035. The analysis projects LPG demand will drop to about 21 million tonnes by 2035, which can be met with increased domestic production and limited imports. The increase in power requirement due to this is about 100 terawatt hours per year or 4% extra, which is easily plannable. In the same way that India switched from earlier cooking fuels to LPG, it must begin an aggressive transition to electricity-based cooking. Ahura Mazda, the 'father of fire,' would approve.

P.S: "If it weren't for electricity, we would be watching television by candlelight," said Comedian George Gobel.

TUESDAY, MARCH 24, 2026

The Indian EXPRESS

NEW DELHI, LATE CITY, 20 PAGES

JOURNALISM of COURAGE

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ISRAEL LAUNCHES NEW WAVE OF ATTACKS ON TEHRAN; IRAN THREATENS TO CUT ALL GULF ROUTES BY LAYING SEA MINES

Trump Twist: Good talks with Iran, 5-day pause on striking energy infra

Rubble being cleared from a residential building that was hit in a US-Israeli strike, in Tehran on Monday. AP

Dialogue and diplomacy only solution, Hormuz blockage unacceptable: PM

Deeptiman Tiwary
New Delhi, March 23

LEADING WITH a strong diplomatic pitch, Prime Minister Narendra Modi on Monday told the Lok Sabha that India is "consistently working" to bring down tensions in West Asia, stressing that "dialogue and diplomacy is the only solution" to the ongoing conflict that has now entered its fourth week.

He said he has personally spoken to leaders across West Asia in two rounds of calls, urging them to de-escalate and end hostilities. "India has opposed attacks on civilians and infrastructure linked to energy and transport. Attacks on commercial ships and blockage of the Strait of Hormuz is unacceptable," he said, adding that New Delhi is actively engaging

Prime Minister Narendra Modi speaks in the Lok Sabha, Monday. PTI

partners to ensure safe maritime corridors.

Positioning India's stance as rooted in "humanity and peace", Modi said the country is encouraging all sides to move towards a peaceful resolution at the earliest, warning that threats to civilian lives are "not in the interest of humanity".

Even as he foregrounded diplomacy, Modi acknowledged that the conflict has triggered "unprecedented challenges" for India — spanning economic pressures, national security risks and humanitarian concerns — given the country's deep trade ties with the region and its reliance on it for energy supplies.

"The region fulfills a large part of our energy needs and is a key trade route. Around 1 crore Indians live and work in the Gulf," he said, underlining the scale of exposure. He noted that commercial shipping routes in the region also employ a significant number of Indians, heightening New Delhi's concerns.

Detailing the government's response, Modi said over 3 lakh Indians have returned home safely since the conflict began, including more than 1,000

»CONTINUED ON PAGE 2

Iran Speaker denies talks; Trump says 'major points of agreement', Hormuz may be 'jointly controlled'

Gram Slattery, Maayan Lubell & Alexander Cornwell
Washington, Jerusalem, Tel Aviv, March 23

US PRESIDENT Donald Trump said on Monday he had given orders to postpone for five days the attacks he had threatened against Iranian power plants, and said the US was in talks with Tehran about ending the US-Israeli war on Iran.

However, Parliamentary Speaker Mohammad Baqer Qalibaf, mooted to be the leader representing Iran in contacts with the US, posted on social media that no talks had been held with the US.

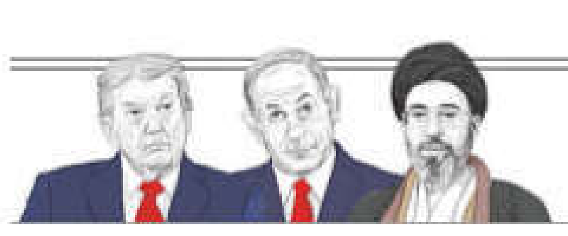
As reciprocal airstrikes continued, financial markets had broadly welcomed the reports of efforts to negotiate an end to the war. Even after Qalibaf's comments, the Brent crude oil benchmark was down around 8% to about $103 a barrel. Iran has effectively closed the key Strait of Hormuz, through which about a fifth of global oil and liquefied natural gas flows.

Trump wrote early in the US morning on his Truth Social platform that the US and Iran had had "very good and productive" conversations over the past two days about a "complete and total resolution of hostilities in the Middle East".

"Based on the tenor and tone of these in depth, detailed and constructive conversations, which will continue throughout the week, I have instructed the department of war to postpone any and all military

»CONTINUED ON PAGE 2

Business as Usual
By EP UNNY



The War
DAY 24

- After Trump announced to pause strikes on Iran's energy sites, oil prices plunged around 8%. Global markets also recovered sharply, with US stock futures gaining around 1.3%
- US Central Command said the campaign against Iran was 'ahead or on plan' even as Tehran and Tel Aviv continued their attacks
- UAE to Saudi Arabia to Kuwait and Bahrain, Gulf countries said they intercepted incoming Iranian projectiles
- The Kremlin said any US military strikes on Iran's Russia-built nuclear power plant could trigger 'irreparable' consequences
- Israeli minister calls for annexation of southern Lebanon, up to the Litani River.
- The death toll from the war rose to more than 1,500 people in Iran, over 1,000 people in Lebanon, 15 people in Israel and 13 US military personnel, as well as a number of civilians on land and sea in the Gulf region

MORE REPORTS, PAGES 12,13,14

Five takeaways: From economic concerns to diplomatic tightrope

Shubhajit Roy
New Delhi, March 23

PRIME MINISTER Narendra Modi's statement in Lok Sabha on the war in West Asia, which marks its fourth week, contained five important diplomatic takeaways — from highlighting the economic fallout to indicating a calibrated approach aimed at safeguarding India's interests.

First, he described the situation in West Asia as "concerning". His choice of Parliament as the platform underscored the gravity of the crisis and its impact on people's lives. Noting the time-frame of the conflict, which has persisted for "more than three weeks", he pointed to the strain it has placed on global supply chains.

Second, he drew attention to the economic fallout of the war, saying it is adversely affecting the global economy and everyday life. Emphasising India's "extensive trade relations" with "countries at war and affected by war", in a reference to nations in West Asia and the Gulf, he flagged the broader economic implications.

In this context, he referred to the Strait of Hormuz, which is an important route for India's

»CONTINUED ON PAGE 2

ISLAMABAD POSSIBLE VENUE FOR TALKS: FINANCIAL TIMES

Munir speaks to Trump, Pak bid to play 'lead mediator' on Iran: Report

Shubhajit Roy
New Delhi, March 23

PAKISTAN IS positioning itself as the "lead mediator" trying to "broker an end" to the US-Israel war against Iran, with Pakistan's Army chief Field Marshal Asim Munir speaking to US President Donald Trump Sunday, UK-based daily *Financial Times* reported Monday.

Pakistan has pitched Islamabad as a possible venue for talks as early as this week involving senior figures from the Trump administration and Iran, the report said.

Pakistan's Prime Minister

Pakistan's Army chief Field Marshal Asim Munir. AP FILE

Muhammad Shehbaz Sharif also held talks with Iranian President Masoud Pezeshkian Monday.

Pakistan's leading English daily, *Dawn*, also reported that Pakistan, along with Turkiye and Egypt, is engaged in "active back-channel diplomacy" to "bridge the gap between" the US and Iran, an official confirmed to the newspaper Monday.

The official said that this also involved US Special Envoy to the Middle East Steve Witkoff and Iran Foreign Minister Abbas Araghchi. A "strategic synergy" between Ankara, Cairo and Islamabad had established a "vital diplomatic conduit, demonstrating that regional cooperation is the most effective antidote to escalation," the *Dawn* report said.

The conversation between

»CONTINUED ON PAGE 2

For women's quota, Govt looks at higher LS strength based on Census 2011

816 in LS, 273 for women; plans to keep same proportion of states in increase, addressing south's delimitation fears

Liz Mathew
New Delhi, March 23

EXPRESS exclusive

THE GOVERNMENT has suggested an increase in the number of Lok Sabha seats, while maintaining the existing proportion among states, and Vidhan Sabha constituencies as a first step towards implementing the women's reservation Act. In talks with Opposition parties to evolve a consensus on this, the government has suggested the increase on the basis of Census 2011 – thus delinking it from the ongoing Census.

An Opposition party MP who was present at a meeting on Monday chaired by Union Home Minister Amit Shah on the Nari Shakti Vandan Adhiniyam told *The Indian Express* that one proposal was to raise the number of Lok Sabha seats to 816 from 543, after delimitation based on Census 2011. Of these, around 270 seats could be reserved for women, said the MP who did not wish to be named.

Shah met leaders of the BJD, YSRCP, NCP (SP), Shiv Sena (UBT) and AIMIM in the Parliament building, and is expected to also hold talks with the Congress, TMC, DMK and SP. Sources said the government may convene an all-party meeting for this Tuesday, with a call on the changes to be taken after that.

Leaders of two other parties whose representatives attended the meeting confirmed the suggestions made by Shah. Sources said the Home Minister suggested that the number of seats in the Lok Sabha as well as Vidhan Sabhas be increased by 50%.

The government apparently hopes to ensure this as well as implement the women quota from the 2029 general elections onwards.

An MP of a party that is an ally of the BJP said the proportion of seats of states would remain the same. This will address the fears of southern states which have been apprehensive about seat increases proportional to population after the ongoing Census and

»CONTINUED ON PAGE 2

DECISION 2026

WEST BENGAL POLL TERRAIN: WHERE COMING FIGHT MAY BE WON OR LOST
PAGE 8

SPEEDBOAT ACCIDENT

Gautam Singhania back in India, families of two missing reach Maldives

Divya A, Nitin Sharma & Shivani Naik
New Delhi, March 23

Ace rally driver Hari Singh; (*right*) Commander Mahesh Ramchandran, Asian Games silver medalist in sailing

FOUR DAYS after a luxury speedboat accident in Maldives in which Raymond Group's Gautam Singhania was injured, two other Indian nationals onboard, ace rally driver Hari Singh and former Navy captain Mahesh Ramchandran, remain missing. As search operations continue, relatives of Singh and Ramchandran have reached Maldives and are being assisted by the Indian embassy in Male.

While 59-year old Hari Singh is a five-time national motor rally champion, Arjuna awardee Commander Mahesh Ramchandran, 60, is an Asian Games silver medalist in sailing.

At the time of the accident late Thursday night, reports said, there were seven people onboard the vessel. Besides Singhania, Singh and Ramchandran, there were two women from UK and Russia, and two other Indian men.

»CONTINUED ON PAGE 2

Happening today

- **DELHI CM REKHA GUPTA**, who also holds the Finance portfolio, will present her second Budget. It will focus on industrial growth, investment and job creation, she has said.
- **DENMARK WILL ELECT** its new Parliament, with opinion polls showing a slim lead for PM Mette Frederiksen's centre-left bloc, boosted by her defiant stand against Trump on Greenland.
- **UNION HEALTH MINISTER** J P Nadda will flag off a 'TB Mukt Bharat Abhiyaan – 100 Days Campaign'.

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NATION

JAISHANKAR SPEAKS TO RUBIO ON 'ENERGY SECURITY CONCERNS'
PAGE 9

• TOOL HAS ALSO PROVED CRUCIAL IN TRACING MISSING CHILDREN, ESPECIALLY AT CROWDED STATIONS

One face, 463 cameras: Facial recognition fights crime on Mumbai rly platforms

Mohamed Thaver
Mumbai, March 23

PAGE 1 anchor

TWO MEN who harassed and stalked a Portuguese tourist were arrested last month; a 14-year-old boy from Goregaon who went missing was traced in Rajasthan weeks ago; the man who stabbed a Mumbai professor was arrested within two days in January; and, nearly a year ago, the alleged attacker of actor Saif Ali Khan was nabbed days after the hunt had hit a deadend.

All of these successful police investigations were aided by one crucial tool: the Facial Recognition System (FRS), linked to 463 of the 3,675 CCTV cameras in the Mumbai Division of the Railway Protection Force (RPF), covering 114 stations from Churchgate to Surat in the north and up to Jalgaon in the east.

According to officials, Western Railway was the first division in the country's rail network to install FRS linked to CCTV cameras. "From last year, they are being installed in other divisions as well — eventually, the entire network will be covered. We get requests for assistance from police forces across the country and Central agencies," a Western Railway official said.

Sources said Mumbai Police units, which operate over 5,000 cameras, are also linking FRS to their network.

In the case of the Portuguese tourist, who uploaded a video showing two men harassing her for a photograph at Dhobi Ghat near Mahalaxmi railway station on February 23, police initially drew a blank after scanning their own camera network. Investigators then took screenshots from the video and approached the RPF, which quickly generated a "hit" on the FRS.

The system found the accused on CCTV footage from

• TRACKING A FACE
- Railway force uploads photo in system.
- System scans stored faceprints to match & alert.
- Checks past footage for other locations, frequency of sightings.

A CCTV camera installed at the Andheri railway station in Mumbai. SANKHADEEP BANERJEE

February 13 at the Marine Lines station. Police then used their own camera network outside the station to track the two men — one of whom had fled to Bihar by then.

The FRS identifies individuals based on facial features such as the distance between the eyes, forehead, nose, ears and cheekbones, converting them into data to create a unique numerical "faceprint".

The FRS-equipped CCTV cameras are connected to software available at Mumbai Central or RPF police stations along the division. If a person on the watchlist passes any of these cameras, the software immediately issues an alert, triggering a chain of actions (see box).

In addition to photographs of accused persons uploaded after incidents, images of suspects are also fed into the system.

In 2024 alone, over 10,000 photographs of individuals being tracked by agencies such as the NIA and CBI were uploaded.

"So, even if the person may not have been spotted till then, once the person enters the platform, we get a hit on the system," the official said.

Another key feature of the system is its ability to map patterns in an individual's movements, which helps investigators track suspects.

In the case of a professor from NM College who was stabbed on January 24 at Malad railway station, RPF obtained footage of the accused, Omkar Shinde, and fed it into the FRS. The system revealed a pattern: Shinde boarded a train every morning from Malad at a fixed time and returned via Marine Lines in the evening. This helped police arrest him. "At times you may know the person

»CONTINUED ON PAGE 2

● WEB EXCLUSIVE
'India chooses silence during war and crises because posturing will have costs.'— *Amit Kumar*
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● WORDLY WISE
The whole is greater than the sum of its parts.
— *Aristotle*

The Indian EXPRESS
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RAMNATH GOENKA
IN 1932
BECAUSE THE TRUTH INVOLVES US ALL

Zone of uncertainty expands each day

THAT PRIME Minister Narendra Modi had to invoke the Covid crisis and the need for national unity in his address to the Lok Sabha Monday shows the scale of the formidable challenges ahead: From the severe dislocation in energy supplies to the unprecedented volatility in financial markets, the shadow of missiles over a one-crore diaspora to the over-heated rhetoric by each of the three sides in the conflict that keeps ratcheting up both temperature and pressure. Underlining the need to be prepared for an eventuality, Modi said, "We faced similar challenges during the Covid crisis with unity. Now again, we must prepare in the same way."

The pain is visible. Disruptions in energy supplies are being felt across parts of the economy — from ceramic tile exporters in Morbi, Gujarat, to restaurants and hotels. The effects are not just limited to oil and gas. The Gulf region is a key choke point — it is a major supplier of fertiliser and inputs such as sulphur and ammonia. Economies could get affected in ways far beyond what the surface-level impact suggests. The second-order effects could soon start showing. Markets are hurting, too. On Monday, the BSE Sensex ended the day down 2.46 per cent. Since the beginning of this conflict, the market has been down around 10 per cent. Foreign investors have withdrawn $11.8 billion in March so far; the rupee was hovering around the 94 level against the dollar.

In the gathering dark, there was a glimmer when US President Donald Trump posted Monday that the US and Iran had held "very good and productive conversations" and that he would hold off military strikes against Iran's energy sites for five days. There has been no reaction from Tehran or Tel Aviv and given Trump's record over the last four weeks, a flip-flop could be just another social media post — or a missile — away. That said, the pause could open the much-needed space for diplomatic manoeuvring. The world is waiting for an off-ramp. Oil slumped, stocks jumped following Trump's post. Domestic political pressure could force a reassessment of the situation as pain thresholds are breached — the national average gasoline price in the US has touched $3.95 a gallon. However, even if there's a let-up, supply disruptions could well persist. Qatar has already said that it could take years to repair its energy infrastructure — in 2024-25, India had imported 41.4 per cent of its LNG from the country. As the zone of uncertainty expands each day, internal adjustments will be needed. The PM's message should go down to all who will bear the costs: Households, firms, and the government.

Surplus heat is new climate challenge

THE WORLD Meteorological Organisation's (WMO) latest State of Climate Report confirms the agency's findings over the last 10 years. The decade that ended last year was the hottest on record. The carbon dioxide in the atmosphere is about 50 per cent higher than pre-industrial levels. In 2025, even as renewable energy edged out coal as the leading source of electricity, global emissions reached record levels. Even though clean power has registered an appreciable increase, it's nowhere close to keeping pace with the world's appetite for energy. All this means that the concentration of carbon dioxide reached its highest level in the last 2 million years, and methane and nitrous oxide in at least the last 800,000 years. In itself, these are sobering figures. The WMO report adds another worrying dimension – the record levels of GHG have upset the Earth's energy equilibrium. Ideally, the amount of radiation entering the planet and the amount of it leaving it should be roughly similar. But a heat surplus has accumulated in the past six decades. That means that even if GHG emissions were to completely stop anytime soon, the planet will continue to warm.

Since water can store large amounts of heat, the seas and oceans have become the main repositories of the extra energy trapped by GHGs. The total effects on oceans are not fully understood. But some of them are apparent. As the authors of the WMO report point out, warmer oceans create conditions for stronger tropical storms. At the same time, melting ice is pushing up sea levels and weakening the planet's ability to reflect solar radiation back into space, aggravating the energy imbalance in the process.

Incremental changes are unlikely to change the trajectory of extreme weather events. Therefore, even as they devote resources towards green methods of production, planners must find ways to make infrastructure more resilient, invest more in early warning systems and shield the vulnerable. The energy imbalance could unfold over decades. Policies must, therefore, be stable enough to guide investments and research over long time-scales. The most important message for governments and the political class worldwide is this: They need to find the resolve and creativity to insulate climate policies from short-term politics.

Banksy is unmasked. And, art is poorer

THE RED balloon of anonymity has finally burst, and the artist known as Banksy has been revealed as one Robin Gunningham from Bristol, England. The elusive figure behind works like 'Love is in the Air (Flower Thrower)' and 'Napalm' was recently identified in a Reuters investigation, nearly 20 years after the British tabloid, *The Mail* on Sunday, arrived at the same conclusion. Who got there first, however, is quite beside the point; Banksy's identity has apparently been one of the art world's most poorly kept secrets. The real question is this: What is gained and what is lost by such an unmasking?

The arguments in favour of pulling aside the veil of anonymity are many: That this is a public figure with a great deal of cultural and artistic influence; that knowing who he is can help deepen the authenticity of a work or add context that was missing; that knowing who exactly made a work of art can lead to greater accountability and more transparency about intentions. Publicity, too, is a factor: While much of the immediate anxiety about the Banksy revelation comes down to economics — would his radical, irreverent works command the same price now that the world knows he's a middle-aged White man? — there are already whispers that the artist himself orchestrated the exposé in the interest of brand-building.

Yet, there is loss too. Anonymity allows a work to remain uncontaminated by biography, with meaning being generated by each viewer through the lens of their own subjectivity. Naming the deliberately unnamed, then, shifts the focus to what the artist intended rather than what the viewer perceived. For an artist who uses a pseudonym in order that he may freely create, the unmasking imposes new constraints, and for art lovers, it diminishes the mystery that gives art its power.

A war lesson: The road to energy security runs on electricity

AJAY SHANKAR AND ARPITA VICTOR

THE WAR on Iran and the disruption in oil and gas supplies have starkly shown India's energy dependence and vulnerability. India imports over 85 per cent of the crude oil it needs and nearly 40 per cent of its natural gas. Being the third-largest importer of crude oil in the world, we are one of the world's most energy import-dependent economies. Over 40 per cent of India's oil supply travels through the Strait of Hormuz, a narrow maritime choke point. About a fifth of global oil flows through it. The war has, in effect, closed the Strait of Hormuz. This is driving up oil and gas prices. The crude oil price is now $109 per barrel, up from $65 in mid-February. The longer the war lasts, the higher will be the price rise. The inflationary impact of this sharp oil price spike would be substantial. LPG gas cylinder prices have already been raised in India by Rs 60 for households and Rs 144 for commercial users, who are also fearing disruption of supply. Forty per cent of our LNG comes from Qatar, which has closed its gas export facilities. The curtailment of gas supply has adversely affected production in industrial units that use imported gas.

These developments have highlighted the critical importance of energy security. For decades, we have tried to find more oil and gas reserves to improve our energy security. Success has eluded us. Import dependence has only increased. Our oil and gas imports cost us over $100 billion annually and constitute 25–30 per cent of our total imports. Is there another way of achieving energy security? What would true energy security look like for India?

Our ongoing successful energy transition has created a potential path for achieving energy security and independence. This is a new development with far-reaching positive implications. The goal of the energy transition to net zero is to give up the use of fossil fuels. As the use of imported oil and gas decreases, our vulnerability declines and energy security rises. Therefore, strategic autonomy and security demand that we accelerate our energy transition away from imported oil and gas.

Accelerating the electrification of our economy is a central pillar of the strategy of energy transition. Electricity would replace the use of fossil fuels in the economy to the extent technically feasible. In parallel, the use of fossil fuels in electricity generation would decline. Electricity generation in India primarily uses domestic coal. However, the share of solar and wind power has been rising rapidly. Power generation is increasing with private investment, and prices are stabilising. The pace of power capacity creation can be raised without difficulty.

Accelerating the transition to electric mobility would reduce our need for imported oil. India has already achieved a critical breakthrough in the domestic production of electric vehicles. There is a competitive industry structure. An increasing number of good electric versions of popular brands are now in the market. Consumers are discovering that the cost of running an electric vehicle is lower. India is becoming a global leader in the use of electric buses. What can be done to increase the share of EVs in the market? A feasible decision would be to mandate that all commercial vehicles be electric from prospective dates, beginning with the NCR. Commercial vehicle operators would save money. No mandate for private vehicles is being suggested, and their freedom of choice may be left undisturbed. In the long run, this would have a decisive impact on reducing air pollution.

With the implementation of this decision, commercial transport would cease to be dependent on imported oil, and this vulnerability would be eliminated. We have enough domestic coal, and a large number of coal-fired stations are being built. These, along with renewables, can fully meet the energy needs of all commercial transport. At a conservative estimate, oil imports would come down by over 30 per cent and further as the market share of EVs among private consumers increases. This is already happening.

Anxiety about the availability of gas is leading to a dramatic surge in the purchase of induction stoves. The switchover to the use of electricity for cooking needs to be promoted in a sustained manner. Induction stoves are more energy-efficient. Using electricity at, say, Rs 8 per unit for cooking, is cheaper than gas. The savings for low-income households would be far higher. The rates of electricity for them are low; in some states, there is free electricity of 200–300 units per month. Then there is the PM's rooftop solar programme for 300 free units per month. Using electricity for all cooking in the country would end dependence on imports and eliminate vulnerability for such an essential necessity.

The key lesson of the present trauma is that we should give up thinking of gas as an intermediate fuel and accelerate our transition away from oil and gas. The sun and wind generate electricity without requiring imports.

However, renewable energy installation requires equipment. For full energy security, India should manufacture all renewable energy equipment and not be dependent on imports. For us, the transition to energy security must also be an industrial transition. Domestic manufacturing of solar panels and batteries with full value-addition should be the goal. In addition to PLI, the government should invite bids for long-term, say 15 years, procurement of solar panels and batteries with full value addition in India from the year the successful bidder can set up a new plant and go into production. It should be prepared to pay a higher price over imports initially. This would lay the foundation for energy-independent manufacturing for the green economy.

For India, the road to energy security runs through the transition away from imported oil and gas and their replacement by electricity.

The key lesson of the present trauma is that we should give up thinking of gas as an intermediate fuel and accelerate our transition away from oil and gas

The writer is distinguished fellow, TERI and author of Rising Ambition: Carving New Pathways – India's Energy Transition. Victor is research associate, TERI

When the workplace wants the worker, not her body

PAROMITA CHAKRABARTI

IN A country where women's pain is routinely trivialised, any resistance to a formal, enforceable entitlement feels like yet another dismissal. And yet, on menstrual leave, the Supreme Court's rejection of a blanket mandate — "...the moment you introduce it as a law, as a compulsory condition, you do not know the amount of damage you will do to their career," the bench said while dismissing the PIL — may have articulated an uncomfortable truth. Not all well-meaning protections liberate.

This is reflected in how labour markets behave when costs are unevenly distributed. According to the Periodic Labour Force Survey 2023-2024, female labour force participation in India stands at around 41 per cent overall, but urban female participation remains around 25 per cent. Salaried women earn roughly about 20 per cent less than their male counterparts. In such an ecosystem, even a marginal increase in the perceived "risk" of hiring women can relocate the inequality from the workplace to the hiring process. Employers may simply opt for the "less complicated" hire: The male worker.

For decades, a similar logic was marshalled against maternity leave — that mandating paid absence would deter employers from hiring women. Yet, over time, the Court has nudged the law towards a more expansive understanding of gender justice. In 2025, in *K Umadevi v. Government of Tamil Nadu*, it held maternity benefits to be a fundamental right. There can be an obvious counterargument to a comparison between maternity leave and menstrual leave: The first is finite, the second is recurring. But this distinction does not alter the constitutional principle. If anything, the recurring nature of menstrual discomfort strengthens the case for structural remedy over episodic concessions.

The Court has recognised this. In *Dr Jaya Thakur v. Government of India* (2026), it declared menstrual health as an integral part of right to life, directing states to ensure access to sanitation and hygiene in all government and private schools. But on menstrual leave, the Court did what an earlier bench had done in July 2024: It placed the onus on the legislature to map the way forward.

This hesitation exposes a deeper tension within feminist policymaking between protection and parity. But the question is not merely whether menstrual leave makes for wise policy — given the inherently masculine nature of workplaces, structural support should not come at the cost of reinforcing stereotypes — it must also address what better policy might look like.

There is, in fact, a credible model worth considering. Employees' State Insurance Corporation scheme already distributes the cost of sickness leave and maternity leave across employers, employees and the state. Extending the same to menstrual leave is not technically inconceivable. But it would require sustained political will and a meaningful expansion of coverage.

There is also the easier step: Extending the SC's own articulation of an infrastructure of menstrual dignity to the workplace. This would require more than provisions of sanitary products. It would need an understanding of women's health as a continuum, and a willingness to balance empathy with the compulsions of productivity.

The conversation around menstrual leave has to take into account another blind spot: The non-inclusion of blue-collar workers. In Karnataka, for instance, which introduced menstrual leave last year — a policy now under challenge in the high court — the mandate applies to working women in government and private sectors. This means it does not reach the domestic worker, the construction labourer, the agricultural worker, women who constitute the overwhelming majority of the female workforce. The discussion has to include them as well.

At its core, menstrual leave is a conversation not just about days off but about whether the workplace is willing to reckon with the body it expects women to leave at the door. The Supreme Court has shown it can hold two truths at once — that policy that subtly signals that women are less available or "reliable" workers risks undermining decades of hard-won progress, and that menstrual dignity is a constitutional value.

The legislature must now do the harder work of holding that balance so that entrenched biases do not get in the way of women just doing their jobs.

For decades, a similar logic was marshalled against maternity leave — that mandating paid absence would deter employers from hiring women

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Japan PM gambles when she buys security in Trump world

GURJIT SINGH

THE SUMMIT between Japanese Prime Minister Sanae Takaichi and US President Donald Trump represents a watershed moment for the Indo-Pacific. Takaichi arrived in Washington not as a junior partner seeking protection, but as a "popular, powerful" leader (in Trump's own words), wielding a historic electoral mandate and a $550 billion checkbook. The summit exposed the transactional and high-stakes nature of the alliance under a second Trump term, specifically regarding Taiwan's stability and the escalating conflict in West Asia.

The centrepiece of the visit was the progress on the 2025 US-Japan strategic trade and investment agreement. This is a geoeconomic blueprint designed to "re-industrialise" America using Japanese capital. Key projects, such as GE Vernova Hitachi's Small Modular Reactor (SMR) plants in Tennessee and Alabama, and the "Portsmouth consortium" led by SoftBank and Hitachi, signal a shift. By funding 9.2 GW of natural gas and AI-ready power infrastructure in Ohio and Pennsylvania, Takaichi is effectively buying "alliance insurance". These investments are tied to US tariff reductions and the revitalisation of the American industrial base. For Japan, this ensures that the "America First" doctrine includes it as an essential partner rather than a trade competitor. The agreement also secures a critical minerals action plan, aimed at breaking China's monopoly.

Takaichi achieved a significant, if delicate, victory regarding Taiwan. She was keen to visit Washington prior to Trump's China visit. She has been more hawkish than her predecessors, suggesting that a Taiwan contingency could constitute a "survival-threatening situation" for Japan. While US initially labelled this a "significant shift", the summit resulted in a reaffirmed commitment to a "free and open Indo-Pacific" and on the importance of Taiwan's stability. But this came with a caveat: Washington remains the ultimate arbiter of when and how to confront Beijing.

The most friction-filled aspect of the summit involved West Asia. President Trump demanded that Japan "step up to the plate" in the Strait of Hormuz. Takaichi faced a "bear trap": Joining a US-led naval coalition against Iran could jeopardise Japan's historical role as a mediator and its energy ties. However, results suggest a tactical compromise. Takaichi avoided a public rift by appealing to Trump's self-image as a "peacemaker" while signalling that Japan would contribute to "appropriate efforts" to ensure maritime safety. The pressure remains high: Trump's praise for Takaichi as a "strong woman" is linked to his expectation that she will deploy Japanese assets, whether financial or military, to support US objectives in the Gulf.

The summit proved that the price of the US security umbrella has risen. Japan is no longer just "hosting" bases, it is "investing" in the US economy to keep it engaged in the Pacific. Takaichi used Japan's financial power to navigate Trump's transactionalism. The "strategic investment" is not just in factories in Tennessee, but in a volatile partnership where Japan's security is increasingly bought with both Yen and geopolitical risk.

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The writer is a former ambassador

40 YEARS AGO
March 24, 1986

INDIAN EXPRESS
Four more Nagaland ministers resign
Jamir Govt faces crisis
Bhajan's resignation sought

Haryana Opposition's demand

THE OPPOSITION-sponsored Samast Haryana convention demanded the resignation of the Bhajan Lal government by April 23 for having failed to safeguard the interests of the state and urged the Centre to abrogate sections 7 and 9 of the Punjab Accord relating to territorial and river water dispute. In case the Bhajan Lal government does not resign, the convention demanded the Centre to dismiss the "anti-people" government, dissolve the Assembly and announce mid-term polls.

Union Carbide case

UNION CARBIDE and American lawyers for victims of the Bhopal gas leak have reached a tentative settlement of $350 million compensation for the disaster. The proposal excludes the Government of India which claims to be the sole representative of the victims. The attorneys representing the Indian government have refused to comment on the matter. But one of those representing the victims has been quoted as saying that the Indian Government has achieved its goal.

Nagaland crisis

THE KILLING of two students in police firing in Kohima is now snowballing into a political crisis for the Nagaland Chief Minister, Senayangba Chubatoshi Jamir, with multiple ministers submitting their resignation to him over the issue. Four ministers, including two belonging of cabinet rank, T A Ngullie and I K Sema, submitted their resignations to Jamir.

Jailbreak in Pakistan

THIRTY-FIVE prisoners condemned to death escaped from the Sukkur jail in the biggest jailbreak in the history of Pakistan. Officials said one prisoner and a jail guard were killed in the shootout between the estimated 50 convicts who organised the escape and evaded the guards. Unofficial reports from Sukkur, a town about 800 kilometres south-west of Islamabad, said the escaped prisoners were on death row for murder and highway robberies with 16 of them allegedly members of dangerous gangs.

Broken UGC needs reform. But the new higher education bill is not the answer

DESHKAAL

YOGENDRA YADAV

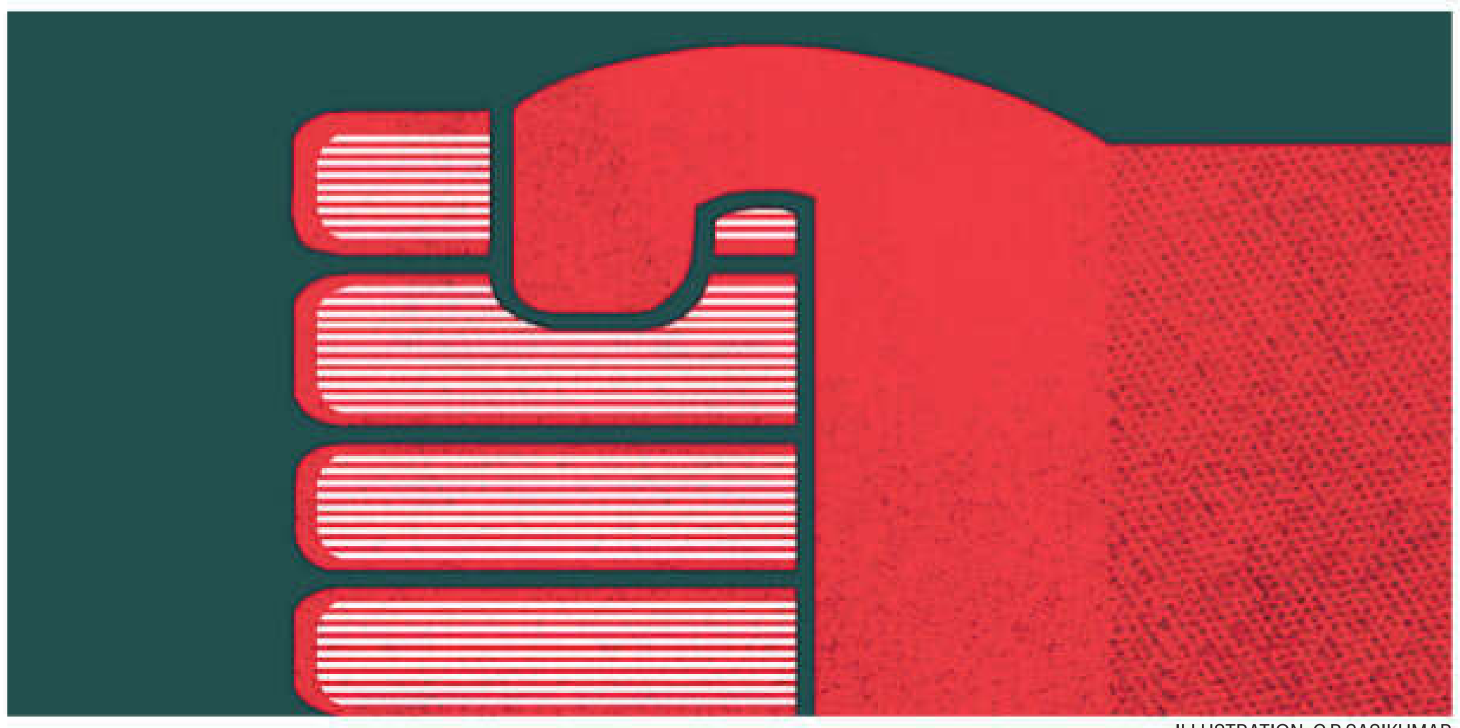


ILLUSTRATION: C R SASIKUMAR

AS A former member of the University Grants Commission (UGC), I would not shed a tear on the abolition of this apex higher education body proposed by the Viksit Bharat Shiksha Adhishthan (VBSA) bill. The UGC is a broken, dysfunctional and counterproductive system. I would have happily supported the new VBSA architecture if the UGC was replaced by a new regulatory architecture promised by the New Education Policy (NEP), 2020: "Independent and empowered" bodies that institutionalise "checks-and-balances in the system, minimise conflicts of interest, and eliminate concentrations of power".

Sadly, the proposed bill, currently under review by the Standing Committee of the Parliament, does exactly the opposite. The new regulatory architecture it proposes is a recipe for concentration of power that can only lead to centralisation, bureaucratisation and commercialisation of higher education — possibly worse than the present UGC system, hard as it is to imagine that.

Between 2010 and 2013, as an honorary member of the Commission, I had a ringside view of all that was wrong with higher education management. The UGC was as corrupt and inefficient as the infamous PWD or DDA, with rampant conflicts of interest. Steered by academic bureaucrats who would stand up to receive a phone call from the joint secretary in the ministry, its autonomy existed only on paper. Even if it had some well-meaning professors on committees, the institution had little capacity or domain expertise needed to regulate the vast ocean of colleges, universities, deemed universities, institutions of national importance and whatnots. The then minister Kapil Sibal tried to clean the system by appointing some members with academic credentials and integrity, but the changes we could bring about were cosmetic, if any.

My singular achievement was to push through the first-ever conflict of interest policy in the UGC. Ironically, that very policy was invoked to remove me from the Commission on the charge of political activism. The system was rotten from the very top. It needed a drastic overhaul.

This was not just my personal impression. This is exactly what a committee headed by Professor Yashpal had concluded in 2009. The report of this "Committee to Advise on Renovation and Rejuvenation of Higher Education" makes for a salutary reading in today's context. The committee had recommended winding up the UGC and the All India Council of Technical Education (AICTE) and replacing these with a National Commission for Higher Education and Research (NCHER), an apex body that would end fragmented regulation and bureaucratic overlap. But the objective of the new regulative structure was to promote academic autonomy by ending "inspection raj", replacing rigid prior approvals with academic self-regulation and by reducing micromanagement and bureaucratic discretion. The Yashpal committee focused on the public purpose of the university by providing for generous state funding and strict regulation of profiteering and commercialisation.

The VBSA moves in the opposite direction. If this becomes the law of the land, it would be a triple whammy: It would take away whatever remains of academic autonomy of the faculty, the institution and of state governments. By doing away with any mandatory provision or institutional design for state funding or social justice, the bill washes its hands of the public purpose of higher education.

Though it is presented as a new piece of legislation, the VBSA bill is actually a reincarnation of The Higher Education Commission of India (HECI) bill that was introduced in Parliament in 2018. The government was forced to withdraw that version, thanks to sharp political push back on the apprehension of centralisation, loss of autonomy and funding. The new bill does not address the key problem of concentration of power; it simply creates a more complex institutional design to achieve the same end. Instead of one all-powerful HECI, now we have an umbrella body called VBSA that supervises three separate Councils in charge of regulation, accreditation and academic standards. But the basic impulse remains unaltered, in some ways stronger than earlier. This can be seen in four key provisions.

> Though it is presented as a new piece of legislature, the VBSA bill is actually a reincarnation of the Higher Education Commission of India (HECI) bill that was introduced in Parliament in 2018

First, there is no autonomy in this new institutional set-up. Four separate institutions are a smokescreen; these are separate but not independent. The VBSA and the three councils are identical in their structure. The chair and members are to be appointed entirely by the central government — including the academics who represent state universities -- which can also remove them and suspend or supersede the entire commission at its will. Some IAS officers will control all the strings at the behest of their political masters.

Second, contrary to the promise of "light and tight" regulation in the NEP 2020, the VBSA now has more coercive intervention powers than the UGC or any other regulatory body before it. On the recommendation of its Councils, the VBSA can exercise draconian powers to penalise institutions and their office bearers, derecognise a degree and recommend the closing down of a college or university. The appeal against any such decision, taken at the behest of the central government, would lie with the central government itself.

Third, and perhaps the worst feature of the new bill, is its complete disregard of the federal structure. State governments, which still fund and supervise higher education institutions that have more than 80 per cent of students, have no say in the running of VBSA, except for one token and rotating seat in a 15-member body and not even that in the three councils. The 2018 bill had provided for a weak consultative structure, an advisory council comprising the Union Education Minister with chairpersons of all State Higher Education Councils. The VBSA bill does away with even that pretence. So, while the state governments will set up colleges and universities, the central government will decide upon their recruitment, their syllabi, their functioning and perhaps their very existence.

Finally, it is worth noticing that the present bill, like its 2018 predecessor, delinks regulation from funding, but does not say a word about where and how the funding will come from. Nor is there any provision for regulating equity, access and non-discrimination in institutions of higher education. This conspicuous silence tells us something about the operational philosophy underlying the new legislation. It is a mechanism for formalising the retreat of the state from the public purpose of higher education at a moment when millions of first-generation college goers need it the most.

The writer is member, Swaraj India, and national convenor, Bharat Jodo Abhiyaan. Views are personal

For TB-free India, we need science, and the power of jan bhagidari

J P NADDA

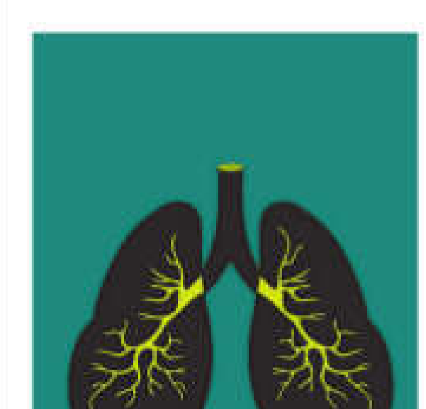
TODAY, AS India launches another TB Mukt Bharat Abhiyaan — 100 days' campaign — I look back at our journey towards eliminating tuberculosis (TB) with pride and optimism. In recent years, India's TB response has been unique — anchored in *jan bhagidari*, the spirit of collective responsibility inspired by Prime Minister Narendra Modi's vision. This whole-of-society approach has proven its strength across sectors, driving the success of initiatives such as Mission Indradhanush and reinforcing the message of Swasth Nari, Sashakt Parivar at the community level.

The experience of the 100-day Intensified TB Mukt Bharat Abhiyaan, launched in December 2024, cemented our belief in the power of people.

The results speak for themselves: India achieved a 21 per cent reduction in TB incidence — nearly twice the global rate — along with 25 per cent decline in TB mortality since 2015. This shows what happens when science, systems and society work together. From central ministries to village panchayats, doctors to public representatives, everyone is involved. Even recovered patients participate as TB *vijetas*, supporting others.

The health ministry has joined hands with 25 other ministries, as well as all panchayati raj institutions and community organisations. Equally inspiring has been the participation of the youth through the My Bharat programme. More than 2 lakh MY Bharat volunteers have registered to provide psychosocial support for treatment adherence and restoring dignity to TB patients.

Our strategy has evolved in response to new evidence and emerging challenges. India's National TB Prevalence Survey revealed that half of the TB patients do not show typical symptoms. This prompted a shift toward proactive screening among the asymptomatic vulnerable population. Silent cases fuel spread — one undiagnosed patient can unknowingly pass the infection to others, making early diagnosis not only a clinical necessity but a civic responsibility.



Equipped with AI-enabled handheld X-ray units, Nikshay Vahans, offering advanced molecular testing, carry diagnostics directly to communities most at risk. More than 20 crore people have been screened, leading to the detection of 32.65 lakh TB cases, including 10.9 lakh asymptomatic ones that would otherwise have remained undiagnosed.

We move forward into the next stage, seeking to deepen gains. Over 3,000 AI-powered handheld X-ray devices will be deployed alongside next-generation diagnostic machines, while data-driven tools will help identify high-risk villages and urban wards where targeted screening can have the greatest impact.

Innovation, however, extends far beyond the deployment of new technologies. India has made sustained investments in research and development, strengthening partnerships between government, academia and industry. These have created a rich and growing pipeline of Made-in-India TB diagnostics. They will not only strengthen India's ability to detect TB earlier and treat patients faster but also position the country as a leader in TB innovation for the Global South.

As India urbanises rapidly, cities have become critical battlegrounds in the fight against TB. To address this challenge, we are focusing on high-risk urban settings such as informal settlements, migrant worker clusters and other vulnerable populations.

Our task is clear: To sustain the momentum and carry it forward with even greater intensity. India has always delivered when we move as one. We stood together to eradicate polio. Under the leadership of PM Modi, we vaccinated over a billion people against Covid in record time. With a record 10-fold increase in funding and a strategy rooting for "Made in India" developments, we have moved from hope to certainty. We will continue to strengthen early detection, expand diagnostic access, and deepen *jan bhagidari* by enlisting every citizen, institution and youth platform in this shared mission.

India's campaign against TB is not just a public health endeavour. It is a testament to our nation's ability to unite for a cause that touches lakhs of lives. A TB-Mukt Bharat will not be built by the government alone. It will be built by all of us — by the choices we make, the habits we adopt, the support we extend and the stigmatising we refuse to tolerate. If we stand together, we can ensure that the next generation will know TB only as a chapter in history, not as a reality in their lives.

I am sure that with our collective efforts, we will achieve the resolution set out by PM Modi, "Yes, We can End TB".

The writer is Union minister of health and family welfare & chemicals and fertilisers

LETTERS TO THE EDITOR

Trump isn't winning

IT IS imperative that President Donald Trump and PM Benjamin Netanyahu terminate their war against Iran with immediate effect; else, the consequences will be quite cataclysmic ('Coming up, war's downstream effects', *IE*, March 23). The dire straits that the world has been plunged into, especially vis-à-vis the Strait of Hormuz, after a resilient Iran has thrown a monkey wrench into the machinery and throttled the safe passage of ships, will not bode well for the region and the world at large.

Hemant Hemmady, Virar

Fire keeps burning

FIRE INCIDENTS revive painful memories of the catastrophe at a private hospital ICU in Bhubaneswar in 2016, which caused 22 deaths, and of the Maharani Laxmi Bai Medical College fire in Jhansi in 2024, in which 18 newborns died ('To prevent fire tragedies, address systemic failures', *IE*, March 23). Compliance checks are often treated as paperwork exercises. Until India treats hospital safety as a matter of criminal accountability rather than bureaucratic paperwork, the fires will keep burning.

S S Paul, Nadia

WE CONSISTENTLY fail to learn from the past ('To prevent fire tragedies, address systemic failures', IE, March 23). The haunting memories of the Uphaar Cinema tragedy should have been a permanent deterrent, yet collective amnesia has become the norm. Citizens routinely prioritise immediate personal gains over collective safety, actively sabotaging urban planning. Planned municipal roads are choked by illegal boundary extensions and haphazard billboards. Unauthorised constructions thrive at a premium, tragically facilitated by the very law enforcers mandated to prevent them. One disaster simply replaces the other, while the pleas of law-abiding citizens are drowned out by profits.

Krishan Chugh, New Delhi

For transgenders, law makes them ask who they are

SHASHANK MAHESHWARI

INDIA'S CONSTITUTIONAL engagement with gender identity reached a transformative moment in NALSA (2014). In the judgment, the Supreme Court recognised the right of transgender persons to determine their gender identity as an intrinsic aspect of dignity, autonomy, and equality. The Transgender Persons (Protection of Rights) Amendment Bill, 2026, raises serious questions by fundamentally restructuring the statutory definition of "transgender person".

At the centre of the amendment lies the redefinition of Section 2(k) of the Transgender Persons (Protection of Rights) Act, 2019. The proposed provision restricts transgender identity to two categories: Individuals belonging to historically recognised socio-cultural communities such as *hijra*, *kinner*, *aravani*, and *jogta*, and persons born with congenital variations in sex characteristics. This formulation marks a significant departure from the 2019 Act, which defined transgender persons as individuals whose gender identity does not correspond with the gender assigned to them at birth, irrespective of whether they have undergone medical transition.

The Statement of Objects and Reasons of the Bill asserts that the law should not extend protection to individuals claiming gender identity based on "self-perceived" characteristics, but only to those who experience discrimination due to biological conditions beyond their control.

This reasoning, however, sits uneasily with the constitutional doctrine articulated in *NALSA*. The SC decisively rejected a biologically deterministic approach, particularly the reasoning derived from the English case of *Corbett vs Corbett* (1970), which treated biological attributes as the definitive marker of sex. Instead, the Court emphasised that gender identity is fundamentally shaped by psychological self-perception and lived experience. By grounding gender identity within the guarantees of Articles 14, 15, 19, and 21, the Court recognised the right of individuals to determine their gender independent of medical or biological validation. By conditioning recognition on congenital variations or membership in specific communities, the Bill privileges biological determinism. This shift potentially transforms a rights-based framework into a regime of state verification.

> 'NALSA' articulated a rights-based framework grounded in autonomy and dignity. The amendment appears to revive a classificatory model rooted in biology and community status

The amendment also reflects a greater conceptual difficulty in the legal understanding of gender identity. In various scholarships, it is emphasised that "transgender" functions as an umbrella category encompassing a wide spectrum of identities and experiences, including persons who may not undergo medical transition but identify with a gender different from that assigned at birth. Legal definitions that rely exclusively on biological markers risk collapsing distinct phenomena — intersex variations, gender identity, and socio-cultural communities — into a single category. Such conflation restructures the legal architecture of recognition around pathology.

Equally important is the legislative process through which such a fundamental redefinition has emerged. Meaningful consultation enables lawmakers to incorporate the experiences of affected groups into the formulation of legal norms, thereby enhancing both the legitimacy and effectiveness of legislation. The evolution of transgender legislation in India, from earlier draft bills to the 2019 Act, has often been criticised for insufficient engagement with transgender communities. Supporters of the amendment may argue that definitional clarity is necessary for the effective administration of statutory benefits and protections. Yet, administrative convenience cannot override constitutional guarantees. The challenge before lawmakers is to design institutional mechanisms capable of implementing rights without undermining the principle of self-determination.

NALSA articulated a rights-based framework grounded in autonomy and dignity. The amendment appears to revive a classificatory model rooted in biology and community status. In doing so, it risks transforming the constitutional promise of self-identification into a regime where gender identity must be verified rather than recognised.

The writer teaches at Jindal Global Law School

Chuck Norris didn't die, God needed a bodyguard

KURIAKOSE SAJU

JESUS CHRIST *could walk on water. Chuck Norris can swim through land.* Carlos Ray "Chuck" Norris died this week after being hospitalised in Hawaii following a medical emergency. The martial artist-turned-action hero was officially 86 at the time of his passing. Unofficially, the near-mythical Norris might be immortal. Chuck Norris "facts" have been an internet staple since the early 2000s. It all started with Vin Diesel. In 2005, a website called "Vin Diesel Fact Generator" popped up, taking inspiration from an array of jokes about Diesel on a *Something Awful* forum. When website creator Ian Spector saw that the Diesel jokes weren't cutting it any more, he ran a poll to see who the top choice for his next tough-guy fact generator should be. Norris won by a landslide. Norris's real-life martial arts credentials and numerous black belts went a long way in establishing him as the alpha of choice. That, and the gift that keeps on giving: Norris's long-running TV show, *Walker, Texas Ranger*.

Chuck Norris once threw a grenade and killed 50 people. Then, the grenade exploded. My first brush with Chuck Norris was his appearance as the titular Walker on the CBS show that ran for nine seasons between 1993 and 2001. As a young boy growing up in '90s India, Norris's small-screen exploits were the closest thing you could get to a steady stream of Rambo-coded machismo on a weekly basis. The camp factor was high, but so was the man's roundhouse kick. Norris's over-the-top heroics were cemented for an entire generation with that one show.

Chuck Norris got bit by a cobra once. After five days of excruciating pain, the cobra finally died. Chuck Norris jokes soon became the prototype for other larger-than-life internet characters like The Most Interesting Man in the World ("His only regret is that he didn't have any regrets"). In India, we had our own version of the Chuck Norris joke, transposing his demigod-like powers on the one and only Rajinikanth. It made perfect sense. After all, with his iconic cigarette-twirling, bombastic dialogue delivery, and truly inimitable charm, even Norris would have had a tough time keeping up with our Superstar. There's a difference in the kind of adulation both receive. Norris is definitely one of the GOATs when it comes to the action genre; his appearance in the second *Expendables* movie is testament to the fact. But the fandom that Thalaivar enjoys is different. A Norrisian joke might claim God prays to Chuck Norris. In India, people actually do pray to Rajinikanth at a temple dedicated to him. *Chuck Norris once high-fived Rajinikanth. That's what scientists called the Big Bang.*

> Chuck Norris jokes became the prototype for other larger-than-life internet characters like The Most Interesting Man in the World ('His only regret is that he didn't have any regrets')

So what makes a man a living myth? Is it the perfect roundhouse kick? Black belts? Mid-air cigarette-catching skills? Or maybe, just a bored teenager with an internet connection and time to spare? Turns out it's all three, plus a battalion of fanboys. So long, Chuck Norris. Sure, your political views might be hard to agree with, and you might have spent way too much time endorsing Glock and Netanyahu. But at least, we'll always have the jokes. *After all, Chuck Norris didn't die. God just needed a bodyguard.*

The writer is an editor and stand-up comedian

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● POLICY

Hyderabad's special taskforce, set to tackle food adulteration

Sreenivas Janyala
Hyderabad, March 23

LAST WEEK, the Hyderabad Police launched the Hyderabad Food Adulteration Surveillance Team (H-FAST) to check adulteration at food manufacturing units, eateries, and restaurants. Police Commissioner VC Sajjanar, who will oversee the team tasked with conducting surprise raids across the city, said that H-FAST is a dedicated, intelligence-driven mechanism to combat food adulteration.

Uptick in incidents

In recent months, Hyderabad has seen an uptick in incidents of food adulteration and manufacturing of spurious food products. During raids, police and health officials have found fake ginger-garlic paste, adulterated paneer, ghee, *kova* (traditional Indian dairy product used to make sweets), pickles, spoilt milk, curd, and dairy products. Spices like turmeric, chilli powder, and mustard have been found to have been prepared using synthetic materials. These products, manufactured by illegal units in unhygienic conditions, are then supplied to roadside eateries, restaurants, and hotels at cheap rates, which could impact the health of consumers.

For instance, in Mallepally, police seized 1,090 kg of fake ginger-garlic paste. Manufactured using acetic acid, synthetic colours, and gum powder, large quantities of it was supplied to shops and popular eateries across the city.

In a separate raid, police seized fake tea powder, detergent powder, and mosquito repellent. From a dairy products shop, police seized 3,000 kgs of fake paneer, adulterated *kova* and ghee, and palm oil.

Incidentally, H-FAST was announced a day after two incidents concerning food adulteration were reported in and around Hyderabad. While the first involved police busting a unit in Jiyaguda that was supposedly manufacturing adulterated pickles, the second saw police raiding six stores for allegedly selling adulterated paneer and dairy products in Secunderabad.

● MODUS OPERANDI

- During raids, police and health officials found adulterated spices and dairy products
- These products, manufactured by illegal units in unhygienic conditions, are then supplied to roadside eateries, restaurants, and hotels at cheap rates. This could impact consumer health

The team's scope

Headed by G Vaibhav Raghunath, Deputy Commissioner (Task Force), H-FAST will initially comprise 28 members, which includes Inspectors, Sub-Inspectors, and supporting staff focusing on surveillance, detection, and prosecution of offences related to unsafe food.

The teams will conduct checks at food manufacturing units and stores and restaurants. Sajjanar said that its main task is food safety enforcement, and the team will work in close coordination with Food Safety Officers to strengthen regulatory action.

Officials emphasised that food adulteration is a serious offence posing a threat to public health. A zero-tolerance approach will be followed, and stringent legal action will be taken against offenders.

Grievance mechanism

In order for the public to reach out to H-FAST, a dedicated toll-free number (8712661212) has been established to report instances of food adulteration and unsafe food practices. Citizens can also reach out to official social media handles to share information and complaints with the taskforce, enabling quicker response and wider public participation.

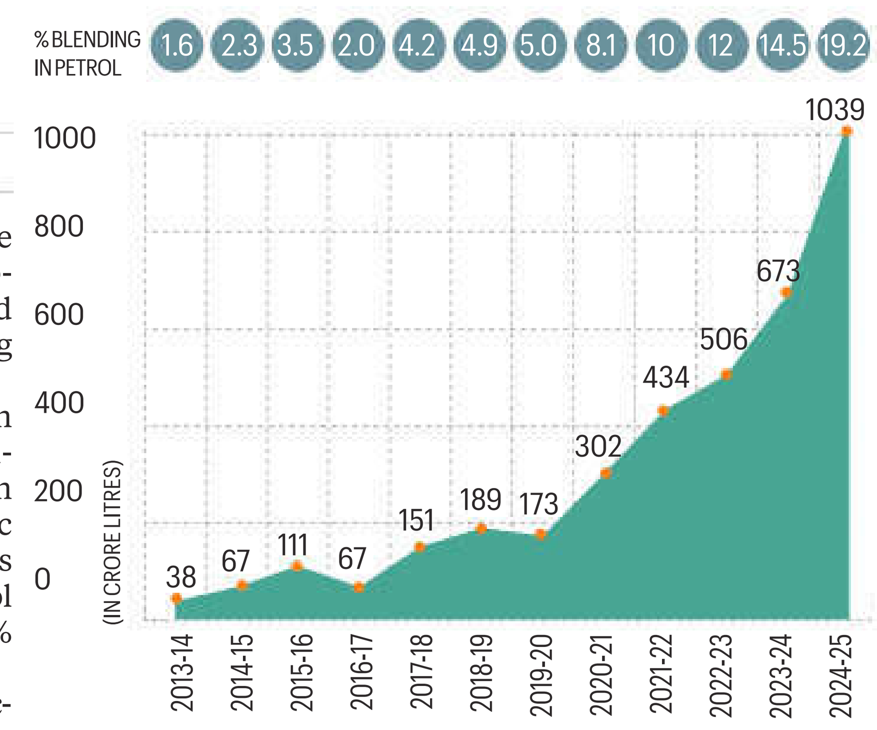
● WAR IN WEST ASIA

Why India must step on the gas with ethanol

Like Brazil after the 1970s oil shocks, India can use the Iran war stress to boost ethanol-blending and reduce fuel import dependence

HARISH DAMODARAN

● Ethanol supply to oil marketing companies



● Ethanol supply by feedstock (in crore litres)

Supply Year	C-Heavy Molasses	B-Heavy Molasses	Cane Juice	Maize	Surplus FCI Rice	Damaged Foodgrain
2013-14	38	0	0	0	0	0
2014-15	67	0	0	0	0	0
2015-16	111	0	0	0	0	0
2016-17	67	0	0	0	0	0
2017-18	151	0	0	0	0	0
2018-19	146	33	1	0	0	10
2019-20	74	68	15	0	0	16
2020-21	39	183	39	0	2	39
2021-22	11	265	85	0	49	24
2022-23	6	235	128	32	74	32
2023-24	58	149	64	286	0	116
2024-25	17	139	165	498	140	80
2025-26*	14	113	165	479	233	54

*CONTRACTED QUANTITY TILL NOW. SUPPLY YEARS ARE DEC-NOV TILL 2021-22; DEC-OCT FOR 2022-23 AND NOV-OCT FROM 2023-24.

IN NOVEMBER 1975, two years after the first global oil shock courtesy of the Arab-Israeli 'Yom Kippur' War, Brazil launched a Proálcool programme, aimed at reducing its dependence on imported fuels.

Proálcool mandated all petrol sold in Brazil to contain a minimum 11% anhydrous alcohol or ethanol. In 1979, when the second oil crisis came with the Islamic Revolution in Iran, Brazil introduced cars that could run on 100% hydrous alcohol (this has 5-6% water as against less than 1% for ethanol).

By 1985, when domestic ethanol production reached nearly 1,200 crore litres, Brazilian fuel stations had two types of pumps: one for ethanol-blended petrol (Gasoline C, as it is called) vehicles and the other for 100% hydrous alcohol (E100) vehicles.

In 2003, the first commercial flex-fuel vehicles were launched in Brazil, with engines capable of running on both Gasoline C and E100 fuel. The vehicles had electronic sensors to detect the particular ethanol-gasoline blend in the tank and automatically adjust the fuel injection volume and spark timing.

The minimum ethanol blend in petrol was hiked to 25% by 2014 and to 27% from March 2015. Since August 2015, the ethanol content in Gasoline C has been set at 30%.

In 2024, Brazil's total fuel alcohol consumption was 3,492.4 crore litres: 2,289.6 crore litres of E100 and 1,202.8 crore litres of ethanol in Gasoline C. While the ethanol blend rate in Gasoline C was 27% (now 30%), the overall mix for fuel used in the country's light-duty vehicles fleet stood at 51.8%. Thus, alcohol constitutes more than half of the fuel powering Brazil's cars and two-wheelers.

Replicating Brazil

Speaking in Parliament on the West Asia war on Monday, Prime Minister Narendra Modi referred to India's own ramped-up efforts on ethanol blending over the last decade that helped address its vulnerabilities from imported energy.

Since 1979, the world has seen at least three oil shocks — in 2008, when Brent crude prices surged to a record $147.5 per barrel on July 11, and in 2022 after Russia's invasion of Ukraine, when they reached $139.13 on March 7.

The latest shock — from the ongoing US-Israel versus Iran war — should be a wake-up call for India and to replicate the Brazil model, said CK Jain, president of the Delhi-based Grain Ethanol Manufacturers Association.

"The government must work with the auto industry and oil marketing companies (OMCs) to gradually increase the ethanol blending mandate from 20% to 30% and incentivise the production of flex-fuel vehicles that can run on 100% hydrous alcohol. Our stations, too, should have separate dispensing units for E30 and E100 fuels, with consumers choosing either depending on the price parity between the two," he said.

An aggressive ethanol use strategy would require a relook at taxation as well.

Currently, ethanol is covered under the Goods and Services Tax (GST) regime. The ethanol used for blending with petrol attracts 5% GST. But petrol remains outside GST, attracting both the Central excise duty and State value added tax. Both ethanol-blended petrol and pure petrol are treated as identical for taxation purposes.

"We need to bring all fuels blended with ethanol – whether E20, E30 or E100 – under GST," added Jain.

The progress so far

India, as the accompanying *chart* shows, has achieved a significant rise in both the quantity of ethanol supplied by distilleries to OMCs and the average blending ratio in petrol — from a mere 38 crore litres (1.6%) in 2013-14 to 1,039 crore litres (19.2%) in 2024-25.

Push to diversify

Policies instituted in the '70s have led to alcohol now constituting over half the fuel powering Brazil's cars and two-wheelers

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India has achieved a rise in average blending ratio in petrol, too, from 38 cr litres in 2013-14 to 1,039 cr litres in 2024-25

Till 2017-18, ethanol was produced by distilleries attached to sugar mills only from so-called C-heavy molasses, a dark brown liquid byproduct of cane processing.

From 2018-19, mills began making ethanol from an earlier 'B-heavy' stage molasses (having higher sucrose content available for fermentation) and also directly from whole sugarcane juice or syrup. They were encouraged to do so by the Centre's decision to pay mills more for ethanol manufactured from the B-heavy and direct cane juice/syrup routes, in order to compensate them for the revenues foregone from lower/nil recovery and sale of sugar.

From 2018-19, the Centre also fixed separate ex-distillery prices for ethanol derived from rice, maize and damaged foodgrains.

Ethanol production involves fermentation of sugar by yeasts. In molasses or cane juice, sugar is present as sucrose. Cereal grains contain starch, a complex carbohydrate that has to first be extracted and broken down into simple sugars before further fermentation, distillation and dehydration to ethanol with 99.99% alcohol concentration.

Out of the 1,039 crore litres of ethanol supplied to OMCs in 2024-25, as much as 718 crore litres or over 69% was from grains, mainly maize and rice. The latter included surplus rice from the Food Corporation of India's (FCI) stocks and broken/damaged grain sourced from the open market.

For the current supply year (November-October), the OMCs have so far contracted 1,058 crore litres — 766 crore litres from grains and 292 crore litres from sugarcane-based feedstocks (*see table*). The ex-distillery price has been fixed at Rs 57.97 per litre for ethanol produced from C-heavy molasses, while at Rs 60.73, Rs 65.61, Rs 60.32, Rs 64 and Rs 71.86 for that from B-heavy molasses, sugarcane juice/syrup, FCI rice, damaged grain and maize respectively.

The road ahead

The latest oil shock presents an opportunity for India to "revisit and review" its existing ethanol blending programme, Tarun Sawhney, vice chairman and managing director of Triveni Engineering & Industries Ltd, pointed out. "We need to first understand that India does not have enough crude oil and imports 90% of its requirement. Second, the internal combustion engine isn't going anywhere and there are limits to how much electric vehicles can replace it. Third, we are surplus in sugarcane, rice and maize and can increase their production much more for E-30 and E-100 fuels," Sawhney, whose company has a capacity to crush 70,500 tonnes of cane, told *The Indian Express*.

The constraint today, he claimed, is not in ethanol production (India already has a capacity in excess of 1,800 crore litres per annum) or retailing. What India needs is a policy push for all new vehicles to be flex-fuel based. The government should also nudge companies to provide conversion kits to modify existing vehicles.

US-Iran pause is a relief for Gulf desalination plants, water supply

Anagha Jayakumar, Harikishan Sharma
New Delhi, March 23

US PRESIDENT Donald Trump on Monday said that following "very good and productive" conversations with Iran, the Department of War would postpone strikes on the country's energy infrastructure for five days, subject to the ongoing discussions.

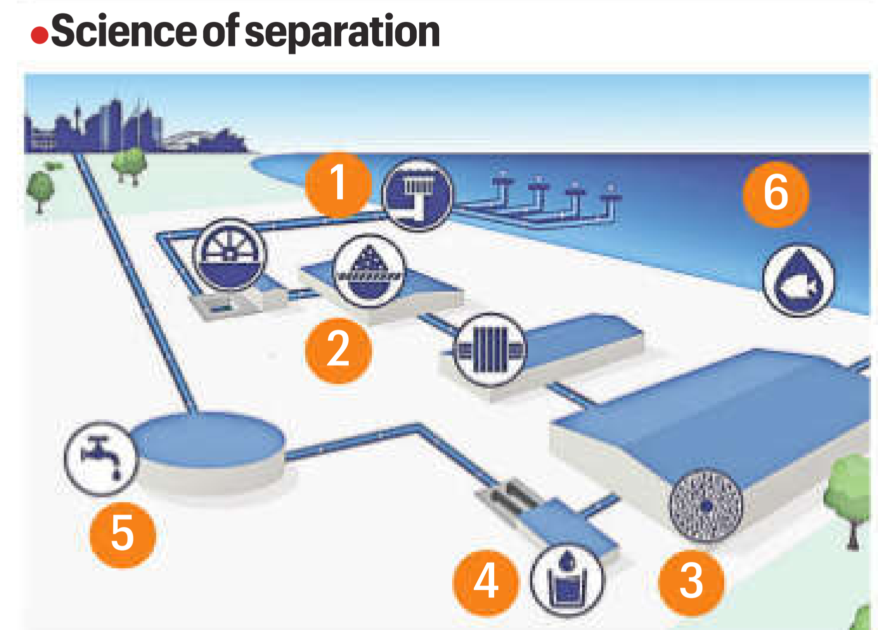
The statement also matters for water desalination infrastructure in the Gulf region, which Iran threatened as retaliation for potential US strikes. Desalination plants are vital for meeting the water requirements of the Gulf Cooperation Council (GCC) countries: Bahrain, Kuwait, Oman, Qatar, Saudi Arabia, and the UAE.

What are desalination plants?

Desalination plants convert saltwater into drinkable water and are a preferred method in coastal regions without natural perennial sources of potable water. It involves separating salt molecules from water molecules either through membrane-based methods, such as reverse osmosis (RO), or heat-based methods.

In RO, saltwater is filtered through a semi-permeable membrane under pressure to trap the dissolved salts and other impurities. The filtered water is further treated to make it fit for consumption. In the thermal method, saline water is heated, and the resultant vapour is collected and recondensed.

● Science of separation



1. Seawater intake 2. Pre-treatment filtration to remove particulate matter 3. Reverse osmosis to remove salt, minerals 4. Post-treatment for consumption readiness 5. Drinking water delivery 6. Seawater concentrate outlet (SYDNEY DESALINATION PLANT AS REFERENCE)

According to the International Desalination Association, over 150 countries rely on desalination to produce freshwater. A 2022 report by the French Institute of International Relations noted that the sector was growing by 6% to 12% annually.

Why does the Gulf need so many?

According to the GCC Statistical Centre, the GCC nations had 172 water desalination stations in 2018, with Oman accounting for the most (65 stations), followed by Saudi Arabia (44) and the UAE (40). The stations have a combined capacity of 22 billion litres per day, with actual annual water production at 6,053 billion litres. The plants are located in coastal areas (*see graphic*).

These countries, covering an area of 2.67 million square km (about 81% of India's geographic area), receive very low rainfall, ranging from 4 to 30 cm. The total precipitation was 281.91 billion cubic metres in 2018 — just 7% of what India receives in a year.

The GCC countries do not have rivers, but wadis (dry valleys or riverbeds), with water availability only during rainfall. The total groundwater abstraction (freshwater extracted from the ground) was recorded at 26,495 million cubic metres across the six countries in 2018. Population and economic growth, as well as increasing urbanisation, have contributed to the rising water demand over time.

With scant rainfall and limited replenishable groundwater, the countries are heavily dependent on desalination plants to meet their water needs. However, desalination plants are highly energy-intensive and run on electricity generated from liquid fuels or natural gas, generating 500 to 850 million tons of carbon dioxide emissions annually.

Another concern is that reverse osmosis, the predominant desalination method, generates large quantities of saltwater brine, typically discharged into the ocean. These can harm marine habitats and sea creatures such as plankton and fish larvae.

How does Iran meet its water needs?

Unlike other West Asian countries, Iran's water needs are mostly met by groundwater sources, like underground aquifers. It operates comparatively few desalination plants, including one on Qeshm Island. However, it has been experiencing a drought for the last five years and has begun expanding its desalination facilities. This has been limited, given infrastructure constraints, high energy costs and international sanctions on the Iranian economy.

● INFRASTRUCTURE

How the Narendra Modi stadium became an economic hub for Ahmedabad

Leena Misra
Ahmedabad, March 23

IN THE days leading up to the T20 World Cup final at the Narendra Modi Stadium in Ahmedabad, a whole bazaar sprung up around it. People sold jerseys, caps, and other cricket merchandise, with some sellers coming all the way from Delhi and raising rates closer to the big day.

On the day of the match, luxury cars carrying fans, some with Maharashtra registrations, crawled in the traffic, two hours before the match. Touts approached them with offers: "Rs 100 for parking the whole evening... can bargain."

The Western and Northern Railways introduced special trains from Mumbai and Delhi, respectively, to ferry fans to Ahmedabad. The Gujarat Metro Rail Corporation extended its working hours past midnight, earning over Rs 10 lakh in revenue.

The "pillarless marvel" that is the Narendra Modi Stadium is fast becoming one of the pillars of Ahmedabad's economy.

Namaste Trump to Coldplay

In February 2020, a month before the Covid-19 lockdown, the Motera Stadium, as it was known then, welcomed US President Donald Trump with a crowd of over one lakh. A year later, it was named after PM Modi and inaugurated by the then President Ramnath Kovind.

Jay Shah, Chairman of the International Cricket Council (ICC), told *The Indian Express*, "The Narendra Modi Stadium is far more than just a cricket venue, it is a powerful example of how sport can become an engine of economic growth and social development. Over the past few years, we have seen how major cricket events hosted here have contributed significantly to Ahmedabad and Gujarat by boosting tourism, supporting local businesses, creating employment opportunities and strengthening the city's global reputation as a premier sporting destination."

The stadium can seat 1.10 lakh people. FILE

Jay Shah was the joint secretary of the Gujarat Cricket Association (GCA) when the stadium was being built, and takes pride in the fact that it was completed in 38 months. The "world's biggest cricket stadium" was built at a cost of Rs 800 crore, doubling the capacity of the existing 49,000-seat stadium to 1.10 lakh.

Five years later, the GCA, which owns the stadium, is looking at it as a potential venue for hosting the opening ceremony of the Commonwealth Games in 2030 and the Olympics in 2036.

Officials of the GCA and the Gujarat government benchmark the January 2025 Coldplay concert as a game-changer for the stadium. A report by EY Parthenon, a global strategy firm, said it directly impacted Ahmedabad's economy by Rs 392 crore, counting the ancillary benefits of what people spent on accommodation, transport, dining, and retail.

Flights, trains and hotels operated at full capacity and hotel rates "soared to historic heights" as the influx of tourists cemented Ahmedabad's position as a "premier entertainment destination", the report said.

Before the Narendra Modi stadium, the Melbourne Cricket Ground, with a capacity of 90,000, was the largest in the world.

According to Jay Shah, "What makes this stadium truly special is not just its scale, but the ecosystem it has helped create, from infrastructure development and hospitality growth to inspiring young people to take up sport."

A GCA official said the stadium hosts about 10-12 major matches in a year and about 30-40 national tournaments.

Riverfront an added attraction

Ahmedabad municipal commissioner Banchhanidhi Pani said the high-profile events at the stadium are playing a part in the development of the city as a whole. The 63-acre Sardar Patel Sports Complex (of which the stadium is a part), located along the Sabarmati Riverfront, is set to be "uplifted". "During events, people also use the city bus services, including the BRTS, to reach the stadium. A serene, beautiful riverfront next to the stadium is an added attraction," Pani said.

Property prices in the stadium's neighbourhood are going up, said an official.

The big-ticket events at the stadium, however, lead to traffic snarls that affect many parts of the city. Ahmedabad city police commissioner Gyanender Singh Malik said, "If 1 lakh people come to a place, traffic will definitely be an issue, but we have solutions. We divert traffic and make arrangements for VVIP movement."

Under scanner

Ahead of the 2022 Gujarat Assembly polls, the Congress party had declared that it would change the stadium's name to Sardar Patel Stadium if it came to power. There have also been claims that many high-profile events go to this stadium because the BJP is in power at the Centre and in the state. A GCA official, however, said, "The fact is that this is the largest stadium. No one can change that."

Opinion

TUESDAY, MARCH 24, 2026

REMEMBERING LOHIA

Prime Minister Narendra Modi

He was a multifaceted personality who played a pivotal role in mobilising people against colonial rule...His unwavering commitment to the cause of empowering the poor and the marginalised continues to inspire generations

Resource rush

Current crisis highlights the need to double down on mineral security, rework policies

THE WAR IN West Asia and the disruption of the Strait of Hormuz offer a vantage point to interpret the increasingly complex supply chain issues haunting a fractured, trust-deficient world. There is a haphazard rush for resource capture and a growing fervour of resource nationalism. India appears to be at the receiving end of this geopolitical churn, perhaps more than any other large nation. The country has only recently surpassed China as the world's largest oil demand driver, with gasoline and diesel consumption in the neighbouring country slowing due to its massive adoption of electric vehicles and gas-fired trucks. As much as 89% of the crude oil consumed by India is imported, as is nearly half of its natural gas. India's persistent exposure to external supply shocks is not restricted to energy; it extends to non-hydrocarbon minerals as well.

For instance, bauxite production has been stagnant over the last decade, and chromite output has even declined. Domestic copper production has been falling, necessitating a tenfold jump in imports of this base metal in the last decade, along with an export plunge. The picture is clearer from aggregated data: gross value added (GVA) in "mining and quarrying" lost its share in overall GVA from 4.8% in 1999-2000 to just 2% in recent years. The country is also heavily import-dependent for critical minerals. While India's import reliance is nearly 100% for lithium and cobalt, 70% of its current supply comes from China.

To be fair, India's policymaking has been responsive to growing mineral constraints in recent years. The last decade saw sweeping reforms in the non-hydrocarbon mineral segment, including the launch of an auction-based allocation system. Composite leases covering prospecting to mining were introduced, along with certainty of tenures and transferability of rights. All this is intended to spur the largely elusive private investor, particularly in extracting deep-seated resources. Administered nature of natural gas pricing has been considerably eased, especially with regard to geographically challenging offshore fields. A National Critical Mineral Mission was launched in January last year, with solid budget and public-sector support. For the first time, exploration licences for critical mineral blocks were auctioned off, with the facility of artificial intelligence-enabled targetting. Coal India, NTPC, Adani Group, Vedanta, and others have laid out plans to venture overseas to explore battery minerals. Bilateral long-term contracts are being finalised with mineral-rich countries. Early this year, Prime Minister Narendra Modi set tall targets, such as $100 billion in investments in the oil and gas sector and the exploration of one million square kilometres by the end of this decade.

Yet, it is clear that the policy initiatives thus far haven't squared up. The current conflict troubling most of West Asia has raised input costs for assorted Indian industries in no time. Scattered resources and the absence of advanced processing technologies have apparently dissuaded private investors from betting big on India's mineral fields. This has limited the overall scale of operations and efficiency gains. Mining intermediaries inflate costs. Policies need to be reworked to address these issues and encourage investments in mineral processing. The latest moves, such as mandating that all solar projects use locally manufactured ingots and wafers, underscore the renewed importance given to indigenisation and upstream linkage. These are welcome. Simultaneously, mining infrastructure must be upgraded and integrated seamlessly with global supply lines and maritime networks. Outbound investments in search of strategic mineral security must accelerate.

TECHPROOF EXPRESS

NEW WAVE IN CODE TEACHING MACHINES TO IMITATE REVIEWERS, TESTERS, & OCCASIONALLY THEOREM PROVERS

An AI layer supervising another

SIDDHARTH PAI

Technology consultant and venture capitalist
By invitation

THERE IS A small irony at the centre of the AI coding boom. The companies that dazzled the software world by letting users generate code from plain English have also created a second market: tools built to inspect, test, and repair the code those models produce. In other words, one layer of artificial intelligence is now being built to supervise another.

That market exists for an obvious reason. Systems such as ChatGPT and Claude can produce working code with remarkable speed, but they are still prediction engines, not infallible programmers. They are good at producing plausible answers, and plausible code is not always correct code. A function may compile and even pass a basic demo while still failing on edge cases, violating internal conventions, introducing security flaws, or breaking dependencies elsewhere in a large codebase.

This is where a newer class of companies has found its opening. Rather than competing only on who can generate more code, firms such as Augment Code, Qodo, Sweep, and Axiom are focusing on what comes after generation: code review, defect detection, test creation, and, in Axiom's case, mathematical verification. The shift is subtle but important. The value is moving from speed alone to assurance.

Augment Code's public pitch is built around what it calls a Context Engine. The idea is that fixing code requires understanding more than a single file or diff. A bug may come from how services interact, how data flows through an API, or how a change in one module ripples across many others. Augment says it semantically indexes a codebase and maps relationships across files, dependencies, architecture, and history. That richer context lets its review system flag logic errors, security vulnerabilities, and breaking changes with more precision than tools that only inspect a narrow patch.

Qodo, formerly Codium, occupies a similar but slightly broader position. Its emphasis is "review first, not copilot first". Publicly, Qodo describes itself as an AI code review and quality platform with deep multi-repo context, agentic workflows, test generation, governance, and standards enforcement. That description matters because it captures the real shape of this industry. The problem is not merely to write a function. It is to decide whether the function is safe, consistent with team practice, adequately tested, and acceptable in a production pipeline.

Sweep is an instructive example of how fluid this category remains. It first drew attention for turning software issues into code changes and pull requests, and is now positioned as a coding agent and autocomplete assistant with explicit attention to editing failures. The practical lesson is that the fixing layer can take several forms: issue-to-pull-request automation, review assistance, or repair suggestions during code editing itself. In each case, the commercial bet is the same: developers will pay not only for code generation, but also for reduction in downstream debugging pain.

> Rather than competing only on who can generate more code, firms are focusing on what comes after generation

What algorithms sit underneath these promises? Public materials support some claims more clearly than others. The first is semantic retrieval. Instead of treating code as ordinary prose, these systems index functions, files, and dependencies so they can retrieve the relevant context for a bug or review task. That usually involves code embeddings, repository indexing, dependency mapping, and search over large codebases. The second is static analysis, the older discipline of inspecting code structure without executing it. Static analysis helps identify suspicious patterns, unreachable branches, API misuse, and missing checks. The third is test generation: the system proposes unit tests or scenario tests designed to expose edge cases, then uses failures as signals for repair.

Qodo has also publicly highlighted multi-agent or agentic workflows, meaning the work is decomposed into narrower review roles rather than handled by one giant model prompt. One agent may check tests, another documentation, another security or standards compliance. That decomposition mirrors good engineering practice. A senior reviewer does not ask one vague question—"is this code good?"—but several concrete ones. Increasingly, these systems resemble small teams rather than single tools.

Axiom takes the most radical path. Its premise is that the long-term answer to unreliable AI code is not just better prompts or broader code search, but formal verification. Axiom's platform, AXLE, is tied to proof verification and proof transformation, reflecting a formal-methods approach that asks whether code-related claims can be machine-checked rather than merely judged plausible. The philosophical leap is enormous. Instead of asking whether code looks right, formal methods ask whether a machine-checkable proof can establish that it is right.

That is a very different standard from conventional code generation. Large language models work by statistical likelihood. Formal verification works by logical necessity. If a proof fails to verify, the system has not merely become uncertain; it has been shown to be wrong. That makes Axiom especially relevant in domains where "probably correct" is not good enough: cryptography, infrastructure, aerospace, medicine, or financial systems with strict correctness and deterministic (meaning absolutely certain, not just the most likely) outcome requirements.

None of this means the code-fixing companies have solved software reliability. Public descriptions from these firms are still, understandably, marketing descriptions, and many internal details remain proprietary. It would be too strong to claim, for example, that every company here uses symbolic execution, satisfiability solvers, or reinforcement learning in the precise ways researchers describe those methods. In several cases, the public record supports broader claims - semantic indexing, context-aware review, test generation, multi-step workflows, formal proof checking - more clearly than it supports specific implementation details.

Even so, the pattern is clear. Code generation is becoming abundant; assurance is scarcer and therefore more valuable. The first wave taught machines to imitate programmers. This new wave is teaching machines to imitate reviewers, testers, and, occasionally, theorem provers. In software, glamour may launch a market. Reliability is what usually turns it into an enduring industry in the end.

Tackling the West Asian curveball

ANUBHUTI SAHAY

Head of South Asia Economic research, Standard Chartered Bank

The government needs to take measures to contain energy demand/import bill with a minimal impact on growth

The escalating West Asia tensions have thrown a curveball at global economic prospects, and India is feeling the heat. The interplay between growth, inflation, and fiscal dynamics has become more complex, leaving policymakers in a tough spot. While inflation is set to jump on elevated energy prices, the key question is whether economic activity can be a bigger casualty. Four out of five global recessions since the 1950s have been preceded by oil shocks.

There's no crystal ball for answers during such geopolitical events. Some fallout is inevitable, but the hit India takes will depend on three things— how long the conflict lasts, the aftershocks in global energy markets, and policymakers' agility during and after the war.

The silver lining is that India has entered the West Asian tensions in a position of strength. Growth over the last 12 months has been close to or higher than trend (partially flattered by one-off measures), inflation has been low, and trade diplomacy is shifting from negotiating tables to delivery mode, albeit with a lag. An import cover of around 10 months (more than 3x the prescribed International Monetary Fund metric to assess adequacy of foreign exchange reserves) assures sufficient external buffers.

Yet, beneath these comforting macro trends is rising discomfort on the ability of the external sector to absorb repeated global shocks amid thinning capital flows. Historically, these buffers have been India's go-to shield for macro stability and growth resilience. If India's growth story is to continue to shine amid the throes of an uncertain world, much will hinge on the resolve to protect these crucial external buffers.

India has posted two years of a balance-of-payments (BoP) deficit for the first time on record, in stark contrast to the $100 billion surpluses in the two years prior to the Russia-Ukraine war. In the past, the BoP impact from a widening current account deficit (CAD) due to a surge in commodity prices was only temporary; recall 2012-13 or the Russia-Ukraine war that started in 2022. As capital flows were steadier then, BoP deficits were rare and a one-year phenomenon.

> While India has little control on elevated global rates or large AI investments, urgent remedial measures are required to boost returns on Indian assets

This is not the case today. Capital flows over the last two years have been less than one-tenth of what was the case two years before the Russia-Ukraine war. We see a risk of a BoP deficit for a third consecutive year, as the comfort from a well-behaved CAD over the last couple of years might not play out this time around; the CAD can be 1.5-2.5x the current expectation of less than $40 billion if crude oil prices stay within $90-110 per barrel for a year.

This implies that India's forex reserves, and thus import cover, run a risk of moving lower unless valuation gains on dollar weakness and higher gold prices offset the drain on foreign reserves on potentially another year of deficit. Between April and December 2025, India's forex reserves were bolstered purely by valuation gains.

With real forex reserve gains hard to come by for now, the right trade-off is to allow the Indian rupee to be the shock absorber. Such an adjustment is likely to keep growth prospects relatively protected by safeguarding against unnecessary policy tightening. Central bank rate hikes aren't just about inflation—they're also used to stabilise the currency if external buffers weaken.

While India isn't anywhere near this point, a relentless defence of the currency amid strong global headwinds and rising global yields can necessitate sharper-than-required rate increases, amplifying downside risk to economic activity. Should oil prices stay above $100 per barrel for a year, GDP growth could slip by almost 100 basis points, with the baseline at 7%, according to a Standard Chartered estimate. This doesn't even account for disruptions in supply, which are already hampering smaller enterprises. Inflation can surge towards 5-5.5% from a baseline of 4% over the next 12 months but is still likely to remain within the mandated band of 2-6%, making rate hikes a less compelling policy measure for now, at least for inflation management.

Along with increased tolerance for the rupee's weakness, more measures to contain demand and thus the import bill are needed as well. India's dependence on West Asia goes well beyond crude oil, extending deeply to gas supplies and fertilisers. While the government has already increased cooking gas prices, rationed gas availability for commercial use, and is identifying new sources of lost energy imports, substituting West Asian energy supplies rapidly is a tall order. Thus, it needs to take measures to contain energy demand/import bill with a minimal impact on growth. Usually, price is a great allocator. However, given that supply is likely to remain a challenge, explicit measures to reduce fuel supply for less productive activity are needed too.

Focus should remain on the crux of the problem-dwindled capital flows. While India has little control on elevated global rates or large AI investment (which the US and China have already made), urgent remedial measures are required to boost returns on Indian assets. Further diversification of energy imports, boosting energy security, and putting the execution capacity to maximise gains from recent trade deals are equally important. Short-term measures can help brave a storm but cannot be a panacea for deeper challenges.

Why young people are taking unwise financial risks

ALLISON SCHRAGER

Bloomberg

THERE ARE AT LEAST two conflicting narratives about Gen Z and young millennials. One is that they are extremely risk averse — afraid even to go out of the house, much less on a date. Another is that they are extremely risk inclined — always betting on sports or speculating on exotic assets. Both of these theories can be true. In fact, they may be equally rational responses to a concern that the world is changing and the old rules don't apply. But they both require an understanding of risk that very few investors, much less young people, possess.

A recent survey from Northwestern Mutual shows that many young people are betting on unconventional high-risk assets because they believe it is the only way to build wealth. It also reveals some good news: Despite a year marked by wild market swings and deep economic uncertainty, more people feel financially secure than they did a year ago, including 39% of Gen Zers (up from 36%) and 52% of millennials (up from 43%). Just a few years ago meme stocks were all the rage, but now it is just straight up gamblin. Some 32% of Gen Zers say they gamble in crypto or sports betting; 35% of millennials own crypto, while 24% bet on sports. Less than 20% of either invest in options or meme stocks. Even more surprising, many of those who don't feel financially secure believe the only way to reach their goals is to take big risks.

It is a core truth of financial markets that higher rewards come with more risk. But some risks are better than others. Betting big on March Madness may mean bigger potential payoffs than investing in the S&P. But the stock market is much more likely to help people build wealth over time.

Of course, young people have been taking dumb risks since Roman teenagers bet on gladiator contests in the Colosseum. And for many younger investors, who have much less experience in markets but also lack a family or a mortgage, now is the time to experiment and learn. My concern is not with risk itself — it's that too many of the younger generation are taking bad risks that offer little potential for growth or upside.

Why are young people choosing bad risks over more productive ones? One answer may be that those more productive risks are out of reach, so they go for broke. Wealth is now more concentrated among the old and the rich, housing is unaffordable, it is hard to find a job. If the economy offers so little in terms of opportunity, betting on a total long shot can feel like the only option.

But even if things are hard, they are not harder than they used to be. It is normal for people in their 20s to feel like the economy is stacked against them. It is a hard time of life, economically if not socially, but there is not much evidence that it is worse than before. The median household under 30 in 2022 had higher wages and wealth than the same group did in 1989. There is also a lower risk of losing your job, and while unemployment for young people has been higher lately, it is still lower than many non-recessionary periods of the past.

Another answer to the question of why may be lack of financial education. Most people don't know how to invest, and people pushing high-risk, high-reward strategies are all over social media. Even financial professionals get swept up in the hype of a bull market — and most investors under 30 have only ever experienced a bull market. (The stock market did fall in 2020, but it came back fast and harder than ever.) A risky market tends to push up the performance of other risky, high-beta assets.

If you think the world is changing so fast that almost anything goes, then you may decide that you have no choice but to try to get in on the next big thing and hope for the best. Or you might decide to own a bit of everything and take advantage of the uncertainty to bet on the one thing you know better than anyone: yourself.

LETTERS TO THE EDITOR

Mulling simultaneous elections

Apropos of "The impact of simultaneous elections" (FE, March 23), conducting Lok Sabha and Assembly polls together could reduce the logistical burden on the state, cut election-related expenditure, and minimise the disruption caused by India's near-continuous election cycle. At the same time, simultaneous elections may allow national issues to dominate the political narrative, thereby overshadowing regional priorities and weakening the federal character of Indian democracy. Such a shift could dilute the space for state-specific debates and reduce the ability of voters to make distinct choices at different levels of governance. The constitutional and structural challenges further complicate the proposal. Thus, while the idea promises efficiency, its implementation must carefully balance administrative gains with preserving India's democratic ethos.
—Narayanan Kizhumundayur, Thrissur

USD's sheen fading?

Apropos "Is $ Losing Reserve Asset Sheen?," (FE, March 23), the dollar's gradual decline in relative share is a long-running trend, not a sudden crisis, and no credible alternative currently meets the structural requirements of a global reserve currency. The euro's fiscal fragmentation and China's capital controls are real constraints, not minor quibbles. India's own reduction in US debt holdings and increase in gold reserves reflects sensible portfolio thinking. Diversification at the margins is prudent; wholesale abandonment of dollar-denominated assets would be neither realistic nor wise for any major economy in the near term.
—M Barathi, Bengaluru

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Upturned by Big ₹upee Movements

RBI may have to delay monetary easing

Rupee's taking a knock as the US-Israel war against Iran enters its fourth week with little signs of likely de-escalation — with the usual Trumpian doublespeak. Oil prices are now 50% higher than when hostilities began, which is straining foreign investor outlook about India's growth prospects. Capital flight out of Indian equities is adding pressure on rupee as the energy shock props up safe-haven demand for dollar. RBI has no preset level to defend for rupee. Yet, the central bank's intervening to ensure an orderly descent. Rupee's not the worst performer among the emerging market pack that's vulnerable to a sustained rise in energy prices. Forex reserves are down on both central bank intervention and higher dollar demand from oil refiners.

The Indian currency's trajectory will be shaped by how the country addresses its vulnerabilities to the energy shock. Dollar is also currently valued beyond a justifiable level as a safe haven. It's reacting to signals of escalation that keeps the situation fluid for RBI — the greenback dipped on Trump's decision not to bomb Iran's power infrastructure a day after he threatened to do so if Iran doesn't let go of its chokehold on the Hormuz Strait. RBI is comfortable with current level of reserves. It will have additional reason to keep interest rates on hold while it assesses the impact of the energy shock on growth and inflation. Incomplete transmission of fuel price increases will weigh on downside risks to growth.

Sustained high energy prices will widen India's current account deficit, and increase the food and fertiliser subsidy burden. RBI may have to delay monetary easing as it smoothes out rupee volatility. Short positions on Asian currencies have moved higher, while speculative activity in the oil futures market could be spilling over into the forex market. RBI will have to tackle heightened volatility as a precondition to stabilising their economies. Indian companies' forex hedges tend to favour limited volatility at a low cost. These are being upturned by big rupee movements.

Transgressive Bill Ignores Individuality

Earlier this month, GoI told Parliament that Census 2027 would record the gender of the head of household under three categories — male, female, and transgender. While this move has been welcomed as a progressive step toward visibility, the proposed Transgender Persons (Protection of Rights) Amendment Bill 2026, placed in Parliament on March 13, signals a troubling shift.

The proposed Bill raises three core concerns: it removes self-identification as the basis of legal gender recognition, departing from the rights framework affirmed in 'NALSA v. Union of India' verdict; it introduces medical and bureaucratic gatekeeping by requiring certification through boards and district authorities; and it narrows the definition of 'transgender', risking exclusion of non-binary, gender-fluid and other identities that fall outside rigid sociocultural or biological categories. This shift replaces a person's internal sense of self with clinical scrutiny, forcing individuals to undergo verification by medical boards and district authorities, thereby transforming a fundamental right into a bureaucratic permit raj, deepening vulnerability and institutional dependence of individuals.

A lack of thought regarding ground realities is apparent. In a country where the transgender community faces systemic abandonment, family rejection and socioeconomic exclusion, adding layers of verification only invites further harassment. Governance is not merely about administrative precision — the 'tightening' in the new Bill is ostensibly to weed out misuse — but also about empathy. Laws must be fundamentally inclusive, reflecting the vast spectrum of identities rather than forcing it into rigid, outdated boxes. This Bill is a regressive step. It should be taken back.

JUST IN JEST

One big fallout of the war in West Asia is abuse of this term

Off with the Head of 'Off-Ramp!' Overuse

Everyone and their uncle's using the word 'off-ramp' in a very off-hand fashion ever since Trump started the war he wants the world to think he's finished. Everyone's 'off-ramp'-dropping — TV and op-ed pundits, fashion show organisers, packers and movers, wheelchair manufacturers…. The expression, used originally by Americans to describe an exit from a highway (our 'exit road' or 'service lane'), has now become *the* geopolitical term of choice — an opening that can get you out of a sticky situation — even when not applicable to geopolitics.

Thanks to the war, the word's metastasised. Your breakup? An off-ramp. Your diet? An off-ramp. Your boss telling you to stop using that hip flask when on the job? Definitely an off-ramp. Whatever happened to the classics? 'Exit strategy' had gravitas, like Napoleon retreating with dignity. 'Withdrawal plan' evoked armies (not just Napoleon's) trudging home in the snow. Even 'pullout' had a blunt honesty — even as it made diplomats blush. Instead, we're stuck with 'off-ramp' in the age of GPS. Somehow, Churchill growling, 'We shall fight on the beaches, but eventually, we'll take the off-ramp provided by the Soviets at Stalingrad,' doesn't quite have a ring to it. So, peeps: easy with 'off-ramp', please. Let wars not (pretend to) end with the imagery of a truck loaded with cauliflower missing a *patli gali*.

STATE OF PLAY From Nowruz to Easter, a bridge for Trump to build — or is it a bridge too far?

Hormuz, Escalation or Exit?

Pranab Dhal Samanta

In the shade, or shady?

Trump may have delayed his plans, but he's marked out the contours of the next level of escalation in the US-Israel war on Iran by identifying power plants as legitimate targets in case Tehran doesn't back off on blocking the Strait of Hormuz. Targeting energy infrastructure is a leaf straight out of the Russia playbook against Ukraine, where the West has had to invest large sums in ensuring the Ukrainian grid keeps functioning despite Russian attacks.

What would happen if Trump were to execute his earlier threat? It's a high-risk strategy, and that's why the reportedly new round of talks, subsequently denied by Iran. Iran has already said it will respond by targeting power utilities of Gulf countries and, possibly, even desalination plants. This would mean the West may have to step in, à la Ukraine, to help these countries economically hit hardest by the closure of Hormuz.

In fact, in an unusual turn of events, Gulf countries have recently floated a Chapter 7 UNSCR against Iran for blocking the strait, calling for UN-backed use of force to ensure freedom of navigation as per international law. This is the second GCC-sponsored resolution this month, the first to talk about 'use of force'. The draft is believed to also mention the option of imposing UN-backed sanctions on Iran if it digs in its heels.

Tehran, for its part, will count on Russia, and China, to veto any such resolution. While that politics plays out in New York, there will be cascading effects — both intended and unintended — of any such escalation.

If the war stretches, consequences will weigh heavy on Trump economically and politically. Hence a revisit to talks

In the first round, militarily the conflict will take an ugly turn as more civilian infrastructure becomes legitimate targets. 'Ukrainisation' of the war will begin with Iran aiming to cause maximum civilian damage in the region, further provoking US-Israel retaliation aimed at securing Hormuz. This won't be easy and may require troops on the ground, which will lead to a longer stand-off, unless Iran relents. But Gulf countries *will* need protection for energy and other civilian infra, raising the cost of war.

Over the past few years, 24 countries have pooled in resources to raise €1.8 bn as part of the Ukraine Energy Support Fund. Last December, through one of the most complex logistical exercises ever carried out, an entire thermal power plant involving 2,400 t of equipment was relocated from Lithuania to Ukraine, adding to the 9,200-odd power generators and 7,000-plus transformers to help Kyiv tide over the crisis following Russian attacks on its energy infra.

The US will have to factor in similar assistance to Gulf countries if Iran were to target their energy infrastructure. It will also need to take into account reluctance of its European allies to get involved in this war. After all, Trump pulled out of many US commitments to Ukraine, including contribution to the energy support fund.

So, the path up the escalation ladder for the US will mean it having to take the call on driving home the ground advantage to achieve military victory *at all costs*. It could work best for US if Iran capitulates quickly through a short ground offensive. But if it stretches, consequences will weigh heavy on Trump economically and politically. Hence a revisit to talks.

Iran, on the other hand, will look for support from Russia and China, as it tries to hold on to Hormuz, its best leverage. In fact, it has, in the past two days, stopped all traffic across the Hormuz on safety grounds, telling trusted interlocutors that it fears attacks could be engineered on foreign ships and wrongly blamed on Iran.

The final pivot of escalation for the US is political with external and domestic dimensions. On the first count, the chasm between the Trump regime and its European allies will grow. Will this force Trump to rethink the US line on the Ukraine war, try talking to Russia, and prioritise a deal in West Asia? This is a strategic imponderable for now. But a recalibration can't be ruled out now that Russian oil is temporarily off sanctions.

On the domestic front, Trump continues to hold on to his MAGA base despite criticism. But a longer commitment into a seemingly 'forever war' will not give him the sort of political impetus he's looking for, given his campaign slogan against long-drawn wars. And for this reason, a 'victory', even if it means a deal on Hormuz, might be an acceptable takeaway as the Iranian government survives to live another day.

But Iran will want more predictability against future attacks in such a deal. Already, military assessments on Iran's hidden capabilities, including nuclear, are being scrutinised again, especially after its missile hit on far-off Diego Garcia, which conveyed an Iranian ballistic missile capability of 4,000 km. Though Iran has not confirmed it, the incident has rattled all who had assessed that Tehran didn't have capabilities to hit beyond 2,000 km. So, Iran will drive a tough bargain.

Chasm between the Trump regime and its European allies will grow. Will this force Trump to try talking to Russia, and prioritise a deal in West Asia?

Either way, Trump's political timeline has spanned itself out between the escalation during Eid/Nowruz and what he can achieve by Easter on April 5. Else, the war is likely to enter a more dangerous, violent and potentially uncontrollable phase — one that will not just disrupt economic realities but also potentially upset political order.

pranabdhal.samanta@timesofindia.com

The Other Ether-Blend Solution

A Prathap Reddy

On Monday, the PM laid out his government's efforts to ensure that petroleum, diesel and gas supply isn't significantly affected by the US-Israel war against Iran and Tehran's response to severely restrict passage of cargo through the Strait of Hormuz. He added that such efforts included blending 'close to 20% ethanol in petrol' along with expanding RE, nuclear power and compressed biogas capacity.

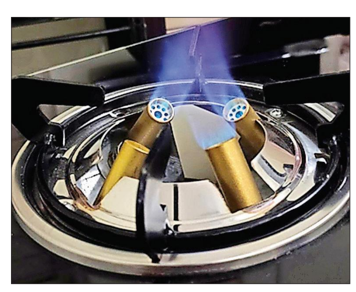
Even with a planned 25% increase in domestic LPG output from refineries, India is still expected to face nearly a 50% shortfall that must be met through imports. This means that India's cooking fuel security and several small-scale industrial operations remain closely tied to stability of global supply chains, an increasingly uncertain situation during periods of conflict.

While ethanol blending has been prioritised, and All India Distillers' Association (AIDA) has offered to supply ethanol blends exceeding 20% with petrol, GoI could also look at another solution to reduce the current energy burden: blending dimethyl ether (DME), a clean-burning fuel that can function as a substitute for LPG.

DME has been adopted in several countries, particularly China, where large-scale production and application demonstrate its practicality. China has installed DME production capacity of around 20 mn MT, and uses it for cooking fuel, industrial heating and transportation applications. Success of this model highlights feasibility of integrating DME into mainstream energy systems.

One of the most significant advantages of DME is its compatibility with existing LPG infrastructure. According to Bureau of Indian Standards (BIS), DME can be safely blended with LPG up to 20%, allowing it to be used across sectors without requiring major changes to current systems.

This blending capability means that transition can begin gradually, reducing dependence on imported LPG, while maintaining continuity in supply and usage. The standard for DME-blending in India has been developed in line with international benchmarks such as ISO 16861:2015, ensuring that safety and performance parameters remain consistent with global best practices.



Flame excuse

Another major advantage is the minimal infrastructure adjustment required for blending. DME can be introduced into LPG bulk storage tanks through controlled mixing or recirculation processes, without requiring extensive new infrastructure. This allows existing LPG storage and distribution networks to be utilised effectively. Even LPG tankers can be adapted to transport DME with minor safety modifications like additional safety valves.

From an industrial standpoint, adopting DME could also help stabilise sectors that heavily depend on LPG. Industries such as food processing, bakery manufacturing, glass production, ceramics and aerosol manufacturing rely on LPG as a cleaner alternative to coal or firewood. Similarly, small businesses — roadside eateries, restaurants, dhabas — depend on LPG for daily operations. Any disruption in LPG supply affects employment, production and livelihoods across large segments of the economy. DME offers a pathway to protect these sectors from energy shortages.

While standalone use of DME may require some modification to burners and stoves to optimise thermal efficiency, research institutions have already begun developing compatible equipment designs. This means that technological barriers to adoption are relatively limited and manageable.

In strategic terms, DME adoption aligns with the broader concept of a 'methanol economy', which promotes alternative fuels derived from domestic resources. This approach places DME within a broader national strategy aimed at building a more resilient and self-reliant energy system that Narendra Modi reiterated in Parliament on Monday.

The writer is founder-executive chairman, Balaji Amines

Not STEM, But STEAMpunk

Rohit K Singh

If you grew up in the 70s-80s, the brightest students were steered into science streams, IIT being the ultimate dream. Engineering or medicine became the gold standard of success. Literature, philosophy or history was admired, but quietly dismissed as an impractical indulgence. In a country emerging from scarcity, STEM was not merely an academic preference, it was a survival strategy.

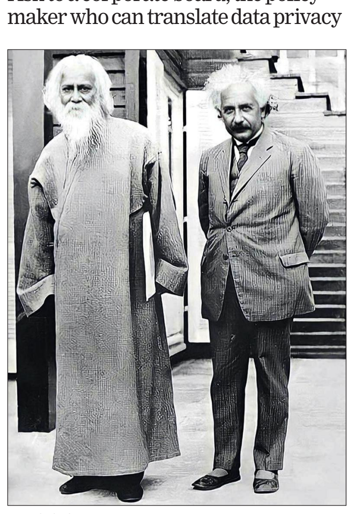
AI is now dismantling that hierarchy. This does not mean science and engineering are losing importance. The AI revolution rests on extraordinary breakthroughs in maths, computing and data science. But the premium is moving away from those who *merely* execute technical tasks towards those who can frame problems, interpret complexity and communicate insight. AI's rise is restoring the importance of capabilities long cultivated by humanities — from STEM to STEM + Arts = STEAM.

For decades, subjects were valued less for what they revealed about the world and more for the salaries they promised. Engineering degrees became the most reliable hedge against economic uncertainty. The system produced exceptional engineers. But it also narrowed our imagination about what intelligence looks like. AI is forcing a rethink. LLMs can generate code, summarise research papers and analyse complex datasets within seconds. The question is no longer, 'Can you code?' It's 'Can you ask the right question?'

Prompt engineering captures this transformation. The quality of an AI system's output depends on clarity and nuance of its instructions. Crafting those requires linguistic precision, conceptual clarity and contextual awareness, capabilities that liberal arts education encourages.

Sam Altman says the effectiveness of LLMs depends on the quality of human prompts and supervision. The most powerful users of AI are not necessarily the best programmers, but those who can frame problems clearly and guide machines towards meaningful outcomes.

Communication is becoming a strategic capability in the AI economy. The engineer who can explain algorithmic risk to a corporate board, the policymaker who can translate data privacy concerns into regulation — these individuals operate at the intersection of tech and society, and increasingly shape the trajectory of innovation.



Artificial, eh?

When introducing iPad in 2010, Steve Jobs argued that the most transformative innovations emerge at the intersection of tech and the liberal arts. Apple's success, he suggested, was not a triumph of engineering but of design, psychology and storytelling. He himself cultivated a deep sense of design after attending calligraphy classes in college.

AI is making that intersection even more important. Machines excel at pattern recognition and prediction. What they lack is judgement. Questions about fairness in algorithmic decision-making, accountability for automated systems or the ethical deployment of AI cannot be answered through maths alone. They require perspectives drawn from philosophy, history, sociology and law. As AI begins to influence hiring, lending, healthcare and governance, stakes of these questions become enormous.

Tension between scientific precision and human interpretation is hardly new. In a celebrated 1930 conversation in Berlin, Einstein and Tagore debated the nature of truth. Einstein argued that scientific truth exists independent of human perception. Tagore countered that truth ultimately acquires meaning through human consciousness and cultural understanding. Nearly a century later, AI's rise is reviving that debate in unexpected ways. Machines may process data efficiently, but the interpretation of meaning, values and purpose remains profoundly human.

India's own technological journey illustrates the importance of this broader perspective. Success of DPI such as Aadhaar, UPI and India Stack did not emerge from engineering excellence alone. It required regulatory imagination, policy design and the ability to build trust among citizens.

The next phase of India's AI journey will require a similarly interdisciplinary approach. Our education system, however, is built around high-stakes exams and narrow academic funnels that privilege technical disciplines while marginalising broader intellectual exploration. While this model produced technical excellence, it often discouraged curiosity across fields. In the age of AI, such rigidity becomes a disadvantage.

Students trained only in routine technical tasks risk seeing those skills automated. What endures is cognitive agility, the capacity to keep learning, think critically and move across domains. Advantage will shift from narrow specialisation to integration: connecting technological capability with human context. A product designer who understands behavioural psychology. A policy expert who grasps ML.

Ironically, rise of intelligent machines is increasing the value of uniquely human qualities — curiosity, creativity, empathy and storytelling. Real AI disruption is not technological alone but also intellectual.

The future will neither belong to STEM alone, nor solely to humanities. It will belong to those who can stand at their intersection, combining analytical precision with human insight. In the age of intelligent machines, the most valuable skill may no longer be solving equations. It will be asking better questions than machines can.

The writer is former secretary, GoI

THE SPEAKING TREE

Divine Feminine

SONAL SRIVASTAVA

We are enveloped by the Divine Feminine. In this cocoon of energy, we are born, live, die and merge into. The Adi Shakti, Divine Feminine, is the primordial source of energy that can neither be created nor destroyed. She is ever-present, energising us to perform our daily mundane tasks to the most creative work. The Divine Feminine is the substratum that shapes us and our creative streaks.

Scriptures say that for a sadhak, spiritual practitioner, engrossed in the worship of the Devi, there is never a dull moment. Not only is his journey full of exciting spiritual experiences — practitioners have reported seeing visions or hearing naad, the unstruck sound — it also culminates in the most satisfactory of ways. The sadhak is assured of the following muktis, forms of liberation, as a consequence of his lifelong sadhana: salokya, samipya, swarupya and sayujya.

The Shaiv and Vaishnav schools of thought offer different interpretations of the four forms of mukti. According to some, salokya means entering the deities' lok; the sojourn in the goddesses' lok depends on the merit that a sadhak has earned through his spiritual practice, the jap; however, the moment his merit is spent, the sadhak is back in the cycle of death and rebirth.

Samipya is when the sadhak develops a closeness with the deity. Swarupya means to attain the physical form of the Devi. Worshipping all forms of the Divine Feminine principle, ultimately leads to sayujja, merging with the Divine Feminine, when the worshipper and worshipped become one and externalities dissolve. It is the ultimate union with the Being.

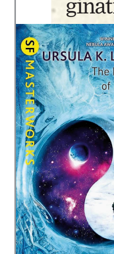
A sincere sadhak can access these through higher states of consciousness by invoking the Devi through spiritual practices.

PARANORMAL ACTIVITY

Paragraphs From Fiction

The Left Hand Of Darkness

Ursula K Le Guin



I'll make my report as if I told a story, for I was taught as a child on my homeworld that Truth is a matter of the imagination. The soundest fact may fail or prevail in the style of its telling: like that singular organic jewel of our seas, which grows brighter as one woman wears it and, worn by another, dulls and goes to dust. Facts are no more solid, coherent, round, and real than pearls are. But both are sensitive.

The story is not all mine, nor told by me alone. Indeed I am not sure whose story it is; you can judge better. But it is all one, and if at moments the facts seem to alter with an altered voice, why then you can choose the fact you like best; yet none of them is false, and it is all one story.

It starts on the 44th diurnal of the Year 1491, which on the planet Winter in the nation Karhide was Odharhahad Tuwa or the twenty-second day of the third month of spring in the Year One.

Chat Room

If Delululemon Were a Person

Apropos the Edit, 'Consider the Madness Of King Donald' (Mar 23), Trump's second term is increasingly marked by concerns that go beyond policy disagreements. At stake is the health of American democracy itself. His pattern of erratic pronouncements, impulsive decision-making and frequent policy reversals has created an atmosphere of uncertainty that weakens institutional stability. In fact, the US has hardly been a beacon of democracy, attacking and bullying less powerful states for decades. Its civil rights movement and treatment of people of colour at home was appalling. Yet, attempts had been made to right some of these wrongs, and create a more level playing field at home. Nonetheless, the US must set its own house in order, instead of launching brutal, senseless wars.

Gregory Fernandes
Mumbai

A thought for today

While you're saving your face, you're losing your ass

LYNDON B JOHNSON

Don Bowls A Doosra

Is his pause the beginning of the end of the Iran conflict, or just the end of the beginning?

Trump's abrupt pivot on targeting Iran's power and energy infra, marked by a five-day pause, has landed like a pressure valve release, on a world bracing for escalation. The reaction was swift and telling. Oil price, the ever-nervous barometer of geopolitical anxiety, exhaled. Brent crude plunged over 7%, slipping to $104 per barrel, while bruised equity markets appear poised for a rebound after weeks of relentless sell-offs. The recent carnage, especially yesterday, with Sensex down 12.7%, and Nikkei shedding 10% over a month, may finally give way to a cautious rally. US stock futures were climbing at the time of writing this.

Any interruption, even a partial and uncertain one, in a conflict that seemed to be hurtling towards a wider conflagration is, undeniably, welcome. Trump's earlier ultimatum to Iran to reopen Strait of Hormuz within 48 hours, had raised the spectre of a critical chokepoint snapping shut. Now, with Trump suggesting Tehran is open to a deal, a narrow diplomatic window may have emerged. That channel, however fragile, must be sustained and broadened. This is where intermediaries like European powers, and India, can step in, shoring up what remains an uncertain backchannel.

Uncertainty, in fact, is the defining feature of this moment. Trump's famously mercurial approach, leaves allies and adversaries alike guessing. The fog of war is thick with "unknown unknowns", to borrow a phrase from American strategic lore. Iran, for its part, denies any talks with Washington, though such denials may well be tactical. There are hints of possible fractures within Tehran, between its military establishment and diplomatic corps. Which faction ultimately shapes policy could determine the trajectory ahead.

Israel's silence adds another layer of ambiguity. Will it align with Washington's pause, or pursue its own strategic calculus? Meanwhile, Trump has conspicuously stopped short of ruling out strikes beyond energy targets, keeping the threat matrix alive.

Three paths now lie ahead. The most optimistic sees an extension of the pause, evolving into a broader de-escalation, contingent on Israeli cooperation. A middle path involves a partial American disengagement while Israel and Iran continue hostilities, containing the crisis but not resolving it. The worst-case scenario envisions continued Iranian strikes in the Gulf, triggering wider regional involvement, and potentially dragging US back in with greater force. Remember, by the end of the five-day period, 4,500 US marines will reach the conflict zone. This moment, then, sits delicately, between resolution and relapse. It may mark the beginning of the end of this conflict, or merely the end of its beginning.

Flight Path

Costly jet fuel may drive up airfares briefly, but lack of competition is the real long-term risk to Indian aviation

India pulled its airfare cap yesterday, without a price surge. Big question now is, what happens next month, when jet fuel prices are revised. If US and Iran achieve "total resolution of hostilities", as Trump has hinted, price pain might not arise. Crude slid below $100 within minutes of Trump's announcement yesterday, and if peace prevails, and Hormuz Strait reopens, it will trend back to its pre-war level. But if fighting resumes, all bets are off.

That's a problem for aviation, because jet fuel isn't seen as a "need" like diesel, or cooking gas. There's no strong political compulsion to cushion its price. Already, Singapore benchmark for jet fuel has touched $225 once – about $50 more than its peak at the start of Ukraine war. Although India is a major producer and exporter of jet fuel, that won't shield Indian airlines from fuel inflation. And since fuel is 40% of operating costs, no airline can absorb its impact for long.

In the worst case, then, flyers would pay a lot more everywhere – not just in India. But India's duopoly market – the Big Two hold over 90% share – is especially suited to price gouging. So, govt must watch airfares. It can also offer some respite by slashing taxes on jet fuel until war ends. More important, it must create conditions for at least one more strong airline to emerge. By govt's own admission, 11 airlines have folded up in the past 10 years. That suggests systemic issues. China's top three airlines together have less than 60% market share. In US, no airline has even 25% share. As the fastest growing aviation market, India should plug this gap. Jet fuel price is a short-term irritant, lack of competition is the real concern.

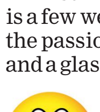
The Khaosan cool

An excursion in the time of war

Anupam Srivastava

By afternoon, the street ceases to exist. Merrymakers swallow it, with as much relish as their beers. A parade of performers – singing Adele to Dylan – traverses the decades with ease. Their audience – including bare-chested European and American men, with their women companions – sways from one end to another. Some hold a bottle or a joint. Khaosan, Bangkok, is unlike any other place on earth. Loud, permissive, garish, yet civilised, not letting its contradictions show an obvious edge against the other. And just around midnight, when you think it would all end, new singers arrive, at the most frequented bars.

Elsewhere, the war grows more intense. The bombing of the school in Minab is a few weeks old. Iran's oil depots were bombed, and a Thai vessel struck. But the passionate conversation about the war in the forenoon, is over with lunch, and a glass of Chang draught, at Molly Bar. The air gets cooler as evening draws near, but Khaosan is feverish, and drenched in its sounds. The music of the eighties, nineties, and today is played by the performers, and several high-wattage concerts are held together. It matters, but not enough – not today. Away at Wat Khuhasawan Worawiharn by the canal, Monk Pramote interrupts his sermon on suffering, and says only unhappy people engage in war.



The air outside King's Cannabis is heavy, as young and old couples sit, on lounging chairs, with long columns of weed ash hanging from their joints. After Wen closes his grocery shop, a 'gas station' comes up, where one can inhale laughing gas for a few Bahts, and have the laugh of one's life.

In this amorphous island of pleasure, signboards try to touch the heart, and make the sale. 'Get a suit before WWIII,' says the board outside Wall Street Tailor, an overnight tailoring shop. A bar urges: 'Stop worrying, start drinking.' And a weed shop remarks thoughtfully: 'Money can't buy you happiness, but it can buy you weed.' The lure of Khaosan made Antonio – who is finally going home to Italy after two months – keep extending his stay.

It is a mystery that the party goes on every day with the same zest. Wen, who retires early for 'a little quiet', says visitors keep renewing, but for regulars, it is just a job. 'We do survive our jobs, don't we?'

Can Trump Bail Out Trump?

His Iran statement is designed to manage tension & reassure Americans. Deliberate ambiguity strengthens his position, but also heightens regional instability

Aydin Sezer

Columnist based in Ankara

Trump's statement yesterday, on talks with Iran, has sent a fresh tremor through the already volatile tectonic plates of West Asia – a region where pressure rarely dissipates, only shifts. His claim that "very good and productive talks have been held towards a full and definitive resolution of our hostilities in the Middle East," coupled with the announcement that military strikes on Iran's energy infra have been postponed for five days, appears, at first glance, to offer a fleeting pause in escalation. Yet, in the theatre of international relations, especially when dealing with a leader who treats unpredictability as a strategic instrument, the real story lies not in what is said, but in when, how, and why it is said.

To understand the weight of this announcement, one must rewind through rapid-fire developments that preceded it. In the days leading up to Trump's statement, Washington had intensified its calls to Nato allies, as well as partners like Japan and South Korea, urging them to ensure that Strait of Hormuz remained open.

At the height of these tensions, reports surfaced alleging that Iran had launched an attack on the US-UK military base at Diego Garcia, located nearly 4,000 kilometres away in Indian Ocean. Tehran's swift rebuttal, dismissing the incident as a "false flag operation", only deepened the fog of uncertainty. In a region where perception often rivals reality in strategic importance, such competing narratives are central to the unfolding drama.

Iran's allegation of a false flag operation introduces a critical dimension to the crisis: the erosion of trust not only between adversaries, but also among allies. Tensions between Washington and Tel Aviv have reportedly been simmering, with differing calculations about how far to push Iran, and when.

In this context, Trump's decision to personally seize the diplomatic initiative and position himself as the manager of the Hormuz crisis, is telling. It is less an act of de-escalation and more a projection of control – a signal that he intends to dictate the tempo of events.

The timing of the five-day postponement is equally significant. It comes as US has deployed approximately 4,500 to 5,000 troops to the region, a move widely interpreted as preparation for a potential ground operation.

Simultaneously, final hours of Trump's earlier 48-hour ultimatum to Tehran, to reopen Strait of Hormuz, were ticking away. In this charged atmosphere, the announcement of a temporary pause, suggests the construction of a fragile diplomatic bridge. Yet, the existence of such a bridge remains uncertain. In high-stakes negotiations, especially those conducted under the shadow of imminent force, appearances can be deceptive.

Indeed, Trump's approach bears all the hallmarks of classic coercive diplomacy. His language carefully blends optimism with menace. While invoking the possibility of a "complete and definitive solution", he also revives the archaic phrase "War Department", a rhetorical choice that evokes the spectre of military action.

The decision to postpone, rather than cancel, strikes on Iran's energy infra, is particularly revealing. It transforms time itself into a weapon, placing Iran under intense psychological and strategic pressure. Five days become not a respite, but a countdown.

This strategy aligns with a broader doctrine of brinkmanship, in which the line between diplomacy and conflict is deliberately blurred. By keeping the military option firmly on the table – repositioned rather than removed – Washington ensures negotiations unfold under constant duress. The conditional nature of the postponement, explicitly tied to the "success of the talks", reinforces this dynamic. Should discussions falter, escalation is not just possible; it is pre-authorised.

At the same time, Trump's calibrated retreat carries implications beyond the immediate crisis. Domestically, it allows him to project the image of a leader committed to peace, particularly at a moment when rising energy prices are straining public sentiment. Internationally, it sends a dual message: reassurance to markets that catastrophe may yet be avoided, and a stark warning to Tehran that the window for compromise is rapidly closing.

This interplay of reassurance and threat, creates a carefully engineered ambiguity – one that strengthens Trump's negotiating position even as it heightens overall instability.

Perhaps the most unpredictable variable in this equation is Israel. Amid the controversy surrounding the Diego Garcia incident and the broader "false flag" debate, the stance of Israeli govt – and particularly the response of Netanyahu to this temporary diplomatic pause – will be pivotal. Israel's security calculations, often more immediate and less tolerant of ambiguity, could either reinforce the current trajectory, or disrupt it entirely.

In the final analysis, Trump's statement is not a declaration of peace, but a manoeuvre designed to manage and manipulate tension. It is a strategic pause, not a resolution; a suspension of force, not its abandonment.

The metaphor is unmistakable: the raised fist has not been lowered, only held in mid-air. Whether it ultimately comes down as a strike, or opens into a handshake, will depend on the substance beneath the rhetoric – do these "productive" talks rest on genuine convergence or merely on the illusion of progress?

Source: The Economist (March 2026)

How America Bails Out Trump

US's underlying strengths are absorbing his erratic choices, including war on Iran. Other countries must work in packs, to hedge against Washington

Ruchir Sharma

Columnist based in New York

Last year America's prowess in AI more than offset the damaging economic effects from Trump's tariffs. Now, different strengths – including energy independence, its army of retail investors, and the dollar's safe-haven status – are saving US economy and markets from the worst effects of his Iran 'excursion'.

Since the first bombing run on Feb 28, gasoline prices have risen more than 20% in US, but much more steeply in many other countries. Natural gas prices have barely budged in US, but have surged in the rest of the world. Across Europe and Asia, the energy squeeze is leading to higher energy prices, factory shutdowns, shorter work weeks, school closures, cancelled flights, and a spike in distressed companies.

Image: AI

So, while no country is gaining from the war, economically, the bigger losers are outside US. This shows up in the mood on the ground. Americans are following events on TV, and grumbling a bit about gas prices, but I see other nations far more absorbed by the war, especially countries in the Indian subcontinent. They are hit harder by the energy shock than Americans are, and many also have closer ties to targeted cities in the Gulf.

Trump is layering a haphazard shooting war, on top of an ill-conceived tariff war, and, so far, largely getting away with it

The signal from global bond markets is that they expect the energy shock to drive up inflation faster in countries other than US. The Trump team is spinning higher oil prices as short-term pain for consumers, but overall a positive for a net energy exporter like US, with producers enjoying a major windfall.

The relative resilience of US economy is feeding the arrogance of American power, which long predates Trump and permeates the establishment. After all, the Biden administration spooked allies as well, by taking the weaponisation of US dollar to a new level in its sanctions against Russia over the Ukraine war.

Lately, when I ask top leaders of Silicon Valley and Wall Street about threats to US economy – from policy unpredictability to declining immigration, and record budget deficits – they typically shrug as if to say, "Where else will the money go?" America's huge lead in AI is reinforcing this narrative.

Superpowerdom naturally breeds hubris. And no country has ever risen up to dominate the financial landscape to the extent US has in recent years. Its share of the widely followed global stock market indices has been mounting for decades, and now tops 60%, while its 25% share of global GDP is the same as it was in the early 1980s.

One mainstay of the US market is its endless army of retail investors. They keep buying stocks at every opportunity, and that enthusiasm has not been dimmed by the grim headlines out of Iran. They have been net buyers every day since the war began, and as a result, US stock market has held up better than international markets. The Indian stock market has fallen twice as much as US market in March.

Though foreigners heap criticism on Trump for a destabilising "war of choice", they don't hold US accountable in financial terms. They too keep buying US stocks, and there is hardly any 'Trump risk premium' visible across US assets.

In fact, the dollar has appreciated this month against every other major currency in the world. This reversed a period of depreciation, which appeared to signal the end of a long dollar bull run.

So, rather perversely, the Iran adventure has at least temporarily rescued the dollar's role as a haven currency.

The mild domestic impact of the war suggests that, once again, Trump is getting bailed out by America's pre-existing strengths. The mighty dollar, the rise to dominance of US markets and revolutionary technologies, the achievement of energy independence – all have roots that predate him.

Trump is layering a haphazard shooting war, on top of an ill-conceived tariff war, and largely getting away with it, in part because, other nations have become too dependent on America – militarily, technologically and, above all, financially.

Countries have recently tried to decouple from US by spending more on defence, or directing central banks to hold fewer dollars in their reserves, or by striking trade deals that don't involve US. But the lopsided economic fallout from the Iran conflict shows that they need to do a lot more.

The law of the jungle prevails, and the best way for smaller powers to survive in an environment where might makes right, is to work in a pack. Nations could form bargaining coalitions to respond to tariff threats, rather than let Trump force deals on one overmatched target at a time.

Europeans could focus more on creating a unified military structure, and spend more on establishing energy independence, rather than building oversized welfare states. Many nations could strengthen their own financial markets, by encouraging domestic investors and sovereign wealth funds to invest more at home, rather than shipping national savings to US.

Until Trump faces serious domestic blowback from his various wars, he will keep exercising American power in an unbridled way. A significant section of MAGA Republicans back the Iran campaign, so other nations can't just hope for his base to revolt. They need to do more to strengthen themselves, collectively and individually, or they will keep suffering more damage from the whims of an erratic US leader.

Calvin & Hobbes

Sacred space

Without deep listening and gentle loving speech, it is very difficult to move towards peace. Peace will only become a reality, when world leaders come to negotiations with the ability to hear the suffering at root of all conflicts.

Thich Nhat Hanh

Poetry Humanises Us In Violent Times

Sumit Paul

Universities around the world are shutting down humanities programmes, saying they don't lead directly to jobs. But education is not just about employment; it's about shaping thoughtful, ethical, and empathetic human beings. Poetry, philosophy, and music humanise us.

Poetry makes us think, and what makes us think guides us. A poem is a hazardous attempt at self-understanding: It's the deepest part of autobiography. Poetry is an ordinary language raised to the nth power. In any language, poetry is the most distilled form of that language. What's stated in a poetic way always strikes deeper and more effectively.

Poets are critics and analysts of life. They're also the chroniclers of lives and times. They're the mouthpieces of the emotions that we all have but are unable to express. Poets do for us what we cannot do for ourselves. There's an English adage: A poet is a prophet. Poetry is indeed prophethood (*Shayari Payambari Ast*: Jami).

The way poetry motivates and inspires all of us, nothing else does. Read Rudyard Kipling's immortal poem, *If*. Recalling the poem makes us feel intrinsically good: "If you can fill the unforgiving minute, With sixty seconds' worth of distance run, Yours is the Earth and everything that's in it, And – which is more – you'll be a Man, my son!" Or read Patricia A Fleming's poem, *What life should be*.

The immortal lines of the great Persian poet Rudaki, which he uttered in ancient Persian, *"Dee az qeen min taftan na nafrat un biqam/ Jaan az shaa'fziyadan pahloo riqam"* – how can i hate anyone when everyone is my extension?

THE SPEAKING TREE

When John Donne says, "Any man's death diminishes me because i'm involved in mankind." We think with the poet and empathise with all, our differences dissolve. Poetry makes us empathetic and humane.

In times of raging global wars, these lines assume much greater significance, and we must ask ourselves how we can hate anyone when all are our extensions and brethren? Are warmongers listening? Are they bereft of compunction? Don't they feel the prick of conscience? Ponder what the Pakistani Urdu poet Jaun Elia says, *"Jang ki itni hi darkaar hai agar/ Insaan, tu apne aap se jang kar,"* – if war cannot be averted at all/ You'd better fight with yourself. This fight is metaphorical; you fight with your conscience and get the reply that any war is futile. After all, the voice of conscience is the voice from the Cosmos. Sahir Ludhianvi's poignant *nazm*, *"Ae shareef insano, jang talti rahe toh behtar hai."* The poet says that war itself is a problem. How can it provide a solution to any problem?

Poetry is a revelation. It enables satori that dawns as an awakening. It is a pursuit of excellence and enlightenment. Poetry is not an ivory-tower production house; it's a manifesto of ground realities. Urdu and Persian poet Nashtar 'Nishapuri' said, *"Main rahoon na rahoon, fikra nahin/ Hoga meri shayari ka zikra Kahin,"* – no matter whether i live or not/ Someday, somewhere, my poetry will be read.

Poetry is a balm for weary souls. In turbulent times, poets have a far greater role to play in assuaging humanity's frayed nerves and providing a definite direction for misguided humans.

Editor's TAKE

West Asia war tests India's economy

As tensions escalate in West Asia, India finds itself confronting the familiar tremors of surging oil prices and currency pressure

Three weeks on, there is no sign of a let-up, only escalation – this sums up the US-Israel-Iran war that started as a regional conflict but is now impacting the whole world, and India is no exception. With our heavy dependence on imported fossil fuels, most of which passes through the Strait of Hormuz, the oil supply is severely hit, and as the war prolongs, things will get worse. The impact is staggering. The oil markets are in a tizzy, oil prices have already reached $119 per barrel. Experts feel that oil prices will touch $150 per barrel within a matter of weeks if the war does not end or some sort of mechanism to supply oil through Hormuz is not reached. The Iran war has once again exposed the fragility of global energy markets-and India's deep vulnerability to them.

Recognising the gravity of the situation, Prime Minister Narendra Modi convened a high-level meeting to review the country's energy preparedness, signalling both urgency and concern. While retail petrol and diesel prices have so far remained stable, this may not hold for long. Industrial diesel prices have already risen by about 25 per cent, reflecting the underlying stress in the system. Such increases inevitably translate into higher transportation and logistics costs, which in turn fuel inflation across sectors. Equally alarming is the falling rupee against the dollar, and it would not be an exaggeration to say that we are now close to breaching the 100 mark in months, if not weeks. The depreciation of the Indian rupee further compounds the crisis by making imports more expensive.

Like every crisis, this too has its own lessons, if we are willing to learn. For long, we have depended upon imported oil for our energy needs without ever caring that this dependency is one of the severest weaknesses any nation can have, especially if it strives to be a regional power. The time to correct this vulnerability is now. It is a structural issue and needs to be tackled with utmost urgency. There cannot be a short-term fix for it, like arranging alternative suppliers or increasing cess to manage demand. It would need investments in exploration and refining facilities. It would need sustainable solutions like biofuels and renewable energy resources – the policy decisions that would alter the energy equation. It is not a unique problem; many countries have successfully become self-sufficient and transitioned from total dependence on oil to a more diversified and self-reliant energy mix. Germany, through its 'Energiewende,' scaled renewables rapidly; Denmark switched to wind energy in a big way; and China dominates solar, wind, and battery supply chains, combining energy security with industrial strategy. Electrification of transport, expansion of public transit, and promotion of energy efficiency can significantly reduce oil dependence. Equally important is macroeconomic resilience. Strengthening the rupee through sound fiscal management and boosting exports can help mitigate the impact of external shocks. The lesson is clear-energy security equals national security, and it should be our national priority. We need a long-term strategy; it will not be fixed with a band-aid!

A new dawn or another illusion?

Nepal's recent election was an eruption of public anger against years of stagnation and corruption. Yet, a pressing question remains: has the country chosen real transformation, or simply been swept up by emotion without a clear path forward?

NILANTHA ILANGAMUWA

The election in Nepal last week was not merely a political exercise; it was an eruption of pent-up fury, a rejection of the old guard that had throttled any semblance of progress for decades. But what now stares back at the country is a stark question: have the people truly changed their future, or simply traded one set of illusions for another?

For years, Nepalis endured the same trio of power brokers-the Nepali Congress, the CPN-UML, and the so-called Communist Party-as these entities pirouetted through government halls, recycled leadership, and maintained an endless cycle of impressive promises and microscopic delivery. Institutions decayed, corruption metastasised, and unemployment worsened further. Youth unemployment stands north of 20 per cent-more than double the national average. Around 1,500 young Nepalis leave their homeland every single day seeking work abroad, a staggering exodus that undermines any future the country might hope to sculpt for itself.

So, when the uprising erupted, when Gen Z and youth frustration boiled over into the streets, it was not just rage-it was despair. Into this void surged Balendra Shah, the rapper-turned-Kathmandu mayor better known as Balen. He became the face of something many claimed they wanted: a break with the past. The Rastriya Swatantra Party (RSP), a party as new as its leader's rise from outside the entrenched political class, swept to an unprecedented majority: 125 of the 165 first-past-the-post seats. A single party holding nearly two-thirds control in Nepal is almost unheard of, a brutal indictment of the old establishment's collapse.

Nepal's new leadership inherited not opportunity but catastrophe. The economic foundation is weak and brittle. Public debt hovers around 40-45 per cent of GDP, but it is the quality of the economy that terrifies: a narrow tax base, enormous dependence on remittances accounting for roughly one-quarter of GDP, and a private sector too fragile to absorb the burgeoning army of young jobseekers. Tourism, once thought a panacea, remains exquisitely sensitive to global disruptions. Agriculture remains archaic and unproductive. Power outages and distribution inefficiencies plague even the most basic enterprises. Crucially, the labour force-the very youth that marched in protests-has no obvious outlet for meaningful employment.

The RSP manifesto, the so-called "2082 Vision", is nothing if not audacious: 1.2 million jobs in five years; GDP expansion to almost $100 billion; per capita income rising to $3,000; 15,000 megawatts of installed capacity; halving LPG imports; digital services exports of $30 billion in ten years; and the construction or upgrade of 30,000 kilometres of national highways. These numbers are ambitious-some might say visionary-but independent observers see them as fantasy built on the emotional reservoir of hope, not on deeply rooted economic analysis. Nepal's energy grid cannot reliably distribute current capacity; transportation infrastructure routinely buckles under seasonal rains; foreign direct investment remains underwhelming; and the digital economy is throttled by regulatory unpredictability and an underdeveloped legal regime for international payments.

FOR A GENERATION RAISED ON UNFULFILLED PROMISES, THE OLD GUARD SIMPLY HAD NO AUTHORITY LEFT TO PERSUADE A BATTERED POPULATION OF ITS RELEVANCE

These are the grim realities. A promise to reduce imports without addressing critical bottlenecks in trade policy or cross-border logistics is a promise destined for frustration. A pledge to build tens of thousands of kilometres of roads without sustained institutional capacity to manage land acquisition, competitive bidding, quality control, and anti-corruption oversight offers little more than ritual ground-breaking and even more ceremonial delays.

This mismatch between aspirational rhetoric and structural capacity points to a far more troubling truth: Nepalis have been deceived not by individuals but by narratives. The uprising was not wrong in its desire for change. But it was driven by visceral emotion-a collective impulse to reject the old, often without a coherent alternative blueprint that could realistically transform the economy and provide stability. Angry protests and street fervour commandeered the engine of politics, and once that engine is running on emotion rather than evidence, it becomes dangerously unpredictable. If Nepal is honest with itself, it must question whether Balen may tread a similar path: overwhelmed by the emotional thunder that elevated him, yet unprepared to deliver the institutional and economic stability the nation desperately needs.

Here is the painful truth: Gen Z politics, fuelled by emotion, creates momentum but not mechanisms. Momentum wins rallies; mechanisms build nations. The current administration's inexperience-not merely in government, but in managing a modern economy under immense pressure-sets the stage for something grim: a crescendo of disappointed expectations.

Worse still, emotional politics is ripe for exploitation by external actors. Nepal is geostrategically hemmed in by its two giant neighbours. India-the largest source of trade, investment, energy supplies, and transit routes-watches with both interest and caution. China, a stakeholder in multiple infrastructure ventures and a central actor in Belt and Road projects, has its own expectations. Both have engaged with the RSP, seeking alignment with their own strategic interests. But emotion is a currency external powers love to leverage: where national confidence is high and institutional clarity is low, foreign influence finds entry points. A government fuelled by public passion-but lacking robust policy anchors-becomes pliable, attractive, and dangerous. The question is: did the electorate truly choose a path to prosperity, or merely a dream of it? Emotional politics gave the people a mirror-a reflection of their hurt, their labour unrecognised, their aspirations denied. But mirrors do not map roads; they only reveal what is already before us.

Balenomics may become a lesson in hubris-not because the goals are unworthy, but because goals without disciplined implementation, institutional reform, and credible governance remain poetry when the country needs engineering. Nepal needs a systemic recalibration of labour markets, transparent rule-making, competitive commerce, legal certainty for investments, and infrastructural credibility-not just slogans that rouse crowds. When citizens see delays, when promised jobs fail to materialise, when inflation stubbornly erodes incomes, and when foreign capital does not flood in simply because of optimism, the inevitable question will surface: was this all just emotional theatre? If the answer is yes, Nepal risks entering a phase worse than the old guard's mismanagement: disillusionment with revolt itself.

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Peace cannot grow from seeds of violence

RAJYOGI BRAHMA KUMAR NIKUNJ JI

2ND OPINION

There has never been a single day in the whole history of the news media when the media reported that peace prevails on the whole planet Earth and that everything is going right in the world. A study of the recorded history of the world gives conclusive evidence that there has constantly been bitter or bloody strife over the last two-and-a-half millennia. This strife may have been social, economic, political, cultural, ethnic, religious, or militaristic, or of any other kind; often, it was bolstered by a claim that it was for a noble goal or a just cause.

Similarly, those who were responsible for the partition of the Indian subcontinent claimed that it was done for the noble goal of seeking justice for one particular community, which was considered the main minority community. Similarly, those who are now waging a struggle for reservation of government jobs for the Scheduled Castes and Scheduled Tribes, etc., claim that they are fighting for social and economic justice for the weaker and backward sections of society.

Of late, even extremists claim that their fight is for obtaining political freedom and for the human right of self-determination. Now, even if we suppose that the struggle waged by the above-mentioned groups or individuals, or by others mentioned in history, has been for a noble goal or a good cause, the question arises about the means used to achieve their professed goal. Have the means adopted in each one of the above cases also been worthy?

Looking at the historical facts, one would find that some of these struggles or protests indulged in gory acts of bloodshed of hundreds of thousands of people, and they created a deep feud against the capitalist world, thus leading to large-scale armament build-up. In their struggle for a classless society, they led to the creation of antagonistic groups or countries locked in global conflicts of various sorts.

The means adopted by the leaders for forcing the partition of India resulted in unparalleled communal carnage and genocide on a mass scale. Similarly, the strategies of those who want reservation for backward castes and the approach and means adopted by militants also have led to great strife and the latter, in particular, have led to killings, abduction, and fear.

It is easy to romanticise revolution or justify conflict in the name of ideals, but one should remember that lasting peace can never emerge from paths paved with violence. In the process of pursuing justice, if we lose our sense of humanity, the very purpose of that justice becomes diluted. So, the analysis of events of history leads us to the conclusion that if we wish peace for ourselves and the world, and if we aspire that bloody strife should end, then we must first review our goal and see if it is really correct and noble, and then we should double-check whether our means are also noble. Let us remember that the means are as important as the goal. The greatest transformations in history have not always come from force, but from the strength of inner conviction and peaceful resistance. Leaders like Mahatma Gandhi and Martin Luther King Jr. taught the world that non-violence, dialogue, and moral clarity have the power to dismantle oppressive systems without bloodshed. In today's fractured world, where lines between activism and aggression are often blurred, we must return to the wisdom of these ideals. True peace is not just the absence of war, but the presence of compassion, understanding, and ethical integrity in both our vision and our methods. So, let our future struggles, if any, be defined not by the wounds they cause, but by the healing they bring.

The writer is a spiritual teacher and a popular columnist

PIC TALK

Students dressed as Lord Hanuman take part in a fancy dress competition ahead of the Ram Navami festival in Patna.

PHOTO: PTI

LETTERS TO THE EDITOR

WHEN EVERY SIDE NEEDS INDIA, IT MUST ACT WISELY

India's carefully cultivated neutrality has gradually transformed into something far more valuable - strategic indispensability. Years of sustaining working relationships with rival powers, without binding itself exclusively to any, have earned India a position that few nations occupy today. The unfolding West Asia crisis is now putting that hard-won standing to its most demanding test yet.

What makes this moment particularly significant is that all sides – the US, Israel, Iran, and the Gulf states – simultaneously need India's ear. That rarely happens. Historically, non-alignment was principled abstention. Today, under Modi and Jaishankar, it has evolved into active, multi-vector engagement. India is no longer watching from the sidelines; it is operating as an indispensable node in a network with few other common points.

But diplomatic access, however valuable, has a shelf life. It must be converted into tangible outcomes – ceasefire facilitation, safe passage for stranded vessels, stable energy corridors, and protection of the Indian diaspora across the Gulf.

India's BRICS presidency adds another lever. Using it purposefully, rather than procedurally, could make a meaningful difference. Back-channel diplomacy, quiet mediation, and confidence-building between Tehran and Washington are areas where India's credibility with both sides gives it genuine utility.

A MYILSAMI | Coimbatore

Lift safety cannot be ignored

Kolkata's RG Kar Medical College is back in the news. A patient's father was killed after he was trapped along with his family inside a lift when it malfunctioned at the hospital's trauma centre. He reportedly was stuck either between the lift doors or between the lift and the wall.

It goes without saying that lifts or elevators should be regularly monitored and their upkeep must be foolproof. Secure electric connections, temperature maintenance, and lubrication of parts cannot be taken for granted. Checking the functionality of doors through periodic check-ups is extremely essential. Door sensors have to be kept in good condition.

Monthly or quarterly servicing must be undertaken. Scrupulous adherence to annual maintenance contracts (AMCs) should be mandatory. Timely audits cannot be bypassed. Traction elevators are more effective in larger buildings.

Old and worn-out elevators should make way for newer ones. Nowadays, lift operators are conspicuous by their absence. Also, the majority of them are untrained. Proper training and supervision are necessary to avoid such tragic incidents in the future. Safety cannot be compromised in public institutions, especially hospitals, where lives already hang in the balance.

GANAPATHI BHAT | AKOLA

CAPF officers deserve leadership roles

In recent times, both social media and traditional media have witnessed active discussion on the CAPF (General Administration) Bill 2026. The Bill has sparked debate, particularly among Indian Police Service (IPS) and CAPF officers, who have publicly expressed their concerns regarding the provisions related to the proposed role of IPS officers.

Having served in the Armed Forces, I have reasonable insight into the functioning of the Central Armed Police Forces (CAPFs). My experience provides practical insight into their structure and deployment in challenging environments. The backbone of the CAPFs lies in their units and sub-units, led by subordinate and CAPF officers, often deployed far from headquarters for internal security, law and order, elections, and protection of vital installations.

It is the cadre officers who are exposed to complex and challenging situations in the field. They coordinate with local administration and other security forces to fulfil both operational and administrative directives. Considering this, the claim that IPS officers alone possess wider experience is far from reality. CAPF officers possess the necessary qualifications and experience to be promoted to higher positions currently held by IPS officers.

PRAKASH BHATT | NOIDA

Nuclear threat remarks are irrational

The remarks of Abdul Basit, the former Pakistan High Commissioner to India, that 'if the US were to ever attack Pakistan's nuclear arsenal, Islamabad would launch retaliatory strikes against New Delhi and Mumbai', are illogical and senseless. His comments are in response to the US National Intelligence Director Tulsi Gabbard's assessment before a Senate committee that "Pakistan's nuclear missiles are a future threat to the US". Basit argues that in the event of a conflict with the US and Israel, since the former is not within Pakistan's immediate nuclear range, India would become the default target.

This reflects a crab mentality, like a man who lost both his eyes finding satisfaction in a neighbour losing one. It is surprising that such remarks come not from a trigger-happy Pakistan general but from a former diplomat. The US has no military base in India, unlike in some West Asian nations targeted in the US-Israel-Iran conflict. Pakistan may partially succeed in its attempt, but if India were to retaliate, it may be wiped out from the face of the earth. Such an eventuality will not be a victory for either, but a colossal failure of diplomacy, statecraft, and human imagination. One hopes Basit is not echoing his government.

V JAYARAMAN | CHENNAI

Please send your letter to the info@dailypioneer.com.
In not more than 250 words. We appreciate your feedback.

The Pioneer
NEW DELHI | TUESDAY | MARCH 24, 2026

Degrees but no job: Is a college education still worth it?

India possesses one of the youngest populations in the world, a demographic advantage that could fuel economic transformation. But this potential can only be realised if the country's education system is closely aligned with the shifting demands of the global economy

FIRST Column

DINESH SOOD

For decades, millions of Indian students have followed a familiar economic script: study hard, earn a college degree, and a stable job will follow. This belief, shaped by the 1990s era of economic liberalisation, evolved into an unwritten social contract between schooling and work.

Yet recent evidence suggests that this contract is rapidly weakening. A recent report by Azim Premji University reveals a striking reality: "around 67 per cent of India's unemployed youth today are graduates. In 2023 alone, roughly 1.1 crore unemployed young people were degree holders, a dramatic rise from 32 per cent in 2004". Over the same period, the proportion of graduates in the youth population rose sharply-from about 10 per cent in 2004 to nearly 28 per cent in 2023.

These figures reveal a structural paradox in India's labour market. While access to higher education has expanded, the economy has not created enough high-quality jobs to absorb the growing number of graduates. The result is an emerging crisis: more degrees, but not enough employment.

For earlier generations, the formula worked. A college degree was rare and therefore highly valuable in the labour market. But the India of 2026 presents a different reality. Higher education has expanded rapidly, yet employment opportunities have not kept pace. As a result, the value of a conventional degree is steadily declining while youth unemployment continues to rise.

Degrees are rising faster than jobs

Data from the International Labour Organisation's India Employment Report 2024 paints a troubling picture. According to the report, "young people account for nearly 83 per cent of India's total unemployed population. Even more alarming is that graduate unemployment is nearly nine times higher than that of individuals with little or no formal education."

These figures raise a difficult but necessary question: does a college degree still guarantee economic mobility in India?

The problem today is not simply a shortage of jobs; it is the hyperinflation of degrees. Over the past two decades, the number of universities and colleges in India has expanded. While this expansion has increased access to higher education, it has also created a mismatch between what students learn and what industries need.

A COLLEGE DEGREE MAY PROMISE POTENTIAL, BUT A PUBLIC PORTFOLIO PROVIDES UNDENIABLE PROOF OF COMPETENCE

The economics of signalling and the declining value of degrees

Economists describe part of this phenomenon via signalling theory. A degree does not simply certify knowledge; it also signals certain qualities to employers-discipline, perseverance, and the ability to complete organised tasks.

When graduates were relatively few, that signal held substantial weight. But when millions of students graduate each year with similar qualifications, the signal becomes diluted.

A Mercer-Mettl India Graduate Skill Index (2024-25) report reveals that only about 55 per cent of Indian graduates are considered employable for modern industry roles. In other words, nearly half of India's degree holders enter the labour market without the skills employers actually require.

One reason for this gap lies in the slow pace of curriculum reform. University syllabi often take years to update, while the digital economy evolves almost weekly. As a result, students graduate with outdated knowledge that does not match contemporary business demands.

Artificial Intelligence and the end of generic knowledge work

Technological disruption, particularly the rise of Artificial Intelligence (AI), has accelerated this shift. Many knowledge-based tasks, once justified by white-collar employment, are now being automated. If a degree primarily trains students to perform routine analytical tasks, basic coding, or standardised reporting, those skills are rapidly losing economic value. AI systems can perform such tasks in seconds and at minimal cost.

Industry projections show this transformation. According to NASSCOM, India will require over one million specialised AI professionals by 2026 to meet growing technological demand. However, these roles require advanced, applied skills that cannot be acquired solely through outdated classroom lectures.

In other words, the modern economy does not simply reward what individuals know. It rewards what they can build, solve, and create.

Proof of Work in the Digital Economy

The internet-driven economy has introduced a powerful shift in recruitment methods. Increasingly, employers and clients care less about formal credentials and more about demonstrable ability.

An aspiring AI engineer is judged by the projects visible on their GitHub profile. A digital writer or marketer is evaluated by their published articles, audience reach, or portfolio. A college degree may promise potential, but a public portfolio provides undeniable proof of competence. This transformation has created a more meritocratic environment where skill, creativity, and execution matter more than institutional pedigree.

The rise of parallel learning ecosystems

Recognising these realities, a parallel learning ecosystem is emerging across India. Web platforms, coding boot camps, digital learning hubs, and professional communities provide learners with practical skills that traditional institutions often fail to deliver.

These platforms emphasise project-based learning, industry exposure, and practical problem-solving. Many global technology firms are increasingly prioritising skills assessments and portfolio-based hiring instead of purely academic credentials.

Government initiatives, including Skill India, Digital India, and various start-up incubator programmes, also reflect a rising recognition that employability depends on practical capability rather than academic qualification alone.

The Real Solution: Degrees Plus Skills

However, the answer is not to abandon formal education altogether. Universities still play an important role in building intellectual foundations, social networks, and academic disciplines that remain essential in many fields.

What needs to change is the way students approach higher education.

Instead of treating a degree as the final destination, students should adopt what investors call a "Barbell Strategy". Completing a college education provides a basic safety net. It satisfies society's expectations and guarantees that graduates are not automatically filtered out by conventional recruitment systems.

At the same time, students must invest in developing specialised, high-value skills, whether in AI, data analytics, design, digital media, entrepreneurship, or emerging technologies. Building projects, internships, and digital portfolios should become an integral part of the college journey.

Redefining the purpose of education

India possesses one of the youngest populations in the world, a demographic advantage that could fuel economic transformation. But this potential can only be realised if the country's education system is closely aligned with the shifting demands of the global economy.

The value of a college degree has not disappeared-but its meaning has fundamentally changed. It is no longer a guarantee of success; it is merely a starting point. In the 21st-century knowledge economy, students must treat their degree as a foundation and make the development of practical skills their true competitive advantage. Those who assume responsibility for their learning journeys and build relevant expertise will define their own success.

To compete and thrive, students must build not just credentials but also skills, practical experience, and a mindset that welcomes change.

By pairing formal education with continuous learning and hands-on expertise, India's youth can turn the degree inflation crisis into an opportunity for long-term growth.

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Decisive Initiative of BBSSL: Towards self-reliance in the seed sector

RAMAN KUMAR

India is an agrarian country where a large section of the population depends directly or indirectly on agriculture. With changing times, agriculture is no longer merely a means of livelihood; it has become a central pillar of self-reliance, food security, and rural economic development. In any agricultural system, seed is considered the most critical input because crop productivity, quality, and farmers' income are directly dependent on seed quality. At the global level, the seed industry is undergoing rapid technological transformation. Advanced biotechnology, artificial intelligence-based agriculture, and organised supply chain systems are taking this sector to new heights. Regions such as North America lead the industry, while the Asia-Pacific region, particularly India, has emerged as a rapidly growing and significant market.

In this context, the establishment of Bharatiya Beej Sahakari Samiti Limited (BBSSL) is regarded as a historic initiative. This institution is not merely a commercial organisation but a national cooperative movement rooted in farmers' participation, aiming to make the country self-reliant in seed production and to provide high-quality certified seeds to farmers. BBSSL has been formed in alignment with the vision of "Self-Reliant India" and "Prosperity through Cooperation."

Global seed market and India's position

In 2024, the global seed market was estimated at approximately Rs 4 lakh crore. North America leads the sector due to precision farming, advanced machinery, and strong supply chain infrastructure. Meanwhile, within the Asia-Pacific region, India is emerging rapidly as a key seed market. India's seed industry was valued at Rs 67,858 crore in 2024 and is projected to reach Rs 1,65,295 crore by 2033. An estimated annual growth rate of around 10.5 per cent highlights the sector's strong potential. Diverse agro-climatic conditions, contributions from research institutions, and government support are accelerating this growth. However, India continues to face challenges such as counterfeit seeds, limited access to quality seeds, and inadequate modern storage facilities. The establishment of BBSSL represents a structured and coordinated effort to address these issues. BBSSL was established with an initial capital of Rs 250 crore, contributed equally by five major cooperative institutions-Indian Farmers Fertiliser Cooperative Limited (IFFCO), Krishak Bharati Cooperative Limited (KRIBHCO), National Agricultural Cooperative Marketing Federation of India (NAFED), National Cooperative Development Corporation (NCDC), and National Dairy Development Board (NDDB). Its authorised share capital has been fixed at Rs 500 crore. Operating under the brand name "Bharat Beej," the organisation promotes indigenous seed production to reduce dependence on imports.

It is a national-level multi-state cooperative seed society, with its registered office located at WTC Tower, New Delhi. Its membership extends from Primary Agricultural Credit Societies (PACS) to state and national-level cooperative institutions. Breeder seeds for production are sourced from the Indian Council of Agricultural Research (ICAR), state agricultural universities, and other national and international institutions.

Objectives and operational framework

The primary objective of BBSSL is to provide farmers with high-quality certified seeds, promote domestic seed production, and reduce import dependence. The organisation places special emphasis on developing climate-resilient and disease-resistant seed varieties. Strengthening farmers economically by integrating cooperative institutions into the value chain is also a key priority. Additionally, increasing the Seed Replacement Rate (SRR) and Variety Replacement Rate (VRR), along with ensuring transparency through a QR code-based traceability system, are among BBSSL's core strategic initiatives.

Strategic operations and achievements

BBSSL procures breeder seeds from reputed research institutions, produces foundation and certified seeds, and supplies them to farmers. The organisation invests in research and development to develop drought-tolerant, pest-resistant, and high-yielding varieties capable of addressing climate change challenges. Through the expansion of its distribution network, BBSSL is ensuring the availability of certified seeds even in remote regions of the country. From a statistical perspective, the organisation has made notable progress. So far, it has received more than 35,524 membership applications and issued 34,078 share certificates. In the financial year 2025-26, it distributed 2,70,167 quintals of seeds, generating an operational revenue of Rs 177 crore. During Kharif 2025-26, seed production was carried out over 11,125.68 hectares, resulting in 1,99,501 quintals of output, while for Rabi 2025-26, production of 7,04,388 quintals is estimated. The organisation has set an ambitious target of achieving a turnover of Rs 25,000 crore by 2035.

Vision 2035: A Self-Reliant Seed Powerhouse

BBSSL aims to become one of the world's top five seed-producing organisations by 2035. To achieve this, it is expanding research based on artificial intelligence and genomics, strengthening local production systems, reducing import dependence, organising farmer training and demonstration plots, and adopting environmentally sustainable production practices.

Conclusion

Bharatiya Beej Sahakari Samiti Limited (BBSSL) represents a historic and transformative initiative in India's seed industry. Through its cooperative-based model, the organisation is providing a new direction to agricultural productivity, food security, and rural prosperity. By integrating quality, transparency, technology, and cooperation, BBSSL is emerging as a strong pillar in realising the vision of a self-reliant India.

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From AI to legal Intelligence: Strengthening governance

NIRMAL SINGH

India's technological rise has entered a decisive phase. For decades, the world viewed India largely as an IT services powerhouse-efficient, reliable, and cost-effective. Today, that perception is changing fundamentally. India is no longer merely supporting global digital systems; it is building and exporting them.

The next frontier in this evolution is not just Artificial Intelligence (AI), but Legal Intelligence (LI). As India's economy expands and regulatory frameworks grow more complex, governance systems must evolve accordingly. A nation aspiring to be a $5 trillion economy cannot depend solely on manual compliance systems, prolonged arbitration processes, and procedural bottlenecks. Speed, transparency, and institutional efficiency are now central to economic competitiveness and investor confidence. Artificial Intelligence, when applied responsibly, can serve as a powerful enabler of governance reform.

However, generic AI models are insufficient for India's layered legal and constitutional architecture. Our federal structure, sector-specific regulations, evolving taxation frameworks, labour codes, and judicial precedents require systems that are context-aware and rooted in Indian realities. Legal Intelligence must be designed with a deep understanding of constitutional values, due process, and regulatory nuance.

Equally important is the question of sovereignty. Legal and regulatory data is strategically sensitive. As India builds digital public infrastructure across sectors — from payments to identity management — it must ensure that AI systems supporting governance are aligned with national priorities and data security imperatives. Dependence on external technological frameworks in such critical domains may create long-term vulnerabilities.

In this context, recent developments signal a promising shift. CognexiaAI recently launched an AI-native Legal Intelligence platform-India's first-in Mumbai, aimed at supporting arbitration, compliance, and institutional processes. The effort reflects an emerging confidence among Indian technologists to address governance challenges through domain-specific innovation rather than generic automation.

This initiative is part of a larger transformation underway in India's technology ecosystem. CognexiaAI has also secured ₹540 crore in enterprise contracts across Australia and New Zealand for its AI-driven ERP and CRM platforms.

While commercial in nature, these global wins highlight a deeper shift: Indian companies are no longer competing only on cost efficiency, but on performance, scalability, and intellectual property ownership.

High-value AI exports contribute directly to foreign exchange inflows, strengthen India's digital trade position, and expand global demand for Indian engineering and domain expertise. Increasingly, Indian talent is travelling abroad not as outsourced support, but as architects of intelligent systems.

Yet technology must serve a higher institutional purpose. Legal Intelligence should augment, not replace, human judgment. Judges, policymakers, compliance officers, and legal professionals bring interpretative wisdom rooted in constitutional balance. AI can assist by accelerating research, identifying regulatory overlaps, streamlining documentation, and reducing procedural delays-while remaining transparent and auditable.

Accountability must remain central. In governance systems, algorithms cannot function as opaque black boxes. Ethical guardrails, explainable AI, and regulatory oversight are essential to preserve public trust.

India has already demonstrated how digital infrastructure can scale securely and inclusively. The next logical extension of that success lies in modernising legal and regulatory systems with equal ambition.

At this juncture, the evolution from Artificial Intelligence to Legal Intelligence aligns closely with the vision of Atmanirbhar Bharat. Technological self-reliance is not confined to manufacturing or strategic sectors; it extends to digital architecture, governance platforms, and institutional frameworks. Building sovereign AI capabilities for legal and regulatory processes strengthens national resilience and reduces strategic dependency. The transition to Legal Intelligence is therefore not merely technological progress. It is an affirmation that India can design systems rooted in its constitutional ethos while competing globally on innovation and excellence. If harnessed responsibly, Legal Intelligence can help India build governance mechanisms that are faster, fairer, and globally respected. In doing so, it will not only enhance institutional efficiency but also reinforce the foundations of a confident and self-reliant Republic.

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